

Who Wins With Up-Zoning? Quasi-Experimental Evidence from California's By-Right Permitting Reforms for Accessory Dwelling Units*

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Abstract

Can up-zoning small units increase housing supply and who captures the benefits? We study California's 2017 de-regulation of accessory dwelling units (ADUs) through state-preemption of local permitting. We exploit a feature of California's regulatory environment to isolate quasi-random variation in the geography of permitting reform: the California Coastal Zone. Leveraging a regulatory discontinuity at the Coastal Zone boundary and pre-registered designs, we find that permitting reforms increased ADU permitting activity by about 30%, although these effects are small in magnitude. However, the reform did increase land values by 7%. With a simple model, we show that the change in land values reflect the decreased permitting costs associated with the reform package. We also find that permitting reforms are likely not associated with large price spillovers, and the increases in land values are associated with increased property tax collections.

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1 Introduction

Housing costs have been increasing as a share of household expenditures in high productivity places for several decades, which has eroded the ability of workers to benefit from agglomeration economies (Ganong and Shoag, 2017; Paulin, 2018; Hoxie, Shoag and Veuger, 2023). This may have substantial macroeconomic consequences through decreased mobility to high productivity places (Eckert, Ganapati and Walsh, 2020; Duranton and Puga, 2023). Economists have linked inelastic housing supply to higher housing costs, which suggests that regulatory reform may be a potential solution to housing affordability concerns (Glaeser, Gyourko and Saks, 2006; Saiz, 2010; Bartik, Gupta and Milo, 2025; Baum-Snow and Han, 2024). In most cases, regulations that govern the construction of new housing units are under local control, and local policymakers may maximize the welfare of current homeowners, perhaps at the expense of local renters, and without considering the cross-jurisdictional spillovers of their housing supply choices (Gordon, 1983; Fischel, 2001; Gyourko, Saiz and Summers, 2008). These considerations have led to increasing interest among state policymakers in circumscribing the powers of local government through state mandated deregulation through up-zoning and permitting reform.¹ Such reforms raise two questions that we address in this paper: how much additional housing supply can up-zoning and permitting reform deliver, and which stakeholders benefit from state-mandated reforms.

This paper quantifies the effects of California’s 2017 state by-right preemption of local zoning and permitting authority for accessory dwelling units (ADUs). After 2017, most residential lots were up-zoned to allow one ADU, a small one unit addition under 1,200 sq.ft., and one junior ADU under 500 sq.ft.. Additionally, if the ADU building plans met state-set specifications, then a permit or denial report must be issued within 60 days (Gray, 2024).² We exploit a unique regulatory discontinuity to identify the causal impacts of permitting reforms. Within the California Coastal Zone the state-wide preemption of local regulatory authority was superseded by the California Coastal Act, passed in 1976. Permitting jurisdictions within this approximately one kilometer coastal strip of land could not be forced to adopt permitting reforms for ADUs (California Coastal Commission, 2022a,b).³ Therefore, we use a difference in discontinuities design to estimate the causal effects of permitting reforms, which we have partially pre-registered (Grembi, Nannicini and Troiano, 2016; Butts, 2023; Cañedo Riedel and Hoxie, 2025).⁴

¹Several states have passed bills to address housing costs. See for example Kahn and Furth (2023), Schuetz (2025), and NYU Furman Center (2026).

²Annual state-wide permitting of ADUs has increased almost 40 fold since 2017, as shown in Figure 1.

³A recent court case, *Riddick v. City of Malibu* (2025), suggests that the Coastal Zone was not allowing owners to take advantage of the new ADU laws, but it also suggests that there may be heterogeneity across jurisdictions within the Coastal Zone about how stringent their regulations are towards ADUs. Although, Monkkonen, Manville and Lens (2024) and Legislative Analyst’s Office (2015) suggest that all coastal jurisdictions may be relatively stringent when it comes to housing policy, even compared to the rest of California.

⁴The pre-analysis plan was posted in August of 2025 before any of our parcel data from Cotality (2025) was opened. We provide a full change-log between the estimates in this paper and the pre-analysis plan following a novel compliance adjudication protocol using Anthropic (2026) in Appendix B (Cañedo Riedel and Hoxie, 2025). The plan specified three main specifications: a difference in discontinuities for identifying the impacts of reform on permitting and land values per acre, as well as a binned spillover regression across the Coastal Zone boundary for transaction prices. We will identify important deviations from our pre-analysis plan in the data and methods sections of this

We measure the impacts of permitting reform on ADU permitting activity, land values, spillovers, transactions, and property taxes using parcel level property and transaction data from [Cotality \(2025\)](#) and permit data from [California Department of Housing and Community Development \(2025\)](#). We find that California’s by-right permitting for ADUs increased permitting activity by about 30%, representing a small absolute increase in a setting where baseline permitting rates are low.⁵ Despite low take-up, we find that permitting reform increased residential land values by 7%. We show that this price increase reflects the *ability* to build an ADU, which is capitalized in the land and accrues to all parcels whether or not the owner decides to build an ADU.⁶

The Coastal Zone boundary difference in discontinuities design relies on a local parallel trends assumption: in the absence of permitting reforms in 2017, parcels just inside and just outside the Coastal Zone boundary would have remained on the same trends as before 2017. We find small detectable level differences in parcel characteristics between parcels inside and outside the Coastal Zone in our pre-period, reflecting the Coastal Zone’s regulatory effects ([Severen and Plantinga, 2018](#); [Kahn, Vaughn and Zasloff, 2010](#)). However, our difference in discontinuities design can account for initial gaps from the pre-2017 discontinuity across the Coastal zone by incorporating them directly into the estimator. We show all estimates on a mean squared error optimal bandwidth around the coastal zone boundary of 450 meters ([Calonico, Cattaneo and Farrell, 2020](#)). We find that local to the Coastal Zone boundary, parcels on either side were following similar trends in total sale prices per acre, land values per acre, and transaction volumes before the 2017 reforms.

Our main results for permitting activity and land values are robust in several ways. First, we present a range of specifications, including alternative running variable polynomials, additional controls, the inclusion of Coastal Zone segment by time fixed effects, and specifications using alternative bandwidths. While we present our pre-registered designs, the richness of the [Cotality \(2025\)](#) data within our analysis bandwidth allows us to extend our analysis to include a repeat sale model, which is common in the literature ([Bailey, Muth and Nourse, 1963](#); [Soltas and Gruber, 2026](#)). We view this as an improvement over our pre-specified models. This allows us to include a parcel fixed effect to capture static, unobserved characteristics of each parcel.

To investigate spillovers from permitting reforms across the Coastal Zone boundary, we use a reference bin approach that compares changes in outcomes across 50 meter bins that span both sides of our baseline analysis sample to changes in a reference bin inside the Coastal Zone. We focus on two channels for spillovers. First, additional units may decrease nearby parcel values by either creating disamenities or by lowering the market value of similar parcels through supply channels. Second, nearby owners exposed to increased disamenities may elect to sell their parcels and move

paper.

⁵The [California Department of Housing and Community Development \(2025\)](#) data begins in 2018, and Figure 1 suggests that about 95% of permitting occurred in 2018 and afterwards. Therefore, we deviate from our pre-analysis plan and present estimates for permitting activity as a regression discontinuity design. We were also unable to credibly identify likely ADU construction events from the property characteristics in the [Cotality \(2025\)](#) dataset, which is another deviation.

⁶This option value may have been particularly valuable as demand for floor-space increased during COVID ([Mon-dragon and Wieland, 2022](#)).

(Cullen and Levitt, 1999; Card, Mas and Rothstein, 2008). Comparing bins just inside the Coastal Zone to changes in our reference bin, we find no detectable evidence of either total price spillovers or transaction frequency spillovers. Our total sale price spillovers also suggest that the increase in land value from the right to build an ADU outweighed the decreases from disamenities and additional unit supply.

To rationalize our low take-up and increased land value results, we build a simple open city model similar to Turner, Haughwout and van der Klaauw (2014), with the addition of property taxes and an endogenous build decision. This simple model allows us to characterize our difference in discontinuities estimates for the impact of permitting reforms on land values using the difference between the land prices on each side of our model city.⁷ This exercise produces three scenarios under which we can interpret our land value results. First, if no lot is of high enough quality to build an ADU, the estimates should be zero. Second, in settings in which it is only worth building an ADU of a given lot quality on the lower-cost side of the city, our difference in discontinuities estimate should increase with lot quality as our estimates will reflect the full option value of building an ADU. Finally, in settings where lot quality is high enough to build on both sides of the boundary, our estimates will reflect the differences in permitting costs between the lower and higher cost jurisdictions and our difference in discontinuities estimates should not show heterogeneity in lot quality. Then, our model suggests that property tax shelter, the extent to which a lot is taxed on a below market basis, may depress transactions and in turn dampen ADU construction. These features motivate our heterogeneity analysis.

In order to better understand how permitting reform increased land values, and motivated by our simple open city model, we explore treatment effect heterogeneity by lot quality for our permitting and land value results. We explore two versions of lot quality: excess square feet on a parcel available for an ADU and distance to the nearest central business district (CBD).⁸ We find stronger heterogeneity for lot quality as proxied by distance to the CBD, reflecting that lots with less excess space not covered by a structure may also be closer to CBDs.⁹ Our estimates of heterogeneity by lot quality suggest that the effects of permitting reform on permitting activity are larger when parcels are closer to CBDs, with a statistically significant interaction term.¹⁰ When we turn our attention to land values, we find that our preferred repeat sale estimates do not have a statistically significant interaction with CBD distance. This suggests that our land value results reflect the changes in permitting costs after ADU reform, rather than the full market option value that embeds the value of the ADU itself net of construction costs. This implies that permitting reforms for ADUs lowered permitting costs by 7% of land values.

Even though cities lose autonomy under state-preemption, our finding that land values increased as a response suggests that cities may have seen a revenue gain. As properties transact and ADUs

⁷This requires assuming that cross-jurisdictional spillovers are negligible, which is consistent with our empirical results.

⁸We get CBD coordinates from Holian (2019). We discretize both heterogeneity variables so that they are dummies for above and below analysis sample medians for excess lot square feet and distance to the CBD.

⁹Marantz, Elmendorf and Kim (2023) suggests that access to jobs is a key driver of ADU permitting.

¹⁰For details on estimating and interpreting the CATE, see Section 4.4.

get built, the value of the permitting reforms enter the property tax base through two channels. First, if a lot transacts, then the reassessment after sale should reflect the value of permitting reform that was capitalized into the sale price. Second, if an ADU is built on a lot, that ADU’s assessed value is added to the existing property tax base, without triggering a full reassessment (San Francisco Assessor Recorder’s Office, 2025). We find that property tax collections increased by about 1.5% in our preferred specification. Although these results suggest that ADU reforms increased local revenues, whether or not these reforms resulted in a net fiscal gain depends on the composition of new residents and their preferences for, and use of local services (Poterba, 1997; Hilber, 2017).

Finally, we provide suggestive evidence that permitting reforms to ADUs didn’t have a detectable effect on rents.¹¹ With so few additional units permitted as a result of the reform package it is unlikely that they substantially lowered rents.

This paper primarily contributes to the literature on regulation, home prices, and housing supply. Several papers link land use restrictions to decreased housing supply and higher home prices (Glaeser, Gyourko and Saks, 2006; Bartik, Gupta and Milo, 2025; Turner, Haughwout and van der Klaauw, 2014). Moreover, Baum-Snow and Han (2024) find that about 40% of the supply response to price changes comes from small renovations, which could include ADUs. However, how effective decreasing local regulations are at lowering housing costs is currently being debated in the literature (Louie, Mondragon and Wieland, 2025; Duranton and Puga, 2023). Our findings help rationalize this divide. Although we find that permitting does increase as a result of reducing local discretion, we don’t see a large shift in units as a share of all parcels. Therefore, it is unlikely that ADUs have meaningfully helped to depress rental prices in the broader market for small rentals, although they may reduce prices locally. In contrast, the increase in land values, created by the ability to build an ADU does impact a meaningful number of parcels. It is likely that aggregate measures of home prices commonly used in macro studies capture the increased land values that can’t be untangled from structure values.¹² Our results suggest that permitting and zoning may not be the the only barrier to increasing housing supply, and other factors like incumbent preferences, construction costs, use restrictions, tax incentives, and other elements of the construction process may warrant additional focus in future research.¹³

Second, we contribute to an active literature on up-zoning and permitting costs. A survey from

¹¹We use the extent to which a block group overlaps with the Coastal Zone as an instrument for changes in ADU permits per housing unit in 2016, and we measure percent changes in average gross rents between 2016 and 2024 (U.S. Census Bureau, 2016, 2024; IPUMS NHGIS, 2026). Our results are insignificant, but our standard errors cannot rule out meaningful effects. Our first stage F-statistic is 19.5, and we include a number of controls for terrain and density, the area age structure of owners, nearby changes in job access, other permitting activity, and access to amenities. These regressions were not pre-registered.

¹²For example, Ganong and Shoag (2017) and Hoxie, Shoag and Veuger (2023) use housing values from the American Community Survey, and Louie, Mondragon and Wieland (2025) uses data from the S&P Cotality Case-Shiller US National Home Price Index and the American Community Survey. In these data sources land and structure values are not differentiated. However, it is also possible that increased housing production from reduced supply constraints could still be consistent with aggregate price increases in general equilibrium.

¹³For example, notice the large gaps between permitting and construction timelines for California cities compared to other jurisdictions in D’Amico, Soltas and Wang (2026).

Freemark (2023) finds that up-zoning is often associated with land value increases, but per unit cost reductions only materialize if units get built. D’Amico, Soltas and Wang (2026) find that permitting and construction timelines can vary widely across localities. The two closest papers to ours also study how land values change with permitting costs. Gold and Tarascina (2026) study ADU up-zoning and permitting reforms in Seattle, WA. Using a difference in differences approach, they find an increase in permitting activity and an option value of 4% associated with lots exposed to reforms. Our estimates suggest a higher land value change from permitting reforms, of at least 7% of land value. This may reflect differences across our settings in the value of building an ADU, the initial and post-reform regulatory environments, and construction costs. Next, Soltas and Gruber (2026) find that developers will pay over 50% more for already permitted land in Los Angeles, CA.¹⁴ Their results are significantly larger than ours in part because they are looking at all land and structure types, not just small units on small residential land, as we do. Moreover, they are valuing the whole permitting process, whereas we are only evaluating how much reforms lowered permitting costs. Therefore, we should expect our results to be smaller in magnitude than theirs. Many of the papers in this literature focus their estimates within a single city or region. Our setting is unique in that the Coastal Zone extends along the California coast from Mexico to Oregon, offering a large and varied quasi-experimental setting to study permitting reform, which is rare in the literature. We also offer an identification improvement. In many applied papers, variation in regulation is chosen by the same jurisdictions whose outcomes are studied (Gold and Tarascina, 2026; Gyourko and McCulloch, 2024; Bartik, Gupta and Milo, 2025; Turner, Haughwout and van der Klaauw, 2014). In our setting, the state imposes reform on localities, and exempt segments of land were determined by legislation passed decades before the reforms that we study.

The economics literature around up-zoning has also focused on cross-plot spillovers from development. In some settings, negative spillovers from up-zoning can be substantial. Diamond and McQuade (2019) find negative spillovers for comparatively large low income housing projects placed in higher income neighborhoods. In a more residential setting, Gyourko and McCulloch (2024) find that 100 homes within 500 meters reduces total home prices by between 1-2%. Blanco and Sportiche (2026) find that smaller housing projects may increase home prices, whereas negative spillovers may be primarily features of larger development projects. In our setting, it is unlikely that any parcel is exposed to an increase in ADU density of the magnitudes in either Diamond and McQuade (2019) or Gyourko and McCulloch (2024), and we do not find evidence of large spillovers.¹⁵

There is a smaller literature linking regulations to land values. Grout, Jaeger and Plantinga (2011) use a regression discontinuity design to link Portland, Oregon’s urban growth boundary

¹⁴On a per acre basis, Soltas and Gruber (2026) find circumventing permitting is worth around \$2 million per acre, whereas our estimates imply a benefits of about \$500,000 per acre.

¹⁵There is small literature on spillovers for ADUs. Tanrisever (2025) estimates a cross-plot spillovers from ADUs of about 3% in Los Angeles, CA. Davidoff, Pavlov and Somerville (2022) finds an equally sized but insignificant spillover effect for all lots, but does find spillovers large spillovers for higher value lots. Our results are not precise enough to reject spillovers of this magnitude. Gold and Tarascina (2026) also do not find evidence of significant cross-plot spillovers from ADUs in Seattle, WA.

to increases in land prices. They attribute these changes to the now limited supply of available land. Similarly, [Cunningham \(2007\)](#) finds that growth controls around Seattle, WA reduced the real option value of land, lowering land values and accelerated development. [Turner, Haughwout and van der Klaauw \(2014\)](#) also find that increased regulation lowers land values. Our findings suggest that the converse may also be true when regulations are relaxed. Our findings show that these results are not perfectly symmetric: we show that in our setting de-regulation for small units can increase land values and increase permitting, albeit modestly.¹⁶

This paper proceeds as follows. In Section 2, we discuss the institutional details surrounding California’s ADU reforms in the context of our Coastal Zone border discontinuity design. We outline our parcel-level dataset and econometric methods in Sections 3 and 4, respectively. We discuss our partially pre-registered main results for ADU permitting, land value changes, and spillovers in prices and transactions in Section 5. Then we build intuition on how homeowners respond to permitting reform for ADUs with an open city model that features endogenous ADU construction, property taxes, and a characterization of our difference in discontinuities estimator in Section 6. This simple model motivates our discussion in Section 7, which explores heterogeneity in our main estimates as well as the fiscal and rental impacts of permitting reform. Finally, we offer brief concluding remarks in Section 8.

2 Housing Policy and ADUs in California

Since at least 1926 cities have had the power to regulate local housing construction through zoning ([Village of Euclid v. Ambler Realty Co., 1926](#)). However, the exact powers that cities possess have largely been determined by the states ([Davidson et al., 2020](#)). In 2016, the State of California passed the first of several laws that aimed to encourage the construction of ADUs by removing local regulatory barriers enforced through zoning, permitting backlogs, parking requirements, and other pre-construction hurdles ([Dillon and Khouiri, 2016](#); [Minott, 2023](#); [Gray, 2024](#)).

Prior to 2016, building an ADU in a particular jurisdiction was largely under local control. Although the State passed legislation aimed at streamlining ADU permitting in 2002, local governments were still able to use the discretion inherent to the permitting process, as well as a myriad of additional requirements, as a *de facto* veto of the attempted state-wide reforms.¹⁷ This left cities with the power to impose additional site requirements and inspections, impact fees, parking requirements, and other cost-increasing measures on proposed ADUs ([Brinig and Garnett, 2013](#)).¹⁸

Starting in 2016, the State has passed a series of reforms that stripped local jurisdictions of their power to impose barriers to ADU construction ([Gray, 2024](#); [Minott, 2023](#)). Assembly Bill (AB) 2299 and Senate Bill (SB) 1069 required “by-right” permitting of ADUs on most residential

¹⁶This literature is also related to the more standard capitalization literature of tax and spending policies, such as [Oates \(1969\)](#), [Black \(1999\)](#), [Linden and Rockoff \(2008\)](#), and [Baylis and Boomhower \(2023\)](#)

¹⁷Moreover, a second bill in 2003 was proposed to address local roadblocks, but Governor Schwarzenegger vetoed the bill since it encroached on local authority ([Brinig and Garnett, 2013](#)).

¹⁸For a more complete discussion of the regulatory environment in the 150 largest California cities surrounding ADUs, please see [Brinig and Garnett \(2013\)](#).

lots with a maximum processing time of 120 days. This meant that if an ADU met state-set requirements and was in a residentially zoned area, then the ADU must be approved. The reforms required jurisdictions to allow ADUs on lots that already have single family homes, and removed a number of local requirements such as street access, new utility hookups, and large setbacks. The reforms also reduced existing parking requirements for ADUs near transit or ADUs built as part of a primary residence (Minott, 2023; County of Humboldt, 2017). These reforms took effect in 2017.

Continuing its efforts in the 2017 legislative session, the state passed SB 229 and AB 494. These bills clarified that local jurisdictions cannot ban homeowners from renting out ADUs separately from the attached residence, that ADUs can be part of proposed single-family housing projects, and limited parking requirements to only one additional space per unit. Then in 2019, the legislature continued its reforms to ADUs with three new bills, AB 68, AB 881, and SB 13.¹⁹ Together, these pieces of legislation removed minimum lot sizes, capped setbacks, eliminated floor area ratios, prohibited owner-occupancy requirements, expanded garage conversions, and allowed ADUs in multi-family areas. These additional limitations on local governments were paired with increased enforcement tools through the California Department of Housing and Community Development. The state also limited the ability of HOAs to restrict ADU rentals in AB 670 (2019) and AB 3182 (2021), clarified that selling ADUs separately from the main house is legal in AB 587 (2019), and required permitting jurisdictions to communicate existing grants and financing programs to encourage ADU construction in AB 671 (2019). Finally, in 2022 the state clarified that all ADUs up to 16 feet high are permitted, further limited the setback restrictions, and required localities to enumerate to applicants why their permits were denied (AB 2221 and SB 897) (Minott, 2023; Gray, 2024).

After implementing the full set of reforms for ADUs, a typical residential lot in California is entitled to build a 1,200 square foot ADU and a 500 square foot Junior ADU with only a 4 foot setback on the side and rear. Moreover, existing detached structures can be converted into ADUs of up to 1,200 square feet. These projects must be approved within 60 days or issued a denial report so they can reapply (California Department of Housing & Community Development, 2026).

However, the set of ADU reforms could not be enforced for land within the California Coastal Zone, a roughly 1 kilometer strip of land along California’s entire coastline.²⁰ The California Coastal Zone’s land use has been governed by the Coastal Act and the California Coastal Commission since 1972 (California Coastal Commission, 2022a). The Coastal Act prohibits the state from imposing the reforms to ADUs that “increase the intensity of” land use on local jurisdictions within the Coastal Zone, such as up-zoning and by-right permitting, which are key features of the state’s ADU reforms (California Coastal Commission, 2022b). Instead, within the Coastal Zone, new development must receive a special coastal development permit, and any of the state-wide reforms may only be implemented by jurisdictions updating their local coastal plans voluntarily (California

¹⁹These bills also expanded the reforms to Junior ADUs, which are smaller than regular ADUs (Minott, 2023).

²⁰The Coastal Zone can extend from 1,000 yards inland to several miles. Some of this variation can be seen in Figure 6.

Coastal Commission, 2022b; Severen and Plantinga, 2018; Kahn, Vaughn and Zasloff, 2010).²¹ In practice, the Coastal Zone is governed by local coastal plans, which cover 145 segments spanning the state’s coastline from Mexico to Oregon. The local jurisdiction (e.g. the City of San Diego, UC San Diego, or the County of San Diego) that overlaps with the Coastal Zone will file a plan that details how the land will be managed in accordance with the Coastal Act.²² These plans must be approved by the Coastal Commission, and can periodically be updated.²³ In Figure 6, we show how the Coastal Zone boundary (in blue) overlaps with real estate transactions (dots), incorporated cities (orange), and unincorporated land (white) in San Diego county.

There are two additional features of California housing policy that will play an important role in how we evaluate the impacts of permitting reforms. First, before the ADU reforms, housing regulations were especially restricted in coastal California. Monkkonen, Manville and Lens (2024) collect data from the 2014 California Housing Element’s Regional Housing Needs Assessment (RHNA) process, in which the state and regional governments set housing targets for housing unit growth over the next 8 years for each city and county. Then, the cities respond to these targets by reporting zoned capacity figures that reflect their assessments of how many new units they can accommodate. Following Monkkonen, Manville and Lens (2024), we view the ratio of zoned capacity to existing housing units as a reflection of both regulatory stringency in a jurisdiction, and that jurisdiction’s stance towards additional housing. More (less) zoned capacity per unit would reflect both a lighter (more stringent) regulatory environment and more (less) favorable local attitudes towards new units.²⁴ In Figure 5a we show a histogram of each jurisdiction’s zoned capacity per housing unit in California. The median jurisdiction’s zoned capacity per housing unit statewide reflects zoning for a 13% expansion in units over 8 years. Figure 5b shows the zoned capacity per unit for each parcel in our analysis sample. Within 450 meters of the Coastal Zone boundary, the median parcel sits in a jurisdiction that only zoned for an 8% increase in units over 8 years, which is reflected in the compressed distribution in Figure 5b. These figures suggest that coastal areas were in general relatively restrictive in their policy attitudes towards new units, so ADU reforms are likely to meaningfully change the regulatory environment in this setting.²⁵

Second, California’s property tax policies likely interact with the effectiveness of the ADU reform package. Proposition 13 allows properties to be assessed at market value after they are purchased,

²¹As of this 2022 memo, only 5 cities have voluntarily adopted parts of the reforms to ADUs. These include Santa Cruz, Santa Barbara, Encinitas, San Mateo, and Pacifica (California Coastal Commission, 2022b).

²²In most cases, cities like San Diego have portions of land within the Coastal Zone and portions outside of the Coastal Zone. In this case, the local coastal plan is only applicable in the fraction of land within the city that overlaps with the Coastal Zone.

²³A recent court case showed that the Coastal Commission has been requiring coastal development permits for ADUs (Riddick v. City of Malibu, 2025).

²⁴In the text, Monkkonen, Manville and Lens (2024) refer to this measure as unbuilt capacity. We also extend this measure to include the unincorporated areas and non-reporting cities by applying the residual RHNA target units to these areas, distributed in proportion to their populations.

²⁵Even if a city zones for units, these units may not be delivered. Glaeser and Gyourko (2025) show that Los Angeles, San Diego, and San Francisco also had some of the lowest housing stock growth between 2010 and 2020, adding about half a percent per year. This also suggests that realized housing construction is likely below what a city is zoned to accommodate.

and then subsequent assessments, on which property taxes are applied, are capped at 2% growth each year. A property will be reassessed above this cap in one of two ways. First, if the property is sold, then a full reassessment takes place, and the property is assessed at or near market value. Second, if major improvements are made on the property, then a partial reassessment takes place on the improvements alone. If an improvement like an ADU is built, then the net value of the ADU is added to the baseline assessed value, but the full land and un-changed structure values are not reassessed ([San Francisco Assessor Recorder’s Office, 2025](#)). This results in most properties being taxed on an assessed value below their market value, which can create lock-in effects ([Wasi and White, 2005](#)). However, in participating counties, owners over the age of 55 can transfer their below market assessed value to a new primary residence through Proposition 90 ([Ferreira, 2010](#)).²⁶ We will leverage these features of California’s property tax system to assess how property taxes may mediate the impacts of the ADU reforms.

3 Data

Our dataset construction, variable definitions, and econometric designs were pre-registered before we began our data-work with the [Cotality \(2025\)](#) parcel data.²⁷ As a result of unforeseen features our dataset, we have deviated in a limited number of ways from our original plan. In this section and the methods section, we will identify deviations from our pre-analysis plan as appropriate. In addition, we provide a full artificial intelligence enabled audit of our compliance, following a novel adversarial protocol, in Appendix B.²⁸

To investigate the impacts of permitting reforms for accessory dwelling units we build two datasets that link parcel-level property characteristics from assessment records, real estate sales data, and building permits. The first dataset is a balanced parcel panel that has one observation for each parcel in each year. The second dataset uses transaction records, which may have multiple observations or no observations for a parcel in a given year. Both datasets are constructed for the 15 coastal counties of California from 2010 to 2024, as data are available.²⁹ Our final dataset contains just over 4 million residential parcels and over 2 million transactions between 2010 and 2024.

The [Cotality \(2025\)](#) (formerly CoreLogic) property data create the backbone of our balanced parcel-year dataset. [Cotality \(2025\)](#) collects and standardizes county assessor records, which are used to levy property taxes. These records include information on assessed values for the structure, building and lot characteristics, and location identifiers including coordinates and a street address. Using property identifiers, we create a balanced panel of about 4 million unique properties between

²⁶The benefit applies to primary residences and may be used one time. Participating counties with Coastal Zone overlap are: Los Angeles, Orange, San Diego, San Mateo, Ventura. Other participating counties are: Alameda, Riverside, San Bernardino, Santa Clara, Tuolumne ([California State Board of Equalization, 2025b](#)).

²⁷The 2025 pre-analysis plan can be accessed on [OSF](#).

²⁸Our adversarial protocol builds on practices outlined by [Clemens and Mahajan \(2026\)](#).

²⁹From north to south, the 15 coastal counties of California are: Del Norte, Humboldt, Mendocino, Sonoma, Marin, San Francisco, San Mateo, Santa Cruz, Monterey, San Luis Obispo, Santa Barbara, Ventura, Los Angeles, Orange, and San Diego.

2010 and 2024. This requires us to summarize building and lot characteristics when a parcel has multiple assessments within a given year, as well as to expand our dataset to fill in gaps between assessments.³⁰

We prune these data, expanding on a process outlined in [Frank et al. \(2025\)](#). First, we limit our dataset to parcels that are ever listed as having a land use or property use that is for small residential uses.³¹ Additionally, the parcel must have an assessed value above \$1,000 (real) USD, have a room to unit ratio within the 1st and 99th percentiles, and have an internally consistent implied age. The impacts of each sample restriction are tracked in Table A.1.³² The final parcel-year sample has just over 66 million parcel-year observations between 2010 and 2024. We use the balanced panel dataset to evaluate the share of parcels that permit an ADU, that transact, or how much property tax each parcel pays in a given calendar year.

In addition to the property records from [Cotality](#), we incorporate data on building permits from the State of California’s housing element progress reports ([California Department of Housing and Community Development, 2025](#)). As part of the post-2016 housing reforms, the State of California began requiring local jurisdictions (cities and counties) to report on the number of land use entitlements, building permits, and certificates of occupancy they issue each reporting year, by project, starting in 2018.³³ This state-wide permitting dataset is then linkable to our property and transaction data using either an assessor parcel number (APN) or matching the reported addresses directly.³⁴ One limitation is that we do not have parcel-level permit data available to us before 2018.³⁵

Figure 1 compares the [California Department of Housing and Community Development \(2025\)](#) post-2018 state-wide counts of ADU building permits that we attached to our parcel dataset with a prior vintage of the same time series.³⁶ The combined series for state-wide ADUs from 2017 to 2024, using the depreciated CA DHCD series (blue) for 2017 and the current housing element

³⁰To expand the data, we take property characteristics from the last available assessment and carry them forward through the panel.

³¹This includes parcels marked as single family residences, duplexes, or with land uses marked as residential uses between 1 and 10 units. The vast majority, over 90% of our final dataset consists of single family homes, as shown in Table A.3.

³²We explore how these sample restrictions interact with our research design in Appendix Table A.2. These restrictions differentially remove observations from inside the Coastal Zone in levels, but these differences appear to be stable over time in our balanced panel dataset.

³³We are unable to account for illegal ADU construction, which may be a meaningful share of ADUs constructed pre-reform. [Jo et al. \(2025\)](#) use satellite data and a machine learning model to detect un-permitted ADUs in San Jose, CA between 2016 and 2020. Their estimates suggest that informal units may outnumber formal units.

³⁴To do this matching, we build a hierarchical matching algorithm that prioritizes matching by APN and then uses the available address information. After we separate all of the listed addresses associated with a project, we attach a permit to all addresses associated with it first using the APN, then an exact match on the address, then on the Google location key from the [Google Maps Platform \(2026\)](#), and finally on previously listed APNs. We describe this process in Appendix E.

³⁵Another issue is that we can only measure permitting. We attempted to measure construction of ADUs using the [Cotality \(2025\)](#), but it was clear that we could only capture the effects of likely construction events in general rather than specifically the construction of ADUs.

³⁶The prior vintage time series comes from the California Department of Housing and Community Development’s (CA DHCD) Housing Element, but we were unable to locate the original official source. The same time series of state-wide ADU permits going back to 2012 is corroborated by [Gray \(2024\)](#), [Minott \(2023\)](#), and [Peterson \(2021\)](#).

(orange) after 2017, shows about 176,000 permits issued in total post-reform. Pre-reform, the depreciated CA DHCD series shows only about 4,000 permits issued between 2012 to 2016. These series suggest that the overwhelming majority of ADU permits issued between 2012 and 2024 are captured by the detailed post-2018 permit data from [California Department of Housing and Community Development \(2025\)](#), shown as an orange dashed line.³⁷ Although we do not have any pre-reform permits to match, Figure 1 suggests that ADU permitting before the reforms for ADUs was generally negligible.³⁸

For analyzing the impacts of permitting reforms on land values and sale prices, we use a dataset based on the [Cotality \(2025\)](#) transaction records. These transactions data also include information on the real property being sold, the date of the sale, an arms-length transaction indicator, and the nominal sale price. We attach building characteristics, assessment information, location, land use categories, and other parcel level information to these transaction records from the balanced parcel panel using the unique property ID provided by [Cotality](#). Next, we trim the sample to include only residential properties and arms-length transactions as well as properties with assessed values greater than \$1,000 (real) USD, (real) sale prices between the 1st and 99th percentiles, plausible transaction volumes within a single year (99th percentile and above are trimmed), room to unit ratios between the 1st and 99th percentiles, (real) land values per acre greater than \$1 USD, and an internally consistent implied age of the structure based on the date of sale and the building date. We describe how these restrictions impact our final 2 million transaction sample in Table A.1.³⁹

Our first outcome variable is a dummy for whether or not a parcel received an ADU permit in a particular year, which is built on our balanced transaction panel. This variable takes a value of 1 if the parcel has an associated ADU permit in a particular calendar year, and a value of zero otherwise. We only use this variable starting in 2018, as shown in Table 2. In the full county sample, only about 0.3% of parcels had an ADU permit in a given year. Although this is a flow measure, it still reflects the fact that ADU permitting, even with reforms, appears to be relatively rare. However, Table 2 does suggest that ADU permits are likely a large share of building permits overall. For legibility in tables, we scale our ADU permit dummy to reflect permits per 1,000 parcel-years.⁴⁰

³⁷There are slight discrepancies between the two series. The detailed data from [California Department of Housing and Community Development \(2025\)](#) include unincorporated permits, and it is unclear whether or not these areas are included in the older series.

³⁸Using only permits from 2018 and after is a deviation from our pre-analysis plan. Given that no state-wide dataset is publicly available before 2018, and collecting permits for all permitting jurisdictions within the coastal counties directly was unlikely to yield a substantial number of ADU permits, we decided to only use permits available in [California Department of Housing and Community Development \(2025\)](#).

³⁹We also show how these restrictions interact with the Coastal Zone boundary on our analysis sample in Appendix Table A.2. These restrictions drop more transactions inside the Coastal Zone than outside, and this relationship strengthens after 2017. However, the magnitude of these changes is small, an additional 1% of dropped transactions get dropped on top of a base rate of over 80% of transaction being dropped. The vast majority of transactions are dropped because they are not registered as arms-length, meaning both parties do not have a prior relationship such as family ties.

⁴⁰We initially planned to build a variable that identified ADU construction in the [Cotality \(2025\)](#) data as part of our pre-analysis plan. However, the building characteristics available to us are not detailed enough to separate construction related to an ADU from construction in general.

Our second outcome variable that we build on our balanced panel dataset is a transaction dummy. We construct a dummy variable using both the panel dataset and transaction records from [Cotality \(2025\)](#) that takes a value of 1 if a parcel has an arms-length transaction in a given calendar year, and 0 otherwise. We will primarily use this variable to investigate transaction spillovers that may result from disamenities associated with ADUs.

We now turn to outcome variables constructed on our transaction dataset. Since land values are not measured directly in our transaction data, we multiply the transaction sale price by the land share of total value from the first available assessment after the transaction to create our measure of (real) land values.⁴¹ We then take the total land price and divide it by the parcel land acreage to get the (real) land value per acre.⁴² As shown in Table 1, on average, before the 2016 permitting reforms, transactions in the full 15 county sample averaged \$3.1 million per acre (2025 USD), compared to \$4.5 million per acre (2025 USD) after the reforms.

How does our measure of land values per acre, built from market transactions and the first post-sale assessed ratio of land and total value, compare to the literature? One common method is to use vacant parcels to estimate land values ([Albouy, Ehrlich and Shin, 2018](#); [Nichols, Oliner and Mulhall, 2013](#); [Haughwout, Orr and Bedoll, 2008](#)). The second method is to try to estimate the value of the land by accounting for the cost of replacing any structures on the land and removing it from the total sale price ([Davis and Palumbo, 2008](#); [Davis et al., 2017](#)). In our setting, using tear downs or vacant lots is difficult because there are very few instances of suitable parcels in coastal California and near the Coastal Zone boundary. Moreover, to estimate the land value under a built parcel requires a form of spatial interpolation that would be ill-suited for measuring the effects of a regulatory environment that changes discretely. When a property is being assessed, and comparable land sales are not available, an assessor will use a “land residual” approach that estimates the value of the structures and assigns the remaining value to the land itself ([California State Board of Equalization, 2025a](#)). As an alternative to our preferred measure of land values built from the total sales price and assessed land value ratio, we will present estimates that reflect land values by using time-updating hedonics to remove the value of current structures from the total sales price.⁴³ As a consistency check, we compare our preferred land value measure, derived from sale and assessment data, to an alternative measure from [Davis et al. \(2021\)](#) directly in Figure 2. [Davis et al. \(2021\)](#) provide a national set of estimates of land values built using appraisal data on new homes. To make our measures comparable, we show correlations of average residential land values per acre for each of the 15 coastal counties in all available years (2012-2024). Each dot represents a county year, with land value estimates from [Davis et al. \(2021\)](#) on the vertical axis,

⁴¹Typically, assessments are updated after arms-length transactions in California, so this is the best available measure of the land share of total value available to us. Moreover, assessors are supposed to estimate the land value by removing the structure value from the total sale price, similar to an approximated hedonic approach ([California State Board of Equalization, 2025a](#)). We view combining the transaction price and assessed land value share as an improvement over using the assessed land value directly, which can have issues in panel settings ([Ahlfeldt, Koster and Vu, 2025](#); [Ihlanfeldt and Martinez-Vazquez, 1986](#)). This change is a deviation from our pre-analysis plan.

⁴²We deflate all monetary variables by the personal consumption expenditures chain-type price index to convert values to 2025 USD ([U.S. Bureau of Economic Analysis, 2025](#)).

⁴³We do not have access to construction data to improve on this method, which is why it will be used as robustness.

and land value calculations from our transaction and assessment data on the horizontal axis.⁴⁴ A simple linear fit shows an R-squared of 0.9 between the two measures, and a coefficient of about 1.1. Although our measure does not perfectly fit the equality line, these results suggest that these two measures of land values are relatively similar.

To capture the impacts of spillovers from increased density across the coastal zone boundary, we use the (real) total sale price per acre. We construct sale prices per acre as the transaction sale price divided by the parcel land acreage from its matched assessment records. In Table 1, we can see that total sale prices increased from on average \$5.6 million per acre (2025 USD) before the reforms to almost \$8 million per acre (2025 USD) between 2017 and 2024.⁴⁵ The typical total sale price for a home in our 450m sample is between about \$2 million USD and \$3 million USD, but there is substantial variation given that our sample spans the entire state.

The core of the regression discontinuity and difference in discontinuities designs is the running variable. Using each parcel’s coordinates, we construct our running variable as distance to the Coastal Zone boundary in meters using shapefiles from the [California Coastal Commission \(2014\)](#).⁴⁶ We then center our running variable at 0 so that values for the distance to the Coastal Zone are less than 0 when a parcel is within the Coastal Zone, and greater than 0 when a parcel is outside the Coastal Zone. We will use this running variable to define the bandwidth for our final analysis sample, as well as to assign parcels to distance bins from the coastal zone boundary. We selected our analysis bandwidth by applying the mean squared error optimal bandwidth selection criteria to a regression discontinuity in land values per acre estimated on pre-period data (2010 to 2016) ([Calonico, Cattaneo and Farrell, 2020](#)).⁴⁷ This gives us an optimal bandwidth of 450 meters from the boundary. We also calculate each parcel’s nearest Coastal Zone boundary segment, which corresponds to the 145 local coastal plan’s that govern each Coastal Zone segment.

We also build several covariates, which we use both to ensure the robustness of our estimates and to characterize pre-period differences across the Coastal Zone boundary. First, we construct a set of fixed effects that are meant to capture hedonic characteristics of the parcel. These fixed effects are a set of interacted bins that capture the parcel’s build year, land use category, number of bedrooms, number of bathrooms, and building square footage before the permitting reforms. We construct fixed effects in a similar style to [Yagan \(2019\)](#), which include the full interaction of the following groups, with bins for missing values in each category. The bins are: above or below county median effective age (and a missing bin), land use category (single family, other, or missing), number of bedrooms (0-2, 3-4, 5 and up, and missing), number of bathrooms (0-2, 2-4, 4 and up, and missing), and within county tertiles of building square footage (and a missing bin). In most cases, we include versions of these fixed effects constructed before our 2017 post-ADU reform period. However, to accompany the alternative land value measure robustness exercise

⁴⁴[Davis and Palumbo \(2008\)](#) have continued to update this series post publication.

⁴⁵Note that if a 3,500 square foot home in La Jolla, CA is on a 0.2 acre (8,700 square foot) lot and sells for \$5 million USD, then its price per acre is \$25 million USD.

⁴⁶To be included in our sample, a property must have a valid identifier and a valid coordinate. If a parcel coordinate is unavailable, we use the block centroid when possible.

⁴⁷We used the `rdrobust` package in R ([Calonico et al., 2017](#)).

we allow these bins to update throughout the panel. We also use parcel coordinates to attach a rich set of spatial data, such as distance to the nearest CBD from [Holian \(2019\)](#),⁴⁸ distance to the nearest coastline from [California Department of Fish and Wildlife \(2026\)](#), and elevation from [U.S. Geological Survey \(2026\)](#) as well as data from the American Community Survey (ACS) by block group ([U.S. Census Bureau, 2016](#)).⁴⁹

We also build additional variables to explore heterogeneity in our estimates. First, the decision to permit and build an ADU is likely a function of lot quality. We identify two major determinants of ADU germane lot quality, although in principle there could be many: the amount of unbuilt “excess” space on the lot where an ADU could be placed and the distance of the lot from the nearest central business district. A lot with additional area not covered by existing structures will be better able to accommodate an ADU without eliminating the existing yard or open space compared to a more space-constrained lot. Lots with more excess space can also accommodate larger ADUs. Additionally, lots closer to the central business district in standard urban models, such as [Alonso \(1964\)](#), [Muth \(1969\)](#), and [Mills \(1967\)](#), will have greater access to jobs and likely command higher rents. Therefore ADUs closer to the city center (CBD) will likely be more valuable to the owner than ADUs built towards the periphery. We will include excess lot square footage and distance to the CBD as two binary variables. The first for excess lot square footage takes a value of 1 when a parcel has above median excess square feet available for an ADU and 0 otherwise. The second variable takes a value of 1 when the lot is above the median distance to the nearest CBD, and 0 otherwise. In turn, these two factors should also interact with the magnitude of the land value jump at the boundary.

Next, we will explore how property taxes interact with permitting reforms. We will use two variables to evaluate the role of property taxes in transmitting or muting the effects of permitting reforms. The first is to use the estimated property tax shelter for each parcel in 2016. We construct each parcel’s property tax shelter as a function of an estimated market value from a hedonic price regression and the imputed assessed value under Proposition 13 in 2016. The formula for constructing our measure of property tax shelter is:

$$PropTaxShelter = \frac{M - A}{M} \quad (1)$$

where M is the estimated market value in 2016 and A is the assessed value in 2016 for each parcel.⁵⁰ The second variable that we use to explore the role of property taxes in our estimates a parcel’s block group level exposure to Proposition 90. We construct this variable using share of home owners above the age of 55 in 2016, attached to each parcel by block group ([U.S. Census Bureau, 2016](#)).⁵¹ Since property tax shelter is likely not independent of build costs, as more tenured owners may both accrue large tax shelters, face more difficulties completing construction projects on their

⁴⁸We choose the Google Earth points, but all choices are highly correlated ([Holian, 2019](#)).

⁴⁹We also use jurisdictional shapefiles from [California Office of Emergency Services \(2015\)](#).

⁵⁰We describe the construction of the property tax shelter variable, including the underlying hedonic estimates in Appendix E.

⁵¹In 2021, this benefit was extended statewide as part of Proposition 19.

lots, or have changing preferences for density due to age, these variables likely provide imperfect evidence of the role that property taxes in the effectiveness of ADU reforms.

In Tables 2 and 1, we present means and standard deviations for variables in our 450 meter bandwidth analysis sample. Columns 1 through 4 present summary statistics for the sub-sample of parcels inside the Coastal Zone from 2010 to 2016, inside the Coastal Zone from 2017 to 2024, outside the Coastal Zone from 2010 to 2016, and outside the Coastal Zone from 2017 to 2024. Across all Columns, around 90% of parcels are single family parcels. Buildings on these parcels tend to be relatively old, with build years around 1970.⁵² The houses on these structures tend to have around 3 bedrooms, 2.5 bathrooms, and have a building square footage of about 2,000 square feet.⁵³

We graphically show the location of our final 450 meter bandwidth sample in Figure 3. We present the density of our Coastal Zone sample by latitude as a solid blue line. Our sample ranges from San Diego in the south, just north of 32 degrees latitude, to the Oregon border at almost 42 degrees latitude. The majority of our analysis sample is located in Southern California, with a large Coastal sample in both San Diego and Los Angeles. As we move north, there is a meaningful sample around Santa Cruz and San Francisco, and then a final set of parcels in Eureka. The density for the remaining (non-Coastal) parts of the 15 counties are shown as an orange dashed line. Even though the ocean-adjacent parts of California include some of the most desirable real estate in the United States, the wide range of this sample will allow us to evaluate the impacts of ADUs in a number of different communities. We will present our methodology for doing so in the following section.

4 Empirical Methods

We can build intuition around our empirical designs using the map of San Diego in Figure 6. This figure displays the Coastal Zone, and its segments, in blue, incorporated cities in orange, and 2016 residential real estate transactions from the [Cotality \(2025\)](#) data as dots. Throughout our analysis period within the Coastal Zone (the blue region), ADUs are subject to a stable, albeit stringent, regulatory environment. However, outside the Coastal Zone (in the orange and white regions) parcels benefit from the suite of up-zoning and permitting reforms for ADUs which take effect in 2017 and after. The core of our spatial difference in discontinuities design is a comparison of changes for parcels just outside the Coastal Zone which benefit from the ADU reform package to changes for parcels just inside the Coastal Zone that are subject to land use rules under their

⁵²Effective build year is the year in which the parcel had its currently assessed features constructed. So, if a new extension to the house was added, such as an ADU, the effective build year would update.

⁵³Full sample statistics, including means, standard deviations, minima, and maxima can be found for the full county sample in Table A.3. It is important to note that the sample is mostly, but not entirely, single family units. We intentionally exclude large apartment or condo multi-family parcels as these types of units would not have their highest economic use increased by the ability to build an ADU. However, given that we have a dataset of tax parcels, rather than lots, it is the case that several commonly owned units may be combined in a single parcel, which can skew the extremes of the distribution for characteristics such as bedrooms, bathrooms, etc. We have tried to remove these parcels from our sample.

local coastal plans.⁵⁴ Critically, this stark change in the regulatory environment represented by the Coastal Zone boundary cuts directly through neighborhoods, which we can see in Figure 6. We formalize this intuition in the remainder of this section.

4.1 A Regression Discontinuity for Permitting Activity

To evaluate the impacts of permitting reform on ADU permitting activity, we will leverage the discrete change in the regulatory environment at the Coastal Zone boundary in a regression discontinuity design. We will estimate the following specification for 2018 to 2024 using our balanced parcel dataset:⁵⁵

$$Y_{ict} = \alpha + \beta_1 OutCZ_i + \beta_2 DistCZ_i + \beta_3 (OutCZ_i \cdot DistCZ_i) + \theta'_{(i)} \gamma + \lambda_c + \tau_t + \varepsilon_{ict} \quad (2)$$

where Y_{ict} is an indicator variable taking a value of 1 if parcel i in county c received an ADU permit in year t . $OutCZ_i$ is an indicator taking a value of 0 if lot i is within the Coastal Zone, and therefore has stable regulations, or a value of 1 if lot i is outside the Coastal Zone and therefore will receive the state-wide permitting reform package in 2016. $DistCZ_i$ is lot i 's distance from the closest point on the Coastal Zone boundary in meters, which we also interact with $OutCZ_i$ to allow the slope to vary across the boundary.⁵⁶ In order to control for pre-reform characteristics, we include a set of hedonic fixed effects using pre-2016 parcel characteristics, $\theta_{(i)}$. We also include county (λ_c) and time (τ_t) fixed effects. The county fixed effects will control for constant characteristics of county assessors offices as well as other static characteristics of the county. The time fixed effects will control for factors impacting all lots in a given period, such as interest rate changes. In some specifications, we replace the time fixed effect with a border-segment by year fixed effect to ensure that all comparisons are within local segments of the Coastal Zone boundary. ε_{ict} is an error term.

The main assumption for our spatial regression discontinuity model is that permitting activity across the Coastal Zone boundary would be the same absent the difference in the regulatory environments. Explicitly, we are assuming that parcels inside the Coastal Zone boundary and parcels just outside the Coastal Zone boundary would have ADU permitting levels if reforms had not taken

⁵⁴There are more pieces of legislation that have been passed by the State of California to address housing affordability. These measures include the ADU reforms, as well as rezoning efforts surrounding transit dense housing, infill housing, and lot splitting among other reforms. To our knowledge, only the ADU reforms and the SB 35 environmental permitting waivers for infill housing conflict with the Coastal Act directly. SB 35 does limit discretionary permit review for developments that classify as infill. There have been about 160 SB 35 projects since its enactment in 2017. These projects are generally much larger projects, with 90% of SB 35 projects containing more than 10 units, and 74% of projects containing more than 50 units. Using SB 35 in a smaller residentially zoned setting would be difficult due to the cost of assembling an economically viable tract of land to build on. This largely puts SB 35 projects farther away from the small residentially zoned lots that are the focus of this paper (Manji and Finnigan, 2023).

⁵⁵This specification deviates from our pre-analysis plan as we were unable to collect enough pre-2018 local permitting data that also distinguished ADUs. Figure 1 suggests that we are likely capturing 95% of all permits issued statewide between 2012 and 2024.

⁵⁶We will use first degree and second degree polynomials, each interacted with the *OutCZ* dummy to allow for slope changes across the boundary.

effect in 2017. Since Figure 1 suggests that pre-reform permitting activity was negligible, we will proceed with assuming that pre-reform ADU permitting activity was essentially zero on both sides of the boundary and therefore there is no pre-2017 discontinuity in ADU permitting to account for. Additionally, the sharp regression discontinuity design demands that the regulatory environment jumps discretely at the Coastal Zone boundary, whereas other relevant features for building ADUs besides regulation change smoothly across the boundary. Severen and Plantinga (2018) and Kahn, Vaughn and Zasloff (2010) both show that land just inside and just outside of the Coastal Zone is similar. Critically, we are identifying the effects of a package of reforms, as we will be unable to separate the impacts of any individual element of the post-2017 up-zoning and permitting reform package.

We explore how covariates evolve across the Coastal Zone boundary in Figure 8a. In this figure, we show estimates of Equation 2 within our analysis sample for the pre-reform years, 2010 to 2016. In orange circles we show standardized coefficients using just county and year fixed effects, and in hollow blue squares we replace the year fixed effect with a segment by year fixed effect. We can see that for sale price per acre and land values per acre, values drop outside the coastal zone, but not significantly.⁵⁷ We also see pre-period balance in transactions. However, several covariates jump across the Coastal Zone boundary before permitting reform, and they are statistically distinguishable from zero. When we undo the scaling on these estimates, however, we find that they are small differences in practical terms. Compared to inside the Coastal Zone, parcels outside the Coastal Zone have houses on them that are about a year newer, have 0.04 fewer bathrooms, are 26 square feet smaller, sit 1.4 meters lower in elevation, are 1% closer to the nearest CBD, and are about 5% closer to the coastline.⁵⁸

4.2 A Difference in Discontinuities Design for Land Value Changes

We use a difference in discontinuities estimator to show how permitting reform changes at the Coastal Zone boundary impact land values. Since we have land values for the full panel starting in 2010, we use our transaction data to estimate models both with and without a parcel fixed effect. We estimate the following equation using our transaction sample from 2010 to 2024:

$$\begin{aligned} \ln(Y)_{ict} = & \alpha + \beta_1(OutCZ_i \cdot Post_t) + \beta_2(DistCZ_i \cdot Post_t) + \beta_3(DistCZ_i \cdot OutCZ_i \cdot Post_t) \\ & + \beta_4 OutCZ_i + \beta_5(DistCZ_i) + \beta_6(DistCZ_i \cdot OutCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict} \end{aligned} \quad (3)$$

where $\ln(Y)_{ict}$ is the log of the real land value per acre for parcel i in county c sold in year t . A difference in discontinuities model stacks two regression discontinuities in one design. The first regression discontinuity, parameterized by β_4 , β_5 , and β_6 , captures the pre-2016 level effects of Coastal Zone regulations, which Figure 8a and prior work suggests are likely present (Severen and

⁵⁷We can see this visually in the pre-2016 series of the binned scatter plots for land values and total sales prices in Appendix Figures A.2 and A.3, respectively.

⁵⁸The Coastal Zone doesn't hug every coastline, just the coastline with the Pacific Ocean. Several bays, coastal inlets, and other features that are classified as coastline extend inland past the Coastal Zone boundary. The primary example of this is the San Francisco Bay.

Plantinga, 2018; Kahn, Vaughn and Zasloff, 2010).⁵⁹ Then, a second set of regression discontinuity parameters are interacted with a post treatment dummy, $Post_t$, to capture the changing effects of regulation at the Coastal Zone boundary after permitting reform is enacted in 2017. Our parameter of interest is the interaction, $OutCZ_i \cdot Post_t$, and the difference in discontinuity estimate β_1 . Just as with Equation 2, we can replace the time fixed effects with segment by time fixed effects that keep the comparisons local to each of the 145 segments of the Coastal Zone boundary. The specification that we pre-registered does not include a parcel fixed effect, which would limit us to including parcels that transact both before and after 2017 in a repeat sale setup. We have a sufficient sample of repeat sales parcels in our dataset to include a parcel fixed effect. Therefore, we present a repeat sales model that includes a parcel fixed effect to control for static characteristics of the parcel, similar to Soltas and Gruber (2026). A repeat sales model also removes our pre-reform regression discontinuity terms, which get absorbed by the fixed effect.

This difference in discontinuity design, developed by Grembi, Nannicini and Troiano (2016), compares changes in land values on parcels just inside the Coastal Zone boundary to parcels just outside the Coastal Zone boundary before and after the reform. The resulting estimator relies on a similar logic to difference in differences and regression discontinuity designs. The core assumption is that, local to the cutoff, land values would have evolved similarly in the absence of the post-2016 reforms to ADUs. However, unlike the regression discontinuity design, stable differences in the characteristics of parcels across the Coastal Zone boundary are accounted for through differencing. In Figure 8, we show level differences in Panel A, and pre-trends compared to our base year, 2016, in Panel B. As with the level differences discussed above, our trends in total sale prices per acre, land values per acre, and transactions shift towards zero when we use segment by time fixed effects instead of the typical year fixed effect. However, for all three outcomes, sales price per acre, land price per acre, and transactions, Figure 8b supports the argument that local parallel trends likely holds near the Coastal Zone boundary.⁶⁰

The second assumption for our difference in discontinuities estimator is that the regulatory environment changes sharply at the Coastal Zone boundary after 2017. Our regression discontinuity estimates for differences in permitting rates for ADUs at the boundary will serve as evidence that the permitting reforms had sufficient bite to justify our design.⁶¹ Finally, we are also assuming that the pre-existing level differences created by Coastal Zone regulations can be accounted for by including our pre-period discontinuity terms (Butts, 2023).

⁵⁹Note that Severen and Plantinga (2018) focuses on multi-family units, and Kahn, Vaughn and Zasloff (2010) use price data from census tracts and Los Angeles property transactions from 2008. So, our baseline balance may not align with either study exactly.

⁶⁰Butts (2023) formalizes the local parallel trends into a continuity assumption of potential outcomes across the cutoff. He also notes that we need to assume the land cannot sort in response to our reforms differentially across the boundary.

⁶¹We also find that our regression discontinuity design doesn't detect changes in larger permitted units, such as multi-family units impacted by SB 35.

4.3 Detecting Spillovers Across the Coastal Zone Boundary

To detect spillovers local to the boundary, we will use a more flexible binned approach following the designs in [Hornbeck and Kenniston \(2017\)](#) and [Severen and Plantinga \(2018\)](#). Parcels just inside the Coastal Zone should be unable to take advantage of the new ADU law, but they will be exposed to any spillovers associated with the new construction. We can estimate a flexible specification as follows for 2010 to 2024:

$$\ln(Y)_{itbs} = \alpha + \sum_{b \neq 1} \beta_b (I\{Bin = b\}_i \cdot Post_t) + \theta'_{(i)} \gamma + \phi_b + \sigma_{s,t} + \varepsilon_{itbs} \quad (4)$$

where $\ln(Y)_{itbs}$ is the log of the real total sale price (land and structures) per acre on parcel i , in county c , in year t , in Coastal Zone distance bin b , and Coastal Zone border segment s . Here, we will construct a series of dummies for 50 meter coastal zone distance bins, $b \in \{1, \dots, B\}$, with an omitted bin 400 to 450 meters within the Coastal Zone as a comparison group.⁶² Our set of estimates, β_b , will capture how prices evolve across a set of bins relative to the comparison bin. We also include segment by time fixed effects, $\sigma_{s,t}$, in our presented estimates. If spillovers are present, then prices should change just on the inside of the coastal zone, reflecting either positive or negative spillovers from the permitting reforms outside the Coastal Zone.

Motivated by our theoretical model, we also use this design to show how transactions evolve across the coastal zone boundary. We show spillover estimates that replace the log of the real total sale price per acre with a dummy indicating if the parcel had an arms-length transaction in a given year. In these transaction specifications, we also include a parcel fixed effect.⁶³

4.4 Heterogeneity

We include heterogeneity variables following [Calonico et al. \(2025\)](#). This requires fully interacting our regression discontinuity variables with the heterogeneity variable of interest. Including heterogeneity gives the modified regression discontinuity equation:

$$Y_{ict} = \alpha + \eta_1(H_i \cdot OutCZ_i) + \eta_2(H_i \cdot DistCZ_i) + \eta_3(H_i \cdot OutCZ_i \cdot DistCZ_i) + \eta_4 H_i + \delta_1 OutCZ_i + \delta_2 DistCZ_i + \delta_3(OutCZ_i \cdot DistCZ_i) + \lambda_c + \tau_t + \varepsilon_{ict} \quad (5)$$

where H_i is our heterogeneity variable of interest. Our new object of interest is the conditional average treatment effect, or CATE. We can write the CATE as a combination of estimates of η_1 and δ_1 at different levels of H_i :

$$CATE(H_i) = \hat{\eta}_1(H_i) + \hat{\delta}_1 \quad (6)$$

⁶²Our selection of the 400 to 450 meters bin as the reference bin is based on our mean squared error optimal bandwidth choice, but in principle we could use any bin as a reference bin.

⁶³We did not pre-register investigating transaction spillovers.

where H_i is once again our heterogeneity variable. [Calonico et al. \(2025\)](#) note that with a continuous H_i , we are required to assume that at the boundary the CATE is linear in H_i . This exercise changes the coefficient of our baseline discontinuity, and it will generally be true that $\delta_1 \neq \beta_1$ from estimates of Equation 2. Rather, δ_1 reflects the treatment effect of permitting reform across the Coastal Zone boundary in the special case when $H_i = 0$. We extend the method to our difference in discontinuity estimates by fully interacting the specification with our heterogeneity variable of interest.

Throughout our empirical estimates, we will use our 450 meter analysis bandwidth. All specifications also include triangle weights, which give additional weight to parcels closer to the Coastal Zone boundary.⁶⁴ In our main tables and figures, we will present parcel clustered standard errors. In practice, at every bandwidth these standard errors are almost identical to the more conservative Coastal Zone segment clustered standard errors. We present our baseline and preferred specifications using parcel clustered, segment clustered, and spatial correlation robust standard errors from [Conley \(1999\)](#) when we explore the bandwidth sensitivity of our estimates.

5 Main Results

We will investigate three primary impacts of the post-2016 set of permitting reforms in this paper. First, we will investigate the reform package’s direct impact on permitting of ADUs. Permitting activity reflects directly how the legal changes are impacting the costs of building an ADU. However, even in the absence of pulling a permit and building an ADU, by lowering the costs of building an ADU the reform package may have created an option value that is capitalized in the land, as suggested by our theoretical model in Section 6. Finally, we will investigate if there are noticeable spillovers onto nearby parcels just inside the Coastal Zone that may reflect a distaste for ADUs that both depresses total sales prices and increases transaction volumes.

5.1 Permitting

First, we will explore differences in permitting activity for ADUs across the Coastal Zone boundary in response to 2017 regulatory changes graphically in Figure 9. We display permits in bins by their distance from the Coastal Zone boundary, where negative values (in blue) show locations within the Coastal Zone that did not receive permitting reforms and positive values (in orange) show locations that were impacted by the state permitting reforms. Each bin covers 50 meters and the Coastal Zone boundary is marked by a line at 0. We highlight bins within our 450 meter analysis sample. In Sub-Figure 9a, the vertical axis shows counts of permits issued per parcel-year. We can see that in terms of volume, ADU permitting activity jumps when moving from the stringent Coastal Zone to the recently reformed area just outside of the Coastal Zone.⁶⁵ In Sub-Figure 9b, we also show how ADU permitting changes as a share of all building permits. In the more lenient non-Coastal Zone

⁶⁴Our pre-analysis plan did not specify triangle weights, but they are standard in the RD literature to make sure that the tails of the sample do not distort the estimate at the boundary [Cunningham \(2021\)](#).

⁶⁵In terms of raw counts of permits, activity more than doubles, but density of parcels also increases across the boundary.

areas, ADU permitting is about 65% of all building permits issued, whereas inside the Coastal Zone ADU permitting is only about 45% of all permits issued.⁶⁶ In both measures, the sharp difference in the regulatory environment is directly reflected in permitting activity.

In Table 3, we use our post-2018 balanced parcel dataset and regression discontinuity design to show how reforms outside the Coastal Zone impacted permitting. Our outcome variable is a dummy taking a value of 1 if a parcel in a given year has any ADU permit associated with it, and 0 otherwise. For display, we present coefficients scaled to be ADU permits per 1,000 parcel-years. In Column 1, we include our baseline specification which only uses the 1st degree polynomial, county fixed effects, and year fixed effects. In Column 2, we add in our pre-2016 hedonic fixed effects to capture differences in parcel characteristics. Column 3 and 4 include segment by time fixed effects to keep our comparisons local to each coastal zone segment. Across our specifications, these coefficients suggest an increase in ADU permitting of about 30% when comparing parcels just outside the Coastal Zone to parcels just inside the Coastal Zone. These results are qualitatively consistent across alternative specifications. Every estimate is significant at the 1% level or higher, and the point estimates range from 0.53 (Column 3) to 0.629 (Column 2) ADU permits per 1000 parcel-years. Our preferred specification is shown in Column 3, which includes a first degree polynomial and segment by time fixed effects. Cumulatively, over the 7 years represented by these data, about 0.5% of parcels outside the coastal zone will get an ADU permit that wouldn't have been pulled absent these reforms.

These results suggest a meaningful change in the regulatory environment for ADUs occurs when moving from the Coastal Zone to the rest of the state. However, if we were to take the direct change in permitting activity as the only impacts of the post-2016 permitting reforms, then we would have to conclude that these impacts were relatively limited, as a very small fraction of parcels have elected to build an ADU. Extrapolating from these estimates, about 60,000 of the 176,000 state-wide ADUs were generated by the up-zoning and permitting reforms that jump at the coastal zone boundary. This means a very small portion of lots ever build an ADU, and the natural question is what might explain the low take-up.

5.2 Land Values

On smaller residential lots, the ability to add an ADU increases the highest economic use of that lot, reflecting the ADU's potential value to the owner. This option value should capitalize in the land regardless of the amount of permitting activity. We explore how the post-2016 permitting reforms impacted land values using our transaction dataset and a difference in discontinuities design in Table 4.

Column 1 of Table 4 shows our baseline specification for the impact of permitting reforms on land values. This column only uses a first degree polynomial in distance to the Coastal Zone boundary, a county fixed effect, and a year fixed effect. In Columns 2 and 4, we add our pre-2016 hedonic fixed effects, which adds precision to our baseline model. Then in Columns 3, 4, and 6 we

⁶⁶Marantz, Elmendorf and Kim (2023) also find that ADUs were a large share of overall permitting in California.

include segment by time fixed effects instead of year fixed effects to keep estimates local to the 145 local coastal segments. Finally, in the last two columns we include a parcel fixed effect and produce estimates for the repeat sale sample. Our preferred specification is the repeat sale model with segment by time fixed effects shown in Column 6.⁶⁷ The estimates suggest that by-right permitting reform increased the value of land by about 7%. Our estimates range from 7% (Column 6) to 14% (Column 1), and all estimates are significant at the 1% level or higher.

We can show this change in land values visually in Figure 10. The graph shows 50 meter bins of within-bin changes in land values between 2010 and 2016 and between 2017 and 2024 by their distance from the Coastal Zone boundary. The vertical axis shows the log difference of the real land value per acre in each bin between the pre-reform and post-reform periods. The effect of the permitting reforms can be seen as the jump in land value per acre changes just before and just after crossing the Coastal Zone boundary. The roughly 10% jump in land value changes visually corroborates our estimates in Table 4.⁶⁸ We also show year by year event studies from our preferred estimates that use a repeat sale model with segment by time fixed effects in Figure 11. The pooled pre-trend estimate is centered around zero, and shows a slightly positive and insignificant point estimate.⁶⁹ After the reforms in 2017, the year-year estimates are a too imprecise to say that any one year had a particularly large impact on land values, which might have suggested parts of the post-2017 reform package that were effective at raising land values. However, the event study does appear to suggest that several post-COVID years saw larger increase in land values across the border. This might reflect increased demand for floor-space, which was common during the work-from-home era (Mondragon and Wieland, 2022). Alternatively, it may suggest that increased state-level enforcement of ADU reforms after 2019 may have had an impact on land values (Minott, 2023).

These results are consistent with previous findings from Ihlanfeldt (2007). He finds that increased land use regulations increase structure prices and decrease land prices. Our findings are the mirror image of Ihlanfeldt (2007); we find that decreased land use regulations increase land prices. Gold and Tarascina (2026) also find an option value of 4% from permitting reforms for ADUs in Seattle, WA using a within-city difference in difference estimator. Although our estimates suggest that land owners that received de-regulation for ADUs outside the Coastal Zone likely benefit from a higher value for their land, dis-amenities from increased density may negatively impact their total sale prices as well as their neighbors.

5.3 Spillovers

The public concern with up-zoning is typically a discussion about spillovers. Neighbor A may use her property in ways that negatively affect the property value or utility of Neighbor B. Dis-amenities from density could arise through decreased street parking, increasing utilization of local

⁶⁷Our pre-registered estimates are in Columns 1 to 4.

⁶⁸The un-differenced version of this figure is available as Figure A.2.

⁶⁹Note that Figure 8b doesn't use the repeat sale sample.

services, building something of low aesthetic value, or simply competing down rental rates for units. To investigate these spillovers, we first show how total sale prices, which include both land and structure values, change on a per acre basis across the Coastal Zone boundary.

We present our estimates of Equation 4 in Figure 12. This figure presents coefficients that capture how the (log of) real total sale prices per acre changed before and after the 2016 reform legislation. All coefficients are relative differences from changes in a reference bin. We use the bin farthest inside the Coastal Zone boundary and within our 450 meter band sample, $[-450, -400]$ meters. The horizontal axis shows the bin's (median) distance from the Coastal Zone boundary, with negative values in blue representing transactions inside the Coastal Zone that did not receive the permitting reforms after 2016. Estimates for bins outside the Coastal Zone that received permitting reforms are shown in Orange. If spillovers are present, they should be shown in the coefficients just inside the Coastal Zone.

In Figure 12a we find no evidence of price spillovers into the Coastal Zone from permitting reforms. The four estimates just inside the coastal zone ($[-200, 0]$ meters) have indistinguishable changes in total sales prices per acre compared to our reference bin. By contrast, a strong jump in estimates can be seen after crossing the Coastal Zone boundary, of between 4-9%. These are broadly consistent with our estimates for land value of a 7% increase in Table 4. Additionally, the estimates outside the Coastal Zone from $[0, 450]$ meters also capture cross-parcel spillovers, suggesting that the net effect on price reflects mostly a change in land values rather than a decrease in structure values. These results suggest that the option value likely dominates concerns over negative spillovers in this setting. These findings are in contrast to Davidoff, Pavlov and Somerville (2022), who find evidence of negative cross-plot spillovers. However, the absence of spillovers is consistent with results from Gold and Tarascina (2026).

We next show how transaction volumes change across the boundary in Figure 12b. Here, we again find no evidence of spillovers. On the reform side of the boundary ($[0, 450]$ meters), we see transaction rate changes that appear similar to those in our reference bin. These results suggest that ADU reforms may not be associated with negative disamenities that induce owners to sell and exit. The magnitudes of the point estimates (-0.2%) are also small in a setting where about 4% of parcels transact at arms-length each year.

5.4 Robustness

In addition to the robustness presented in our main tables, we provide several additional exercises to further probe our results. First, we explore how bandwidth selection and choices around the appropriate standard errors impact our results for both estimates of Equations 2 and 3 for ADU permits and land values, respectively. We present these alternative choices in Figure A.4, with ADU permit results sensitivity shown in Sub-Figures A.4a and A.4b and land value results sensitivity shown in Sub-Figures A.4c and A.4d. Panels A and C show our baseline estimates and Panels B and D show our preferred estimates. For both sets of permitting estimates, with and without county by

time fixed effects, point estimates are remarkably stable at 0.6.⁷⁰ For land value estimates in Panels C and D, we see a slightly different pattern. In Panel C, point estimates are steadily increasing as the bandwidth gets closer to the boundary, with estimates from the 100 meter bandwidth reaching almost a 20% increase in land values. However, in Panel D using our repeat sale sample and segment by time fixed effect, we see much more stable estimates around 7%.

In all panels of Figure A.4, we show standard errors three ways. First, we show our preferred parcel-clustered standard errors. Then, we show segment-clustered standard errors in orange. Since the Coastal Zone is administered by 145 local coastal programs, if these segments differed substantially in their pre-reform treatment of ADUs this would vary our treatment along the boundary and warrant clustering (Abadie et al., 2023). In all specifications, these segment clustered standard errors are almost identical to the parcel-clustered standard errors. Finally, we also show spatial correlation robust errors from Conley (1999) with a 200 meter cutoff in light blue. These standard errors are larger in our repeat sales model, and render estimates from bandwidths of 350 meters and smaller insignificant. Across all specifications, our 450 meter bandwidth remains significant across all three standard error choices.

Next, we probe the land value results in Table 4 in two additional ways. In Appendix Table A.4 we show estimates that use a second degree polynomial. In general, these results increase both the point estimate and the standard errors associated with permitting reform. Even so, these estimates are quantitatively and qualitatively similar to those presented in Table 4.⁷¹ Finally, we explore trying to isolate changes in land values using updating hedonics in Appendix Table A.6. We estimate a modified version of Equation 3 where we use the log of the real total sale price per acre on the left hand side and allow the binned hedonic fixed effects to update as characteristics of the structures on the property change over time. This mimics the types of adjustments done by assessors when assigning land values where they try to simply remove the structure value, making the land value a residual California State Board of Equalization (2025a). This alternative approach again produces qualitatively similar estimates to those in Table 4 that are all significant at the 1% level.⁷²

6 Theoretical Framework

To build intuition about our difference in discontinuities estimates, we present a model of an open city with two permitting jurisdictions. Within each jurisdiction, incumbents start with lots of heterogeneous quality and without ADUs. Lots are evenly distributed across the two jurisdictions,

⁷⁰We use county by year fixed effects here because segment by time fixed effects drop a number of segments without enough variation in permitting with bandwidths below 200 meters from the boundary. At 450 meters, the county by year fixed effects and segment by year fixed effects give almost the exact same point estimate.

⁷¹We also show robustness to polynomial choice for our regression discontinuity results for permitting in Appendix Table A.5.

⁷²Following our pre-analysis plan, we also show estimates that drop segments of the Coastal Zone that overlap with jurisdictions that voluntarily adopted ADU reforms inside their Coastal Zone segments. These cities are: Santa Cruz, Santa Barbara, Encinitas, San Mateo, and Pacifica. These estimates are shown in Table A.7 and are consistent with our main estimates.

with one side having lower permitting costs than the other. Incumbent owners can elect to build an ADU on their lot or sell their lot to an entrant who will elect to build an ADU. This choice is characterized by an indifference condition for incumbents between selling and staying, where the extent to which an incumbent benefits from California’s below-market property tax assessment regime, the disamenities from other lots building ADUs within the city, and the gap between the value placed on the option to build an ADU by the entrant and the owner (net of taxes) are the prevailing factors. With some assumptions, we can pin down the market value for the ability to build an ADU on a lot. Here, we abstract away from pre-existing differences across the jurisdictional boundary and our difference in discontinuities estimator is simply a regression discontinuity across the permitting jurisdiction boundary.

We can characterize the land price jump across the boundary as the difference between the market value of the option to build an ADU between the low permitting cost side and the high permitting cost side. This difference is a step function in lot quality for an ADU. If it is never profitable to build an ADU on either side of the boundary then both options on either side of the boundary are valued at zero. Similarly, if lot quality is high enough that it is profitable to build on both sides of the boundary, then the difference in discontinuities estimator will pick up just the difference in permitting costs between the jurisdictions. However, in the likely more common case where an ADU is only profitable on the lower permitting cost side of the boundary, the difference in option values embeds the value of the ADU to the owner net of permitting and construction costs on the lower permitting cost lots. The size of this gap is driven by lot quality for an ADU and the entrant’s construction cost.

There is a literature that uses open city models to characterize estimators of discrete regulatory effects in regression discontinuity settings. [Turner, Haughwout and van der Klaauw \(2014\)](#) organizes their model as two adjacent open linear cities and decomposes the effects of regulatory boundaries on land prices into an own-lot and external effect that changes with distance to the boundary. Similarly, [Gyourko and McCulloch \(2024\)](#) characterize the effects of discrete changes in minimum lot sizes on home prices. Here, as in [Turner, Haughwout and van der Klaauw \(2014\)](#) density exposure decays as lots on the high minimum lot size side of the border get farther from the denser jurisdiction. Finally, [Severen and Plantinga \(2018\)](#) build a model of how multi-family unit values and rents responded to the Coastal Zone’s permitting regime. In their model, the sign of the price effect of regulation local to the boundary hinges on whether or not the effect of limiting construction by neighbors outweighs the increased compliance costs from regulation, assuming amenities are similar across the border.⁷³ Our model abstracts away from the running variable, which can be embedded in our variable “lot quality”, but it endogenizes the construction and sale decisions to characterize the price and creates an explicit “option value” from deregulation that may be valued by incumbents and entrants differently. Additionally, we show how California’s property tax system can distort both the choice to sell a parcel, the decision to build an ADU, and the value of constructing an

⁷³[Severen and Plantinga \(2018\)](#) notes that if regulation reduced supply, then prices would increase for all lots. It is, in practice, hard to separate the aesthetic effect of neighbor’s construction from the supply effects.

ADU.⁷⁴

6.1 An Open City Model in Coastal California

The city has two permitting jurisdictions under a common fiscal authority. The left jurisdiction is inside the Coastal Zone, with high permitting costs, and the right jurisdiction is outside of the Coastal Zone with low permitting costs. There is a single measure of parcels that can hold a main structure and an ADU. Every parcel we will assume starts without an ADU, and we will assume only owners benefit from an ADU, ignoring the possibility of tenants for simplicity.⁷⁵ Each lot has one owner. A lot is characterized by its quality, with CDF $l \sim F_l$ and an assessed base $B \sim F_B$ on which the parcel is initially taxed at a fixed rate τ . The parcel also has amenity value, $v(l)$ which is increasing in l .

Each parcel can build an ADU. A lot's net present value for an ADU is given by an increasing function in quality, $G(l)$, with a flow value of $rG(l)$, where r is a discount rate.⁷⁶ Owners also draw construction costs from a distribution $k \sim F_k$, which is bounded below by k_{min} . k_{min} is the cheapest cost at which an owner can build an ADU.⁷⁷ Owners that want to build an ADU also face a jurisdiction-specific permitting cost, κ_m , with $m \in \{L, R\}$. These costs are not just permitting fees, but also reflect the hassle, construction, and carrying costs that accompany permitting requirements.⁷⁸ Reflecting the Coastal Zone's stringency, we impose $\kappa_L > \kappa_R$.

ADUs can be built by their incumbent owners, or the parcel can be sold to the market to build an ADU and sell to an entrant. An owner can elect to build an ADU after paying the permitting cost κ_m and their build cost k . Alternatively, the owner can also sell their lot. We assume that these entrants are sufficiently dense that all owners that want to sell, can sell. When an incumbent sells, their parcel is bought by a firm first, which momentarily holds the parcel and can build an ADU at the most competitive cost, k_{min} . These firms then sell the bundled parcel and ADU to a pool of (marginal) entrants, whose utility is pinned down by the outside option O . However, when an ADU is built, all residents value the disamenity of the total count of ADUs in the city following $D = \phi N$, where N is the count of ADUs built.

We model entrants as buying parcels built by a competitive firm primarily for expositional purposes. For parcels that ever build an ADU, we show the probability that they transact in a given year around their permit year in Figure 7. Almost 25% of parcels that build an ADU transact in the year before a permit is pulled, about 17% transact in the same year that an ADU permit

⁷⁴For a full discussion of taxes in simple urban models, see Glaeser (2013).

⁷⁵Tenants, or rental demand, enter by increasing $G(l)$, and these effects can be included in this model by thinking about l , lot quality, as strongly correlated with rental demand. We provide some evidence for this when we explore heterogeneity of our empirical estimates by lot quality.

⁷⁶In general, flow values will be lower case and net present values will be upper case. A net present value can be converted to a flow values using the discount rate r . The exception to this rule will be the flow property tax, τB , where B reflects the assessed base, which can be viewed as a present value from a previous period as it reflects the original sale price of the parcel.

⁷⁷This cost distribution can be thought of as reflecting an owner's differential access to financing, characteristics of the parcel that may make building more costly, or the owner's ability to secure a low cost contractor.

⁷⁸We also assume that permits are budget neutral and we abstract away from a city using them as a revenue source.

is pulled, and about 12% transact in the subsequent year, with declining shares transacting going forward. This suggests that it is possible that firms buy parcels to build ADUs and sell them, but that it is more common for entrants to buy parcels and then build ADUs themselves. For now, we will assume away that entrants buy and then build, but we will include this channel in future work.⁷⁹

Parcels in the city are taxed following California's unique property tax system, Proposition 13.⁸⁰ If a parcel doesn't sell, it pays (flow) property tax equal to τB . If the parcel builds an ADU and doesn't transact, it will pay a total tax bill of $\tau B + \tau G$. Finally, if a parcel transacts and is sold with an ADU, it pays property tax on the full sale price, losing its under-market rate base, paying τP^{Fin} .⁸¹

We will now work through the incumbent's decision to build an ADU. An incumbent parcel owner will build an ADU if the post-tax value of the ADU is above the total build cost, including both construction k and permitting costs κ_m . We can then write the owner's option to build as:

$$\begin{aligned} OwnOpt_m(l, k) &= \max\{G(l) - \frac{\tau}{r}G(l) - k - \kappa_m, 0\} \\ &= \max\{(1 - \frac{\tau}{r})G(l) - k - \kappa_m, 0\} \end{aligned} \quad (7)$$

The improvement rule for Proposition 13 means that the owner enjoys the entire improvement value of an ADU, $G(l)$, but that value is fully taxed, giving the present value of the property tax burden equal to $\frac{\tau}{r}G(l)$.⁸²

We can now write down the net present value welfare for an incumbent of owning a lot in the city. This value will be a discounted value stream, net of taxes, and the option of building an ADU. We can express incumbent welfare as:

$$W(l, \alpha, B, k, m) = \underbrace{\frac{v(l) + \alpha + u(g) - h - \theta D - \tau B}{r}}_{\text{Value of living on the incumbent's lot}} + OwnOpt_m(l, k) \quad (8)$$

Welfare for incumbents on their own parcel is a function of their type, characterized by lot quality (l), their attachment to their parcel (α), their below market assessed property tax base (B), their building costs (k), and their permitting jurisdiction (κ). Their welfare is a positive function of the housing amenity value of the their lot $v(l)$, their attachment α , their utility from city services $u(g)$ and their own option value of building an ADU on their lot $OwnOpt_m(l, k)$. Owners are also faced with two forms of taxes, the head tax, h , and the property tax as applied to their assessed base B . Finally, as more ADUs are built, owners internalize the city-wide density

⁷⁹Appendix Figure A.1 shows that just 2.5% will permit for an ADU at all, and 1% of parcels both build an ADU and transact.

⁸⁰The property tax rate, τ , is also not directly controlled by the city, and therefore we take it as a parameter. For an example of modeling Proposition 13 see [Green et al. \(2026\)](#).

⁸¹In a version of this model where an entrant builds herself, the market rate on the parcel without an ADU would be taxed with the ADU added following the assessment rule, $\tau P^{lot} + \tau G$.

⁸²We require $1 - \frac{\tau}{r} > 0$ in order for this option to have positive value.

disamenity, θD .

In addition to staying on a parcel without an ADU, and building an ADU on their own lot, incumbent owners can also sell their parcels to a mass of entrants. When an owner sells their parcel, they receive the sale price, $P_{0,m}(l) + \Omega_m(l)$, and the outside option utility O . When an owner sells, the parcel is purchased by a firm that holds the parcel momentarily, builds an ADU, and then sells to an entrant. The firm builds an ADU at the lowest possible price, k_{min} . The firm values the option to build an ADU, $\Omega_m(l)$, which is given by:

$$\Omega_m(l) = \max\left\{\frac{r}{r+\tau}G(l) - k_{min} - \kappa_m, 0\right\} \quad (9)$$

Critically, firms sell the parcel and ADU right away, which means that the taxes capitalize into the price of the ADU. So, the value of a new ADU to a firm is higher than to an owner, creating a wedge.⁸³ Additionally, the market option is greater than the owner's option, $\Omega_m(l) > OwnOpt_m(l, k)$ for all $\{l, m\}$, when both options are non-zero. The gap between the market and owner options arises because of the tax wedge between sold ADUs from the firm and ADUs built by owners, and the fact that firms can build at lower costs than owners.⁸⁴ Free entry of firms that flip un-built parcels bids down any profits generated to zero.

The marginal entrant sets the price of the un-built parcel, and she is indifferent between entering and the outside option. This allows us to write down the indifference condition and solve for the baseline parcel price:

$$\frac{v(l) + \alpha_m^e + u(g) - h - \theta D - \tau P_{0,m}}{r} = P_{0,m} + O \implies P_{0,m} = \frac{v(l) + \alpha_m^e + u(g) - h - \theta D - rO}{r + \tau} \quad (10)$$

This baseline price and the ADU value together then gives the built lot price:

$$P_m^{Built}(l) = P_{0,m}(l) + \frac{rG}{r + \tau} \quad (11)$$

Since by our assumption every transactable and buildable lot is purchased by a firm and flipped, the final price of a sold parcel is given by:

$$P_m^{Fin}(l) = P_{0,m}(l) + \frac{rG}{r + \tau} \mathbb{1}\{l, m : \Omega_m(l) > 0\} \quad (12)$$

If the parcel is not purchased and flipped, but only sells to an entrant, the price is then the market price without an ADU, $P_{0,m}(l)$.

We can now characterize the incumbent owner's stay or sell decision by setting the benefits of staying equal to the benefits of selling the parcel. Notice that the value of selling to either the firm or the entrant is equal to $P_{0,m}(l) + \Omega_m(l) + O$, because the market option's value determines whether or not the firm or the entrant buys the parcel. Setting this equal to incumbent welfare

⁸³To see this, subtract $\frac{r}{r+\tau} - (1 - \frac{\tau}{r}) = \frac{\tau^2}{r(r+\tau)} > 0$.

⁸⁴Relaxing these assumptions will be a focus of future work.

from Equation 8:

$$\alpha_m^*(l, k, B) = \alpha_m^e + \underbrace{\tau(B - P_{0,m}(l))}_{\text{Prop-13 Lock-In}} + \underbrace{\frac{\tau}{r + \tau}\theta D}_{\text{Density Push-out}} + \underbrace{r[\Omega_m(l) - \text{OwnOpt}_m(l, k)]}_{\text{Option Wedge}} \quad (13)$$

where $\alpha_m^*(l, k, B)$ is the cutoff attachment level for the incumbent. Incumbent owners with $\alpha \geq \alpha_m^*(l, k, B)$ stay on their lot, and $\alpha < \alpha_m^*(l, k, B)$ sell. Here, we can see three characteristics of the incumbent's lot that drive the decision. Owners whose property tax bills are more sheltered by Proposition 13 through under-valued assessments B are less likely to sell to an entrant. Similarly, a higher market price for the un-built parcel $P_{0,m}$ also increases the attachment necessary to not sell the parcel. In Figure 4, we show how the Proposition 13 assessment rule correlates with parcel level transactions and ADU permit rates. In both panels, a Proposition 13 gap over market value of 0 reflects a parcel that is taxed at full value, whereas a value of 1 reflects a completely sheltered parcel from the property tax.⁸⁵ In Panel A, we show that the share of parcels that transact at all between 2010 and 2024 is declining in the amount a parcel is sheltered by Proposition 13.⁸⁶ In Panel B, we can see that a depressed transaction rate is also correlated with a lower rate of permitting an ADU. At the population level, these figures support the idea that tax shelter created by Proposition 13 can depress transactions, and in turn ADU permits.

Equation 13 also points to additional density from built ADUs as a driver of incumbent owner exit.⁸⁷ This is consistent with findings that owners dislike density, although if ADUs preserve neighborhood character this effect may be muted (Broockman, Elmendorf and Kalla, 2026). Finally, if the market values the build option more than the owner, this also increases the attachment needed to not sell the parcel to a firm for construction. Notice that Proposition 13 enters the decision in two ways. First, it creates a lock-in effect for incumbent owners on lots assessed farther below market value as shown in Figure ??.⁸⁸ Second, it increases the option wedge between firms and owners on buildable parcels and incentivizes owners to sell.⁸⁹ The option wedge is non-linear in build costs for the owner. When the owner's own option is smallest, with a high draw of k , only the market option is positive and the pull to sell is larger and decreasing in permitting costs κ_m . However, when the owner's option is positive, the wedge is driven by the difference in build costs between the owner and the flippers as well as the size of the ADU assessment wedge, $\frac{\tau^2}{(r+\tau)}G(l)$.⁹⁰

⁸⁵We discuss how we construct parcel level property tax assessment bases and market values in Appendix E.

⁸⁶A transaction rate of 70% necessarily implies a non-transaction rate of about 30%. We can write $P(\text{No transaction in 15 years}) = (1 - h)^{15} = 0.3$. This implies that: $h = 1 - 0.3^{(\frac{1}{15})} \approx 0.077$, which gives the annual turnover rate, h . The expected tenure on a property is then given by $\frac{1}{h} \approx 13$, which matches plausible tenure ranges from recent data (Anderson, 2026).

⁸⁷The density level D is the result of a fixed point contraction mapping. We are implicitly assuming that given the city's policy mix $\{h^*, g^*\}$ owners internalize this process.

⁸⁸Ferreira (2010) and Wasi and White (2005) find evidence of lock-in effects from Proposition 13. Similarly, Kopczuk and Munroe (2015) finds evidence that transfer taxes can also decrease transaction volumes and Rauh and Shyu (2024) finds that tax increases can cause out-migration.

⁸⁹In a scenario in which entrants value the public service more than incumbents, a fourth wedge emerges, $(\chi - 1)u(g)$, where $\chi > 1$ indicates that entrants value the service more than incumbents. This may be plausible according to Hilber (2017). If we allow entrants to build themselves, the tax driven option wedge will disappear.

⁹⁰The wedge is increasing in τ for $\tau > 0$.

Finally, both policy levers of the city, h and g , work to push out and retain residents, respectively.⁹¹ So far, we have only focused on how incumbents and entrants respond to the policy variables h and g . Next, we will introduce the city that sets h .

The city maximizes welfare by selecting a head tax rate h , applied per parcel, to supply a service g , subject to a balanced budget constraint. The city budget consists of revenues from the head tax, h , and a piecewise property tax regime reflecting California's assessment rules under Proposition 13 and state-set property tax rate τ . The city then spends these revenues on a service flow, g , which agents value with the concave function $u(g)$.⁹² We present this as a Stackelberg game, where the city moves first and the residents respond, and we solve the game by backward induction. We solve the city's problem in Appendix F. The city's optimal choices over h and in turn g follow the usual Samuelson Rule, and assuming that g is distributed to residents uniformly across the permitting boundary these choices do not influence our characterization of the difference in discontinuities estimator.

6.2 Characterizing the Difference in Discontinuities Estimator

Under the assumption that the attachment of the marginal entrant is equal on both sides of the cutoff, $\alpha_L^e = \alpha_R^e$, we can write the difference in prices between parcels in the left and right permitting jurisdictions as a function of lot quality.⁹³ First, define the minimum quality level at which the firm would build an ADU on each side of the city as l_m , with $l_R < l_L$. Then we can write the difference in market option values across the permitting jurisdictional boundary, the Coastal Zone, as a step function in lot quality.

$$\Omega_R - \Omega_L = \begin{cases} 0, & \text{if } l \leq l_R, \\ \frac{r}{r+\tau}G(l) - k_{min} - \kappa_R, & \text{if } l_R < l \leq l_L, \\ \kappa_L - \kappa_R, & \text{if } l > l_L. \end{cases} \quad (14)$$

On lots that would never build under either the Coastal Zone (κ_L) or non-Coastal Zone (κ_R) regimes due to their low quality, the difference in the option value is zero. When a lot could build an ADU on the reformed side of the boundary, the difference in the option value is the full market value on the reformed side of the boundary. Finally, when an ADU is buildable on both sides of the boundary, the option value is the difference in the permitting costs.

Shifting to our full difference in discontinuities estimator, we only need to assume that attachment of the marginal entrants doesn't change over time local to the boundary. Our estimate will then be the difference in the market option values across the boundary in the post period, where pre-existing heterogeneity in attachment or other characteristics across the boundary can be controlled for under a local parallel trends assumption (Grembi, Nannicini and Troiano, 2016).

⁹¹With the derivatives $\frac{\partial \alpha_m^*}{\partial h} = \frac{\tau}{r+\tau} > 0$ and $\frac{\partial \alpha_m^*}{\partial g} = \frac{-\tau u'(g)}{r+\tau} < 0$

⁹²Here, $u(\cdot)$ is well behaved, with $u'(\cdot) > 0$ and $u''(\cdot) < 0$.

⁹³If $\alpha_L^e \neq \alpha_R^e$, then these entrant demand terms for side m would also be included in the difference in discontinuity estimator.

The model suggests permitting reform will have the following impacts on incumbents and entrants in the case where no ADUs are worth building on the high permitting cost (κ_L) side of the boundary. First, there will be more transactions and in turn more ADUs on the reformed side of the boundary (κ_R). Second, land values on the reformed side of the boundary will be higher than on the un-reformed side, *ceteris paribus*, as reflected in Equation 14. As more ADUs are being built on the reform side of the boundary, spillovers from the nuisance created by ADUs will both increase transactions and possibly even decrease sale prices, even on the un-reformed side of the boundary. Finally, the reformed side of the boundary should generate more tax revenues per lot as lots that transact and / or build an ADU contribute more through the property tax to supply the public service.⁹⁴

This simple open-city model also motivates a few heterogeneity exercises. First, the option value in the model is a function of lot quality for ADUs, where lot quality should weakly increase our estimates of the option value if it is generally the case that only building on the reformed side is viable. We can explore this by using the lot distance from the central business district (CBD) and the amount of available square footage on the parcel as measures of lot quality for an ADU. Next, we will turn to Equation 13, which suggests that the amount that Proposition 13 shelters a parcel's property tax base should decrease transaction volumes on both sides of our discontinuity equally. However, if Proposition 13 decreases transaction volumes then the model still suggests this may decrease ADU permits as entrants and firms can build at a lower cost than incumbents. Due to their ability to transfer their property tax shelters, owners over the age of 55 should be more likely to transact, as the effect of the Proposition 13 lock-in should be diminished for them and the gap between their build option and the market option may be greater than for other owners. The model would suggest that this in turn should spur ADU construction.

7 Discussion

7.1 Heterogeneity

7.1.1 Does Lot Quality Matter?

We investigate how lot quality, as measured by a parcel's distance to the nearest CBD, and suitability for an ADU, as measured by excess land on the lot, may influence the impacts of permitting reforms. We discretize both outcomes into dummy variables, where a value of 1 indicates a value above the analysis sample median. Our theoretical model in section 6 suggests that lots of higher quality for an ADU should build them at higher rates. Additionally, Equation 14 implies that if we are capturing the full option value for just the reform side of the boundary, then our difference in discontinuities estimates should increase with lot quality. If we do not see heterogeneity by lot quality, then our estimate is capturing differences in permitting costs across the Coastal Zone boundary.

⁹⁴In Appendix F, we work through the city's problem.

Our conditional average treatment effect estimates by lot quality are presented in Table 5. We present estimates in odd columns for our baseline specification interacted with the heterogeneity variable of interest following Equation 5. Then, in the even columns we present our preferred specification with segment by time fixed effects. Column 2 presents our preferred heterogeneity specification showing how ADU reform impacts on permitting activity vary by excess square footage on the lot. The coefficient on the Coastal Zone boundary dummy is interpreted here as the treatment effect of permitting reform on lots with below 4,212 sq.ft. of available space on their lot for an ADU. The coefficient is significant at the 5% level and comparable to our main estimates in Table 3, 0.618. Although, our coefficient on the interaction of the Coastal Zone dummy and the excess lot sq.ft. dummy suggests that lots with additional space permit additional ADUs, the coefficient itself is insignificant. The insignificant interaction likely reflects the fact that the vast majority of lots in our sample are suitable for an ADU, according to Table 2.

Estimates in Panel A Column 4 of Table 5 suggest that lots within 26 kilometers (16.2 miles) of a CBD have larger treatment effects from permitting reform. Our coefficient on the Coastal Zone boundary dummy suggests that permitting reform within 26 kilometers of a CBD increased by almost 1 unit per thousand parcel years. These effects are significant at the 0.1% level and are almost 50% larger than our baseline estimates. Moreover, the interaction term coefficient suggests that for parcels more than 26 kilometers from a CBD, the impacts of permitting reform were just 30% of our main estimates. This suggests that a large component of lot quality for an ADU is related to access to urban agglomerations. Moreover, if smaller lots are in general located near CBDs, then Table 5 suggests that a lot's relationship to urban agglomerations may be a more important measure of quality than unbuilt area on the lot, which is abundant in our sample.

Heterogeneity by lot quality may help us understand if our difference in discontinuities estimates are picking up mostly differences in option value or differences in permitting costs across the border. Recall from Equation 14 that differences in land values across the boundary can reflect the full market option value if after reform it is only viable to build outside the Coastal Zone or our estimates can reflect differences in permitting costs across the boundary if there is no heterogeneity in lot quality.

We present our estimates of heterogeneity in treatment for land values by lot quality in Panel B of Table 5. In Columns 1 and 2 we show difference in discontinuities with heterogeneity using the above and below median excess lot square footage dummy. In columns 3 and 4 we show estimates that use above and below median dummies for distance to the nearest CBD. Odd columns use our baseline specification without parcel fixed effects and segment by time fixed effects, and even columns use our preferred specification with a parcel fixed effect and segment by time fixed effects. In Column 3, our estimate on the Coastal Zone boundary dummy suggests that on lots within 27 kilometers (17 miles) of a CBD permitting reform increased lot values by 17%.⁹⁵ These estimates are significant at the 0.1% level and more than double our reported main estimates of about 7%.

⁹⁵Since the sample varies between Panels A and B, the medians used to build the binary heterogeneity variables change with the sample.

The coefficient on the outside the Coastal Zone boundary and farther than 27 kilometers from a CBD is also negative and significant at the 5% level. This suggests that as parcels got farther away from the CBD, the impact of permitting reform on land values decreased. These estimates suggest that there is likely a gradient in the effects of permitting reform on land values in lot quality as measured by distance to the CBD. However, when we add in segment by time fixed effects and a parcel fixed effect the picture becomes more complicated. The estimate in Column 4 for the coefficient on the Coastal Zone boundary dummy is positive and significant at the 5% level. The coefficient is consistent in magnitude with our main estimates, showing that parcels within 27 kilometers of a CBD saw a 6% increase in land values from permitting reforms. The interaction coefficient shows a null effect. In contrast to Column 3, when we add segment by time fixed effects we appear to be detecting differences in permitting costs across the boundary more than differences in the full option value.

How can we reconcile the differences in treatment effect heterogeneity in lot quality for our land value estimates? The estimates in Panel B Column 3 of Table 5 capture variation not just within the 145 Coastal Zone segments but between the segments as well. In contrast, the estimates in Column 4 only capture variation within each coastal zone segment. Across segments there are likely large differences in the value of an ADU along many dimensions, not just distance from the CBD or available space to build an ADU. Moreover, in the overwhelming majority of Coastal Zone segments there is ADU permitting on both sides of the boundary. These results could also reflect differences in baseline stringency from jurisdictions bordering the Coastal Zone. Figure 5 suggests that there isn't much variation in zoned capacity across coastal jurisdictions, but zoned capacity may not perfectly correlate with the local permitting factors that are most relevant for ADUs. Overall, the heterogeneity estimates suggest that our repeat sale estimates with segment by time fixed effects are likely detecting mostly differences in permitting costs across the boundary rather than full market option value.

7.1.2 Do Property Tax Shelters Mitigate the Impacts of Permitting Reform?

Our second set of heterogeneity exercises unpacks how property taxes interact with ADU permitting through their impacts on transactions. The owner's buy or sell indifference condition, Equation 13 in Section 6, suggests that additional property tax shelter should decrease the likelihood that a parcel transacts equally on both sides of the boundary, *ceteris paribus*. However, both our model and Figure 7 suggest that ADU permitting typically occurs around a transaction event. Across all parcels, we see that additional property tax shelter is correlated with both lower transaction rates and lower permitting rates for ADUs in Figure 4.

We display our heterogeneity estimates exploring how property tax shelter influences the effects of permitting reform on transactions in Table 6 Panel A. Columns 1 and 2 interact the Coastal Zone boundary dummy with the 2016 estimated property tax shelter for each parcel. Our preferred estimate is in Column 2, which includes segment by time fixed effects.⁹⁶ The coefficient on the

⁹⁶Note that for transactions we can now always include a parcel fixed effect.

Coastal Zone boundary dummy suggests that parcels that were assessed at market value in 2016 saw smaller changes in transaction volumes than similar parcels inside the Coastal Zone. This effect is a decrease of about 1% and significant at the 5% level. The interaction term suggests that as tax shelter increases, transaction volume differences barely move. These estimates suggest that a parcel with average shelter outside the Coastal Zone only transacts about 0.5 percentage points less than a similar parcel inside the Coastal Zone. In the entire sample, about 10% of parcels transact each year. These estimates support the functional form of Equation 13, which suggests that property tax shelter decreases transactions independently of permitting reforms.

Table 6 Panel B shows heterogeneity estimates for how the property tax interacts with treatment effects for permitting activity. In Column 1 we show our baseline regression discontinuity design fully interacted with the property tax shelter in 2016. In Column 2 we show our preferred specification that adds segment by time fixed effects. The coefficient for the Coastal Zone border dummy here captures the effect of permitting reform on ADU permitting for parcels assessed at market value (no shelter). The estimates show that these parcels with no tax shelter in 2016 build ADUs at higher rates between 2018 and 2024. The coefficient is larger than our main estimates and significant at the 5% level. The coefficient on the interaction term suggests that parcels with more tax shelter permitted fewer ADUs, and the estimate is of a meaningful magnitude, but the coefficient is not significant at traditional levels. These estimates suggest that property tax shelter may decrease ADU permitting, and mute the impacts of permitting reform to an extent. Although it is difficult to disentangle property tax shelter from preferences that reduce moving and ADU construction that likely correlate with sufficient amounts of tenure to accrue large amounts of tax shelter on a parcel.

To provide additional evidence on what role California’s property tax system played in the low take-up of ADU construction, we leverage a special rule that allows homeowners over 55 years of age to transfer their below-market property tax base to a new primary residence. Proposition 90 allows owners over 55 to transfer their tax base if they move within one of ten participating counties, which include all of the Southern California coastline.⁹⁷ Then, in 2021, the state extended these privileges to all owners over the age of 55. The goal of these estimates is to leverage the fact that owners over the age of 55 essentially won’t have property tax shelter depressing their transaction rates if they are willing to move within California. We do not have data on owner ages, so we instead use the share of homeowners above the age of 55 in 2016 in a block group from the 2016 5-year pooled ACS (U.S. Census Bureau, 2016). We also break this exercise up into estimates stopping before COVID and estimates on the full dataset. We do this because COVID-19 likely decreased the willingness of older adults to move.⁹⁸

We show our estimates of how property tax base mobility for owners over 55 impacted the effects of permitting reform on transaction rates in Panel A of Table 6. Columns 3 and 4 show pre-COVID estimates, and Columns 5 and 6 show estimates for all available years, including the COVID

⁹⁷Ferreira (2010) provides evidence that Proposition 90 did induce owners over 55 to move.

⁹⁸Foster, Fiorio and Ellis (2024) use US Postal Service change of address data to show that mobility among older natives declined in the US in 2020 and 2021 relative to 2019, likely in response to the COVID-19 pandemic.

years. Column 4 shows our preferred pre-COVID heterogeneity specification which uses segment by time fixed effects. These estimates suggest that a block group with no homeowners above the age of 55 would be about 2% less likely to sell their home or transact the parcel compared to a similar parcel within the Coastal Zone. The coefficient is significant at the 5% level and more than double the magnitude and opposite sign of estimates without heterogeneity from Figure 12b.⁹⁹ The interaction term of the share of owners above 55 and the coastal zone boundary and post dummy show a strong upward slope in transaction rates with block group age. The coefficient implies that for every additional 10 percentage point increase in area owners over the age of 55, the transaction rate increases by 0.4 percentage points. This coefficient is significant at the 5% level. The average parcel in our sample has a share of owners in its block group above 55 of about 61% implying a CATE on changes in transactions of about 0.1 percentage points, a bit lower than our main estimates in Figure 12b. However, the full sample estimates from 2010 to 2024 in Column 6 reflect an overall increase in transactions, but a decline in the gradient of the CATE with area owner shares above 55 years of age. Together, these results suggest that allowing wider property tax base transfer rights could have increased transactions post reform.

Our estimates in Panel B of Table 6 will shed light on whether or not access to property tax mobility impacted the effects of ADU reforms on permitting activity. Across specifications, pre-COVID and on the full sample, in Columns 3 through 6, we see only suggestive evidence that parcels within block groups with fewer owners above the age of 55 may build ADUs at higher rates, since the coefficient on the interaction term between the share of owners over 55 and the Coastal Zone dummy is negative and insignificant across specifications. These estimates could be consistent with differing preferences over density and ADUs across age groups, rather than property tax mobility, as influencing transactions and permitting. The vanishing significance of the interaction terms in the full sample suggests that older owners' preferences may have changed during the pandemic. Parcels in block groups with more owners over 55 transacted more before COVID, but this likely didn't lead to additional ADU permits as a result.

Overall the evidence that the property tax lowered ADU permitting by depressing transaction volumes is only suggestive. The best evidence comes from the heterogeneity using the property tax shelter itself. These estimates support that property tax shelter alone didn't meaningfully impact the effects of permitting reform on transactions, but the least sheltered properties did permit ADUs at higher rates. This pattern is supported in the aggregate by Figure ??.

7.2 Why Was Permitting So Low?

We have explored treatment heterogeneity in permitting by distance to the CBD and property tax shelter. Although it might be the case that low take-up is a story of a lack of suitable lots or that Proposition 13 is depressing ADU construction, the presence of option value in the land points to an alternative story.

The literature on option value in housing development typically finds that it is associated with

⁹⁹It should be noted that there are no block groups with zero owners above the age of 55 in our dataset.

low rates of development. [Cunningham \(2007\)](#) studies growth controls around Seattle and finds that when the uses for lots are constrained the option value of waiting to develop decreases and more owners develop their land. In a low-demand setting, [Brooks, Rose and Veuger \(2024\)](#) find that owners of lots in DC destroyed by the MLK riots of 1968 allowed these lots to remain vacant for decades. They rationalize this finding with an option story: owners delayed development until uncertainty over future prices decreased. In contrast, [Hornbeck and Kenniston \(2017\)](#) find that large redevelopments can have positive spillovers that incentivize redevelopment of nearby parcels. There are also several other papers that view real estate development from an options perspective ([Titman, 1985](#); [McMillen and O’Sullivan, 2013](#); [Yao and Pretorius, 2014](#)).

Our theoretical model in Section 6 and a simpler real option model we show in Appendix C both point to take-up as a function of the expected benefits from an ADU and the costs of building an ADU. Permitting reform directly lowers one type of cost: the cost, in time or money, of getting the permission to build an ADU. However, permitting reform cannot address construction costs which can be substantial. Industry quotes for a new ADU range from \$200 to \$300 per square foot ([Henson, 2023](#)). Construction costs have been rising generally, and they may be a part of the story we will need to leave for future work ([Gyourko and Saiz, 2006](#); [Brooks and Liscow, 2023](#); [Eriksen and Orlando, 2024](#)).

A second channel that may dampen construction and increase option value is uncertainty about future rents. In the open city model this would be equivalent to assigning a distribution to how ADU value might scale with lot quality. In the traditional option model in Appendix C, uncertainty is modeled as a distribution across a high state and a low state in the future. Increased uncertainty over the future value of an ADU should decrease development and increase the value of waiting to invest.

There are several reasons why landowners in California may feel like their ability to benefit from their ADU is uncertain. First, California in 2019 passed state-wide rent control, and several cities also have their own forms of rent control, which may lower the supply of ADUs for rent ([Asquith and Reed, 2021](#); [Diamond, McQuade and Qian, 2019](#)). Second, a number of jurisdictions also impose restrictions on short term rentals, which might reduce the number of ADUs permitted as a potential source of rental income is restricted ([Koster, van Ommeren and Volkhausen, 2021](#); [Seiler, Siebert and Yang, 2024](#); [Bibler, Teltser and Tremblay, 2025](#)). We will leave investigating the role of rental income uncertainty on ADU construction for future work.

7.3 Fiscal Impacts

Although the permitting reforms may not have put an ADU in every backyard, it did increase the value of these backyards, which could have significant fiscal impacts through property taxes. Property taxes in California are artificially low as a result of Proposition 13; taxable value updates after a sale or when there is a significant addition to the property ([San Francisco Assessor Recorder’s Office, 2025](#)). Even so, if the land values across the state have increased, then jurisdictions that rely on the property tax will see revenue gains through two channels. First, as ADUs get built,

assessments on the new property will add the value of the ADU to the property tax base without re-assessing the whole parcel. Second, when a parcel transacts the entire parcel gets reassessed. Both of these channels are impacted by ADU permitting reforms, as shown in our theoretical model and in the empirical results above. The overlap of these channels makes a simple back of the envelope style estimate of the impacts of permitting reforms on property tax collections non-trivial.

We estimate the impact of ADU reforms on property tax collections in Table 7. Here, we estimate Equation 3 after replacing the outcome variable with the log of real property taxes paid per acre. Our preferred specification is in Column 7, which includes segment by time fixed effects. Our estimates suggest that permitting reforms for ADUs increased property tax revenues per acre by 3% and our estimates are significant at the 0.1% level and robust to alternative specifications.

However, from these revenue increases alone we cannot make conclusions about the overall impact of ADU reforms on local government’s fiscal health. The net fiscal impacts could be positive if the composition of residents by demand for local services (police, fire, education, etc.) remains constant and the resident population stays constant. However, our estimates suggest that parcels in older areas, at least before COVID, were more likely to transact in response to permitting reform. This may suggest that these areas are getting younger, which may come with a shift in preferences of spending and associated changes in local expenditures that may shift the fiscal gain in revenues into a net neutral fiscal impact or even a fiscal loss (Poterba, 1997; Hilber, 2017).

7.4 Rents

Basic economic models and empirical evidence suggest that more elastic housing supply should reduce rents (Glaeser, Gyourko and Saks, 2006; Molloy, 2020; Molloy, Nathanson and Paciorek, 2022). Naturally, since our estimates suggest that permitting reform for ADUs led to additional permitting, then the open question is whether or not permitting reforms reduced rents. We use an instrumental variables approach based on the Coastal Zone boundary and block group data on gross rent from the ACS 5-year Samples from 2016 and 2024 to probe this question (U.S. Census Bureau, 2016, 2024). We use the share of block group area that overlaps with the Coastal Zone as an instrument for the cumulative change in ADU permits from 2018 to 2024 per housing unit in 2016. We then use this instrument to show how changes in ADU permits impacted percent changes in block group gross rents between 2016 and 2024. Our first stage F-statistic for our preferred specification with controls using block groups within 2,000 meters of the Coastal Zone boundary is 19.5. We show details about these estimates in Appendix D.

Our estimates in Appendix D suggest that permitting reform for ADUs had no detectable impact on area rents. For the average block group in our sample, rents increased between 2016 and 2024 by 28%. Our point estimate suggests that a one standard deviation increase in ADU permits (16.5 permits per 1,000 units in 2016) decreased rent changes by 4 percentage points. These results are imprecise and not statistically significant at traditional levels. These estimates are only suggestive, as the low uptake of ADU permitting as described above is likely the culprit for the imprecision of

these, and related, estimates.¹⁰⁰

The evidence presented here is not sufficient to conclude that ADU reforms didn't lower rents. However, given the low take-up associated with ADU permitting, about 176,000 units statewide since 2017, it is unlikely that ADUs had a large macro effect on rents. However, Figure 13 suggests that there are some areas with quite concentrated areas of ADU permitting. It might be the case that ADUs meaningfully decreased rents in jurisdictions where ADUs made up a significant portion of the supply of units. Outside of our 15 coastal county sample, the San Fernando Valley and Bakersfield appear to have enough ADU permitting as a share of existing pre-reform units to see meaningful rent declines.

8 Conclusion

Who benefits from permitting reforms for ADUs? We deployed a plausibly exogenous boundary discontinuity that captures discrete changes in where permitting reforms applied. Our results suggest that reforms increased permitting activity by about 30%, local to our boundary. However, the baseline permitting rate is so low that just 1 in 200 parcels were induced to permit an ADU. The principal effect of the reform was to raise the land values by about 7%. This increased land value applied to every reformed parcel, whether the owner applied for an ADU permit or not. This increase of about \$500,000 per acre (\$100,000 on a 0.2 acre lot) accrued to incumbent land owners, who could either exercise their option to build an ADU or realize the benefits by selling their parcel. Local and county governments also saw about 3% higher property tax collections, although the full net fiscal impact depends on several factors outside the scope of this paper. We do not find evidence of significant negative spillovers from permitting reform, which is consistent with our finding of low overall take-up. However, our results do not support the finding that permitting reforms for ADUs lowered rents, but these estimates are imprecise.

Our simple open city model characterizes the increase in land values with permitting reform as a step function in lot quality. In an environment where it is only economically viable to build on one side of the boundary, our difference in discontinuities estimates should capture the full market option of building an ADU and exhibit heterogeneity by lot quality. However, if it is viable to build on both the low permitting cost side and high permitting cost side of the boundary, then our estimates will capture the difference in permitting costs and not show heterogeneity in lot quality. Our results suggest that we are in the latter case, and our estimates suggest that permitting reform for ADUs was valued at 7% of land value. Our estimate for the value of permitting reforms is meaningfully smaller than recent estimates from Soltas and Gruber (2026), who estimate the value of circumventing permitting entirely.

Although permitting reform was valuable, it likely did not induce take-up at a scale large enough to broadly reduce rents. High construction costs, of about \$200 to \$300 per square foot, and uncertainty about future rental income may be factors that depress take-up, although exploring

¹⁰⁰Bhatt (2023) also uses ACS data at the tract level and finds no evidence that ADUs lower rents in major California cities using a difference in differences approach that compares single family zoned tracts to multifamily zoned tracts.

both factors empirically is left for future work ([Henson, 2023](#); [Seiler, Siebert and Yang, 2024](#); [Bibler, Teltser and Tremblay, 2025](#)). Our estimates suggest that permitting reform is only one component of the barriers to ADU construction.

Our results speak to why gentle, or “light touch” up-zoning for small units have been so popular. Nationally, as of April 2026, 96 land use bills for ADUs have been passed across 24 states since 2016 ([NYU Furman Center, 2026](#)). Homeowner resistance to up-zoning typically reflects a desire to maximize home values ([Fischel, 2001](#)). However, ADU reforms have the opposite effect: they raise home values, and too few units are built for negative spillovers to manifest, at least in our data.¹⁰¹ Our results should be interpreted as local to our setting: the California Coast. We also are only able to measure permitting activity, rather than construction. There may be steps between permitting and completion that our results cannot speak to. Our results suggest that ADU reforms created many winners: incumbent homeowners and possibly even government budgets.¹⁰² Even so, reforms aimed at addressing the high cost of housing may need to be less “light touch”.¹⁰³

¹⁰¹Figure [13a](#) shows areas in which ADUs make up a meaningful supply of pre-reform housing units. These areas might experience meaningfully different dynamics than areas near the Coastal Zone.

¹⁰²In our setting, we model entrants as indifferent between purchasing a built parcel and the outside option for simplicity. However, this masks ways in which entrants may benefit from ADU reform.

¹⁰³For examples, see [Anagol, Ferreira and Rexer \(2026\)](#) and [Asquith, Mast and Reed \(2023\)](#).

References

- Abadie, Alberto, Susan Athey, Guido W. Imbens, and Jeffrey M. Wooldridge.** 2023. “When Should You Adjust Standard Errors for Clustering?” *Quarterly Journal of Economics*, 138(1): 1–35.
- Ahlfeldt, Gabriel M., Hans R. A. Koster, and Tu Giang Vu.** 2025. “Mind the Lag: Using Assessed and List Prices as Proxies for Housing Market Values.” CESifo CESifo Working Paper 12290.
- Albouy, David, Gabriel Ehrlich, and Minchul Shin.** 2018. “Metropolitan Land Values.” *Review of Economics and Statistics*, 100(3): 454–466.
- Alonso, William.** 1964. *Location and Land Use*. Harvard University Press.
- Anagol, Santosh, Fernando Ferreira, and Jonah Rexer.** 2026. “Estimating the Economic Value of Zoning Reform.” *American Economic Journal: Economic Policy*, 18(3): 235–274.
- Anderson, Dana.** 2026. “The Typical U.S. Homeowner Hangs Onto Their House For 12 Years. In Los Angeles, It’s 20 Years.” *Redfin Real Estate News*. Updated March 4, 2026.
- Anthropic.** 2026. “Claude Opus 5.” Anthropic.
- Asquith, Brian J., and Shane M. Reed.** 2021. “Rent Control in California: Policy Review.” W.E. Upjohn Institute for Employment Research Technical Report, Kalamazoo, MI.
- Asquith, Brian J., Evan Mast, and Davin Reed.** 2023. “Local Effects of Large New Apartment Buildings in Low-Income Areas.” *The Review of Economics and Statistics*, 105(2): 359–375.
- Bailey, Martin J., Richard F. Muth, and Hugh O. Nourse.** 1963. “A Regression Method for Real Estate Price Index Construction.” *Journal of the American Statistical Association*, 58(304): 933–942.
- Bartik, Alexander, Arpit Gupta, and Daniel Milo.** 2025. “The Costs of Housing Regulation: Evidence From Generative Regulatory Measurement.” *Available at SSRN*.
- Baum-Snow, Nathaniel, and Lu Han.** 2024. “The Microgeography of Housing Supply.” *Journal of Political Economy*, 132(6): 1897–1946.
- Baylis, Patrick, and Judson Boomhower.** 2023. “The Economic Incidence of Wildfire Suppression in the United States.” *American Economic Journal: Applied Economics*, 15(1): 442–473.
- Bhatt, Ishan.** 2023. “Yes, Literally, In My Backyard: The Effect of “Gently” Upzoning Single-Family Neighborhoods.” Mossavar-Rahmani Center for Business and Government, Harvard Kennedy School M-RCBG Associate Working Paper Series 203.
- Bibler, Andrew J., Keith F. Teltser, and Mark J. Tremblay.** 2025. “How Do Short-Term Rental Regulations Affect Market Outcomes? Evidence from San Francisco’s Airbnb Registration Requirement.” *Real Estate Economics*, 53(4): 841–868.
- Black, Sandra E.** 1999. “Do Better Schools Matter? Parental Valuation of Elementary Education.” *Quarterly Journal of Economics*, 114(2): 577–599.

- Blanco, Hector, and Noémie Sportiche.** 2026. “There Goes the Neighborhood? The Local Impacts of State Policies that Override Municipal Zoning.” *American Economic Journal: Applied Economics*. Forthcoming.
- Brinig, Margaret, and Nicole Stelle Garnett.** 2013. “A Room of One’s Own? Accessory Dwelling Unit Reforms and Local Parochialism.” *The Urban Lawyer*, 35(3): 519–570.
- Broockman, David, Christopher S. Elmendorf, and Joshua L. Kalla.** 2026. “How Sociotropic Aesthetic Judgments Drive Opposition to Dense Housing Development.” Niskanen Center.
- Brooks, Leah, and Zachary Liscow.** 2023. “Infrastructure Costs.” *American Economic Journal: Applied Economics*, 15(2): 1–30.
- Brooks, Leah, Jonathan Rose, and Stan Veuger.** 2024. “Destruction, Policy, and the Evolving Consequences of Washington, DC’s 1968 Civil Disturbance.” *Review of Economics and Statistics*, 1–45.
- Butts, Kyle.** 2023. “Geographic difference-in-discontinuities.” *Applied Economics Letters*, 30(5): 615–619.
- California Coastal Commission.** 2014. “Coastal Zone Boundary [ds990].”
- California Coastal Commission.** 2022*a*. “Our Mission.” California Coastal Commission.
- California Coastal Commission.** 2022*b*. “Updates Regarding the Implementation of New ADU Laws.” California Natural Resources Agency.
- California Department of Fish and Wildlife.** 2026. “Biogeographic Information and Observation System (BIOS).” Accessed August 30, 2026.
- California Department of Housing and Community Development.** 2025. “Housing Element Annual Progress Report (APR) Data by Jurisdiction and Year.”
- California Department of Housing & Community Development.** 2026. “ADU Handbook.” California Department of Housing & Community Development.
- California Office of Emergency Services.** 2015. “Enriched CalTrans Incorporated Cities 2015.” California Open Data Portal.
- California State Board of Equalization.** 2025*a*. “Lesson 17: The Land Residual Techniques of Income Capitalization (The Income Approach to Value).” California State Board of Equalization Appraisal Training.
- California State Board of Equalization.** 2025*b*. “Transfer of Base Year Value for Persons Age 55 and Over — Propositions 60/90.” Accessed August 30, 2026.
- Calonico, Sebastian, Matias D. Cattaneo, and Max H. Farrell.** 2020. “Optimal Bandwidth Choice for Robust Bias-Corrected Inference in Regression Discontinuity Designs.” *The Econometrics Journal*, 23(2): 192–210.
- Calonico, Sebastian, Matias D. Cattaneo, Max H. Farrell, and Rocio Titiunik.** 2017. “rdrbust: Software for Regression Discontinuity Designs.” *The Stata Journal*, 17(2): 372–404.

- Calonico, Sebastian, Matias D. Cattaneo, Max H. Farrell, Filippo Palomba, and Rocio Titiunik.** 2025. “Treatment Effect Heterogeneity in Regression Discontinuity Designs.” *arXiv*.
- Cañedo Riedel, Gabriel, and Philip G. Hoxie.** 2025. “Living with Density: The Impacts of California’s By-Right Permitting for ADUs.” Center for Open Science Pre-Analysis Plan.
- Card, David, Alexandre Mas, and Jesse Rothstein.** 2008. “Tipping and the Dynamics of Segregation.” *The Quarterly Journal of Economics*, 123(1): 177–218.
- Clemens, Jeffrey, and Anwita Mahajan.** 2026. “The Utility of AI Tools in Auditing Adherence to Pre-Analysis Plans.” National Bureau of Economic Research NBER Working Paper 35719.
- Conley, Timothy G.** 1999. “GMM Estimation with Cross Sectional Dependence.” *Journal of Econometrics*, 92(1): 1–45.
- Cotality.** 2025. “Historic Property and Transfers.” Corelogic Incorporated.
- County of Humboldt.** 2017. “Summary of Recent Changes to ADU Laws.” County of Humboldt.
- Cullen, Julie Berry, and Steven D. Levitt.** 1999. “Crime, Urban Flight, and the Consequences for Cities.” *The Review of Economics and Statistics*, 81(2): 159–169.
- Cunningham, Christopher R.** 2007. “Growth Controls, Real Options, and Land Development.” *The Review of Economics and Statistics*, 89(2): 343–358.
- Cunningham, Scott.** 2021. *Causal Inference*. Yale University Press.
- D’Amico, Leonardo, Evan Soltas, and Audrey Wang.** 2026. “Where Is Housing Slow to Build, And Is It Getting Slower?” Working paper.
- Davidoff, Thomas, Andrey Pavlov, and Tsur Somerville.** 2022. “Not In My Neighbour’s Back Yard? Laneway Homes and Neighbours’ Property Values.” *Journal of Urban Economics*, 128: 103405.
- Davidson, Nestor, Richard Briffault, Paul Diller, Sarah Fox, Laurie Reynolds, Erin Adele Scharff, Richard Schragger, and Rick Su.** 2020. “Principals of Home Rule in the 21st Century.” National League of Cities.
- Davis, Morris A., and Michael G. Palumbo.** 2008. “The price of residential land in large US cities.” *Journal of Urban Economics*, 63(1): 352–384.
- Davis, Morris A., Stephen D. Oliner, Edward J. Pinto, and Sankar Bokka.** 2017. “Residential land values in the Washington, DC metro area: New insights from big data.” *Regional Science and Urban Economics*, 66: 224–246.
- Davis, Morris A., William D. Larson, Stephen D. Oliner, and Jessica Shui.** 2021. “The Price of Residential Land for Counties, ZIP Codes, and Census Tracts in the United States.” *Journal of Monetary Economics*, 118: 413–431.
- Diamond, Rebecca, and Tim McQuade.** 2019. “Who Wants Affordable Housing in Their Backyard? An Equilibrium Analysis of Low-Income Property Development.” *Journal of Political Economy*, 127(3): 1063–1117.

- Diamond, Rebecca, Tim McQuade, and Franklin Qian.** 2019. “The Effects of Rent Control Expansion on Tenants, Landlords, and Inequality: Evidence from San Francisco.” *American Economic Review*, 109(9): 3365–3394.
- Dillon, Liam, and Andrew Khouri.** 2016. “How to solve California’s housing shortage? Build ‘Granny Flats’ In Homeowners’ Backyards.” *Los Angeles Times*.
- Duan, Naihua.** 1983. “Smearing Estimate: A Nonparametric Retransformation Method.” *Journal of the American Statistical Association*, 78(383): 605–610.
- Duranton, Gilles, and Diego Puga.** 2023. “Urban Growth and Its Aggregate Implications.” *Econometrica*, 91(6): 2219–2259.
- Eckert, Fabian, Sharat Ganapati, and Conor Walsh.** 2020. “Skilled Scalable Services: The New Urban Bias in Economic Growth.” CESifo 8705h.
- Eriksen, Michael D., and Anthony W. Orlando.** 2024. “A Cost Decomposition of Break-Even Rents for New Multifamily Housing Development.” *Journal of Housing Economics*.
- Ferreira, Fernando.** 2010. “You Can Take It with You: Proposition 13 Tax Benefits, Residential Mobility, and Willingness to Pay for Housing Amenities.” *Journal of Public Economics*, 94(9–10): 661–673.
- Fischel, William A.** 2001. *The Homevoter Hypothesis: How Home Values Influence Local Government Taxation, School Finance, and Land-Use Policies*. Harvard University Press.
- Foster, Thomas B., Lee Fiorio, and Mark Ellis.** 2024. “Internal Migration in the U.S. During the COVID-19 Pandemic.” U.S. Census Bureau, Center for Economic Studies CES Working Paper 24-50.
- Frank, Eyal G, Maximilian Auffhammer, David McLaughlin, Elisheba Spiller, and David L Sunding.** 2025. “The Cost of Species Protection: The Land Market Impacts of the Endangered Species Act.” National Bureau of Economic Research Working Paper 33352.
- Freemark, Yonah.** 2023. “Zoning Change: Upzonings, Downzonings, and Their Impacts on Residential Construction, Housing Costs, and Neighborhood Demographics.” *Journal of Planning Literature*, 38(4): 548–570.
- Ganong, Peter, and Daniel Shoag.** 2017. “Why has Regional Income Convergence in the US Declined?” *Journal of Urban Economics*, 102: 76–90.
- Glaeser, Edward L.** 2013. “Urban Public Finance.” *Handbook of Public Economics*, 5: 195–256.
- Glaeser, Edward L., and Joseph Gyourko.** 2025. “America’s Housing Supply Problem: The Closing of the Suburban Frontier?” *Brookings Papers on Economic Activity*, 2025(1): 375–425.
- Glaeser, Edward L., Joseph Gyourko, and Raven E. Saks.** 2006. “Urban Growth and Housing Supply.” *Journal of Economic Geography*, 6(1): 71–89.
- Gold, Daniel, and Anya Tarascina.** 2026. “Much ADU About Something?” *SSRN Electronic Journal*.
- Google Maps Platform.** 2026. “Places API.” Google.

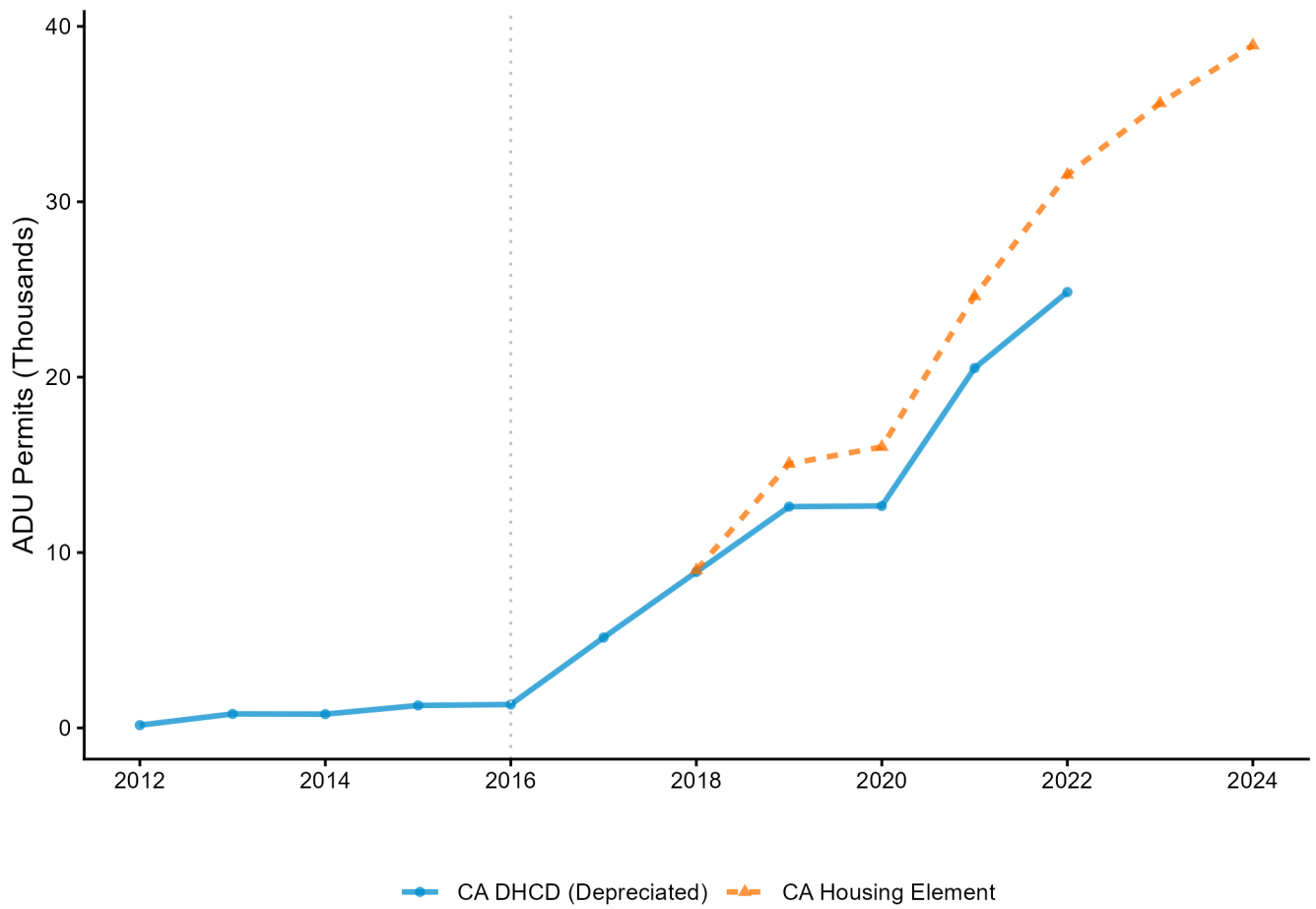
- Gordon, Roger H.** 1983. “An Optimal Taxation Approach to Fiscal Federalism.” *Quarterly Journal of Economics*, 98(4): 567–586.
- Gray, M. Nolan.** 2024. “California ADU Reform: A Retrospective.” California YIMBY.
- Green, Daniel, Vikram Jambulapati, Jack Liebersohn, and Tejaswi Velayudhan.** 2026. “Fiscal Externalities of Transaction Taxes: Evidence from the Los Angeles Mansion Tax.” *Journal of Public Economics*, 260: 105678.
- Grembi, Veronica, Tommaso Nannicini, and Ugo Troiano.** 2016. “Do Fiscal Rules Matter?” *American Economic Journal: Applied Economics*, 1–30.
- Grout, Cyrus A., William K. Jaeger, and Andrew J. Plantinga.** 2011. “Land-use regulations and property values in Portland, Oregon: A regression discontinuity design approach.” *Regional Science and Urban Economics*, 41(2): 98–107.
- Gyourko, Joseph, Albert Saiz, and Anita Summers.** 2008. “A New Measure of the Local Regulatory Environment for Housing Markets: The Wharton Residential Land Use Regulatory Index.” *Urban Studies*, 45(3): 693–729.
- Gyourko, Joseph, and Albert Saiz.** 2006. “Construction Costs and the Supply of Housing Structure.” *Journal of Regional Science*, 46(4): 661–680.
- Gyourko, Joseph, and Sean E McCulloch.** 2024. “The Distaste for Housing Density.” National Bureau of Economic Research Working Paper 33078.
- Haughwout, Andrew F., James Orr, and David Bedoll.** 2008. “The Price of Land in the New York Metropolitan Area.” *Current Issues in Economics and Finance*, 14(3).
- Henson, Trevor.** 2023. “Demystifying The Costs: How Much Does it Really Cost To Build An ADU in California?” Beach Front Property Management Incorporated.
- Hilber, Christian A. L.** 2017. “The Economic Implications of House Price Capitalization: A Synthesis.” *Real Estate Economics*, 45(2): 301–339.
- Holian, Matthew J.** 2019. “Where Is the City’s Center? Five Measures of Central Location.” *Cityscape*, 21(2): 213–226.
- Hornbeck, Richard, and Daniel Kenniston.** 2017. “Creative Destruction: Barriers to Urban Growth and the Great Boston Fire of 1872.” *American Economic Review*, 107(6): 1365–1398.
- Hoxie, Philip G., Daniel Shoag, and Stan Veuger.** 2023. “Moving to Density: Half a Century of Housing Costs and Wage Premia from Queens to King Salmon.” *Journal of Public Economics*, 222: 104906.
- Ihlanfeldt, Keith R.** 2007. “The Effect of Land Use Regulation on Housing and Land Prices.” *Journal of Urban Economics*, 61(3): 420–435.
- Ihlanfeldt, Keith R., and Jorge Martinez-Vazquez.** 1986. “Alternative Value Estimates of Owner-Occupied Housing: Evidence on Sample Selection Bias and Systematic Errors.” *Journal of Urban Economics*, 20(3): 356–369.
- IPUMS NHGIS.** 2026. “Geographic Crosswalks: 2010 Block Groups to 2020 Block Groups.” 2010–2020 block group crosswalk.

- Jo, Nathanael, Andrea Vallebuono, Derek Ouyang, and Daniel E. Ho.** 2025. “Not (Officially) in My Backyard.” *Journal of the American Planning Association*, 91(1): 30–45.
- Kahn, Eli, and Salim Furth.** 2023. “Breaking Ground: An Examination of Effective State Housing Reforms in 2023.” *Mercatus Policy Brief*.
- Kahn, Matthew E., Ryan Vaughn, and Jonathan Zasloff.** 2010. “The Housing Market Effects of Discrete Land Use Regulations: Evidence from the California Coastal Boundary Zone.” *Journal of Housing Economics*, 19(4): 269–279.
- Kopczuk, Wojciech, and David Munroe.** 2015. “Mansion Tax: The Effect of Transfer Taxes on the Residential Real Estate Market.” *American Economic Journal: Economic Policy*, 7(2): 214–257.
- Koster, Hans R. A., Jos van Ommeren, and Nicolas Volkhausen.** 2021. “Short-Term Rentals and the Housing Market: Quasi-Experimental Evidence from Airbnb in Los Angeles.” *Journal of Urban Economics*, 124: 103356.
- Legislative Analyst’s Office.** 2015. “California’s High Housing Costs: Causes and Consequences.” California Legislative Analyst’s Office.
- Linden, Leigh, and Jonah E Rockoff.** 2008. “Estimates of the Impact of Crime Risk on Property Values from Megan’s Laws.” *American Economic Review*, 98(3): 1103–1127.
- Louie, Schuyler, John A Mondragon, and Johannes Wieland.** 2025. “Supply Constraints do not Explain House Price and Quantity Growth Across US Cities.” National Bureau of Economic Research.
- Manji, Shazia, and Ryan Finnigan.** 2023. “Streamlining Multifamily Housing Production in California: Progress Implementing SB 35.” The Turner Center for Housing Innovation at UC Berkeley.
- Marantz, Nicholas J., Christopher S. Elmendorf, and Youjin B. Kim.** 2023. “Evaluating California’s Accessory Dwelling Unit Reforms: Preliminary Evidence and Lessons for State Governments.” In *Learning from Land Use Reforms: Housing and Regulatory Change.*, ed. Noah M. Kazis. New York City, USA:NYU Furman Center.
- McMillen, Daniel, and Arthur O’Sullivan.** 2013. “Option Value and the Price of Teardown Properties.” *Journal of Urban Economics*, 74: 71–82.
- Mills, Edwin S.** 1967. “An Aggregative Model of Resource Allocation in a Metropolitan Area.” *The American Economic Review*, 57(2): 197–210.
- Minott, Owen.** 2023. “Accessory Dwelling UNits (ADUs) in California.” Bipartisan Policy Center.
- Molloy, Raven.** 2020. “The Effect of Housing Supply Regulation on Housing Affordability: A Review.” *Regional Science and Urban Economics*, 80.
- Molloy, Raven, Charles G. Nathanson, and Andrew Paciorek.** 2022. “Housing Supply and Affordability: Evidence from Rents, Housing Consumption and Household Location.” *Journal of Urban Economics*, 129: 103427.
- Mondragon, John A., and Johannes Wieland.** 2022. “Housing Demand and Remote Work.” National Bureau of Economic Research NBER Working Paper 30041.

- Monkkonen, Paavo, Michael Manville, and Michael Lens.** 2024. “Built out cities? A new approach to measuring land use regulation.” *Journal of Housing Economics*, 63: 101982.
- Muth, Richard F.** 1969. *Cities and Housing: The Spatial Pattern of Urban Residential Land Use*. University of Chicago Press.
- Nichols, Joseph B., Stephen D. Oliner, and Michael R. Mulhall.** 2013. “Swings in Commercial and Residential Land Prices in the United States.” *Journal of Urban Economics*, 73(1): 57–76.
- NYU Furman Center.** 2026. “Land Use Reform Tracker.” <https://www.furmancenter.org/land-use-reform-tracker/>, Data accessed April 13, 2026.
- Oates, Wallace E.** 1969. “The Effects of Property Taxes and Local Public Spending on Property Values: An Empirical Study of Tax Capitalization and the Tiebout Hypothesis.” *Journal of Political Economy*, 77(6): 957–971.
- Paulin, Geoffrey.** 2018. “Housing and Expenditures: Before, During, and After the Bubble.” *Beyond the Numbers*, 7(10).
- Peterson, Kol.** 2021. “California ADU Growth by City from 2012-2019, Charted.” Building An ADU.
- Poterba, James M.** 1997. “Demographic Structure and the Political Economy of Public Education.” *Journal of Policy Analysis and Management*, 16(1): 48–66.
- Rauh, Joshua D., and Ryan Shyu.** 2024. “Behavioral Responses to State Income Taxation of High Earners: Evidence from California.” *American Economic Journal: Economic Policy*, 16(1): 34–86.
- Riddick v. City of Malibu.** 2025. California Court of Appeal, Second Appellate District, Division Five, No. B323731.
- Saiz, Albert.** 2010. “The Geographic Determinants of Housing Supply.” *The Quarterly Journal of Economics*, 125(3): 1253–1296.
- San Francisco Assessor Recorder’s Office.** 2025. “Learn About Real Property Assessments.” City and County of San Francisco.
- Schuetz, Jenny.** 2025. “State Housing Policy Changes Are More Random Than You Think.” The Mercatus Center.
- Seiler, Michael J., Ralph B. Siebert, and Liuming Yang.** 2024. “Airbnb or Not Airbnb? That Is the Question: How Airbnb Bans Disrupt Rental Markets.” *Real Estate Economics*, 52(1): 239–270.
- Severen, Christopher, and Andrew J. Plantinga.** 2018. “Land-use Regulations, Property Values, and Rents: Decomposing the Effects of the California Coastal Act.” *Journal of Urban Economics*, 107: 65–78.
- Soltas, Evan, and Jonathan Gruber.** 2026. “How Costly Is Permitting in Housing Development?” Working paper. Available at https://evansoltas.com/papers/Permitting_SoltasGruber2026.pdf.

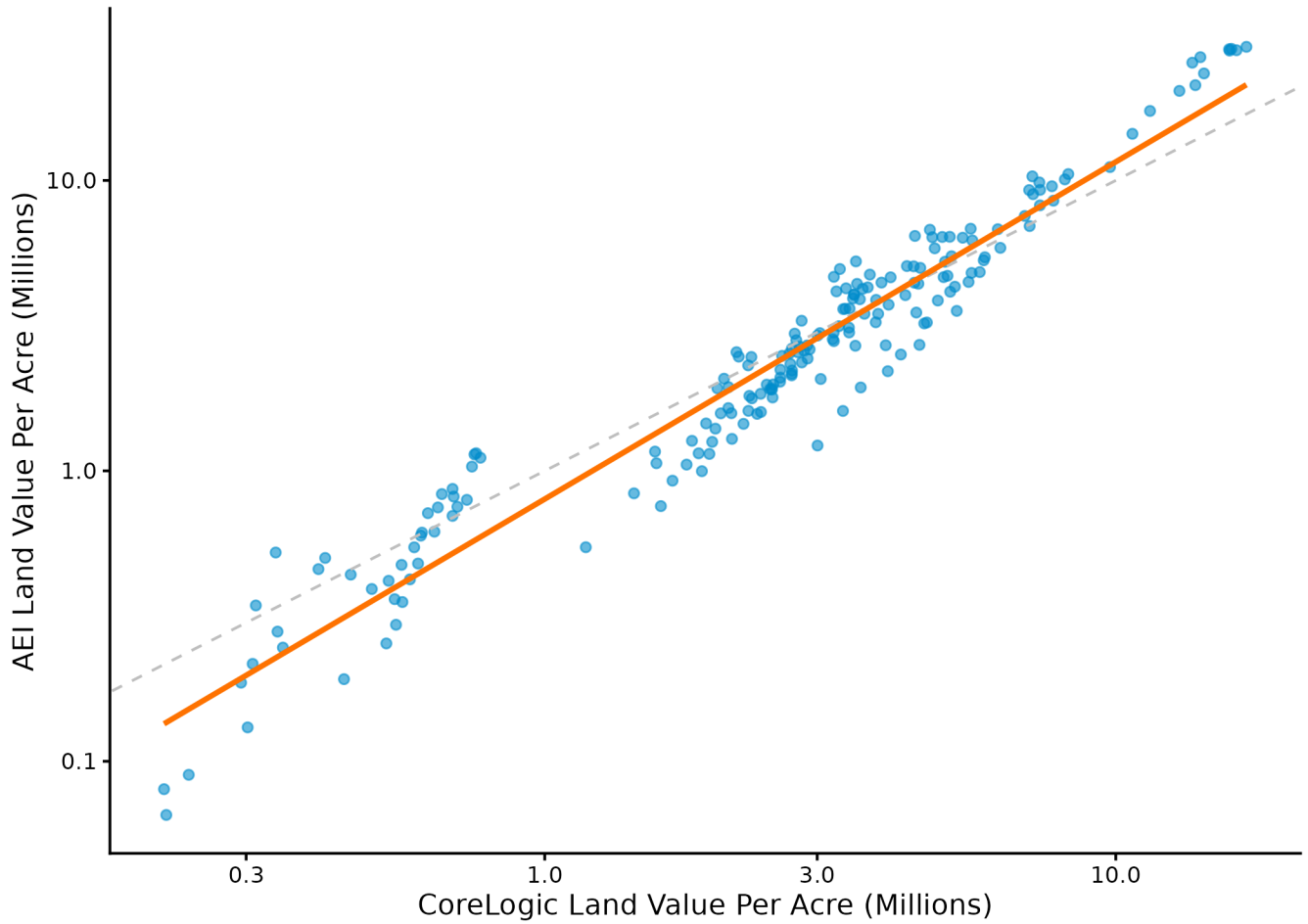
- Tanrisever, Idil.** 2025. “Spillover Effects of Accessory Dwelling Unit Development.” *Regional Science and Urban Economics*, 114: 104136.
- Titman, Sheridan.** 1985. “Urban Land Prices Under Uncertainty.” *The American Economic Review*, 75(3): 505–514.
- Turner, Matthew A., Andrew Haughwout, and Wilbert van der Klaauw.** 2014. “Land Use Regulation and Welfare.” *Econometrica*, 82(4): 1341–1403.
- U.S. Bureau of Economic Analysis.** 2025. “Personal Consumption Expenditures: Chain-type Price Index.” Retrieved from FRED, Federal Reserve Bank of St. Louis.
- U.S. Census Bureau.** 2016. “American Community Survey 2012–2016 5-Year Estimates.” American Community Survey.
- US Census Bureau.** 2023. “Longitudinal Employer Household Dynamics.” United States Department of Commerce.
- U.S. Census Bureau.** 2024. “American Community Survey 2020–2024 5-Year Estimates.” American Community Survey.
- U.S. Census Bureau.** 2026. “Census Geocoder.” <https://geocoding.geo.census.gov/geocoder/>, Accessed September 2, 2026.
- U.S. Geological Survey.** 2026. “3D Elevation Program.” Accessed August 30, 2026.
- Village of Euclid v. Ambler Realty Co. 1926. 272 U.S. 365.
- Wasi, Nada, and Michelle J. White.** 2005. “Property Tax Limitations and Mobility: The Lock-In Effect of California’s Proposition 13.” *Brookings-Wharton Papers on Urban Affairs*, 2005(1): 59–97.
- Yagan, Danny.** 2019. “Employment Hysteresis from the Great Recession.” *Journal of Political Economy*, 127(5): 2505–2558.
- Yao, Huimin, and Frederik Pretorius.** 2014. “Demand Uncertainty, Development Timing and Leasehold Land Valuation: Empirical Testing of Real Options in Residential Real Estate Development.” *Real Estate Economics*, 42(4): 829–868.

Figure 1: State-Wide Accessory Dwelling Unit Permits Issued in California from 2012 to 2024



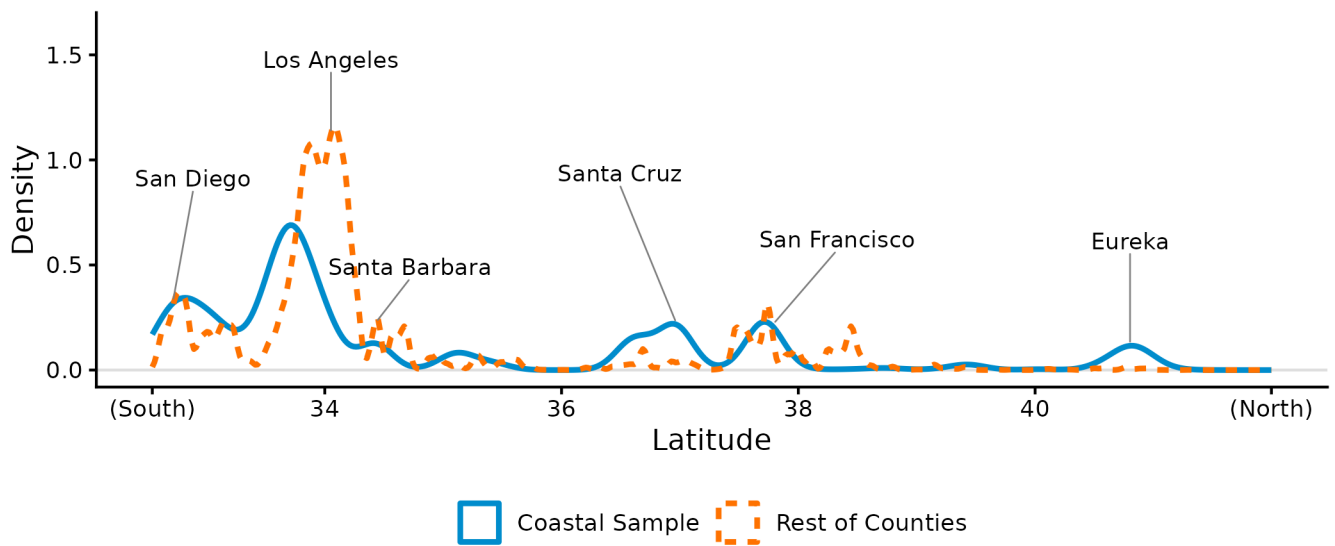
This figure shows a time series of accessory dwelling unit permits issued in the state of California from 2012 to 2024. The figure uses two data sources. The blue line shows permits counted by [Peterson \(2021\)](#), which collects data from the California Department for Housing and Community Development. The orange and dotted line shows permits according to the California Housing Element Progress Report (Table A2) data ([California Department of Housing and Community Development, 2025](#)).

Figure 2: Comparison of County-Year Average Land Values Between [Cotality \(2025\)](#) and [Davis et al. \(2021\)](#) from 2012 to 2024



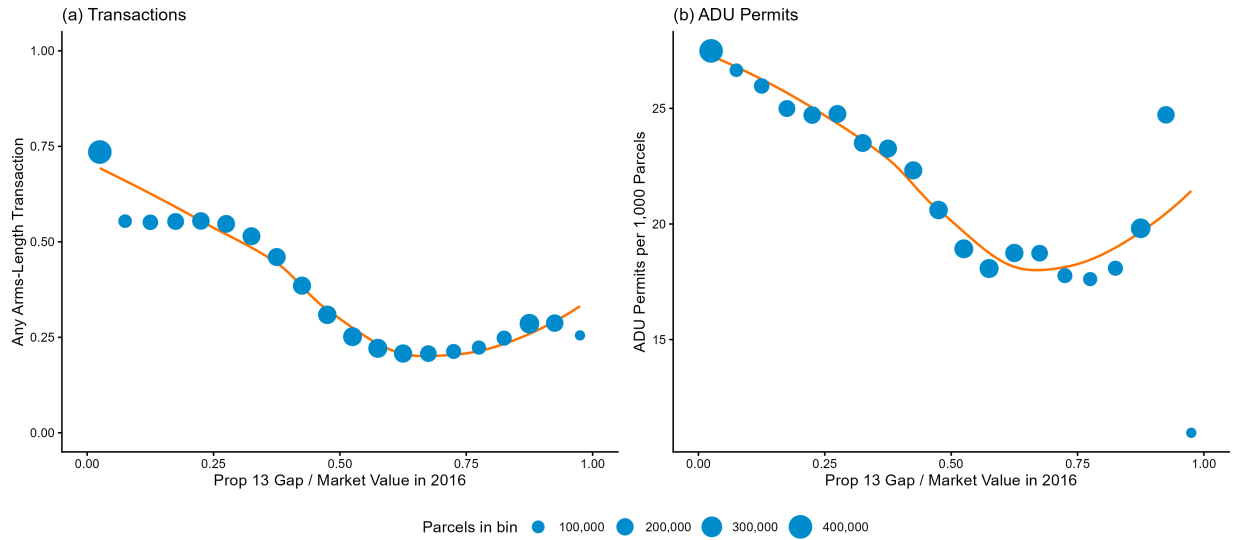
This figure uses data from [Cotality \(2025\)](#) and [Davis et al. \(2021\)](#) to compare county-year average land values. The horizontal axis shows the county average of the (real) land value per acre value used throughout this paper calculated using all residential parcels. The vertical axis uses the county-year land value data from [Davis et al. \(2021\)](#) as a comparison. The equality line is shown as a dashed line, and the best linear fit shown in solid orange. Both axes are on log scales. The regression fit has an R-squared of 0.9 and a regression coefficient of 1.1.

Figure 3: Density of [Cotality \(2025\)](#) Sample Parcels by Latitude



This figure shows the density of the [Cotality \(2025\)](#) property sample by latitude. The figure shows unique residential parcels from the year 2016. The solid blue line shows the sample within 450 meters of the Coastal Zone boundary, and the dashed orange line shows the remaining sample in 15 coastal counties outside of 450 meters from the Coastal Zone boundary. The horizontal axis shows parcel latitude, with Mexico abutting the southern portion of the sample on the left hand side, and Oregon abutting the northern portion of the sample on the right hand side. Selected cities are marked for reference.

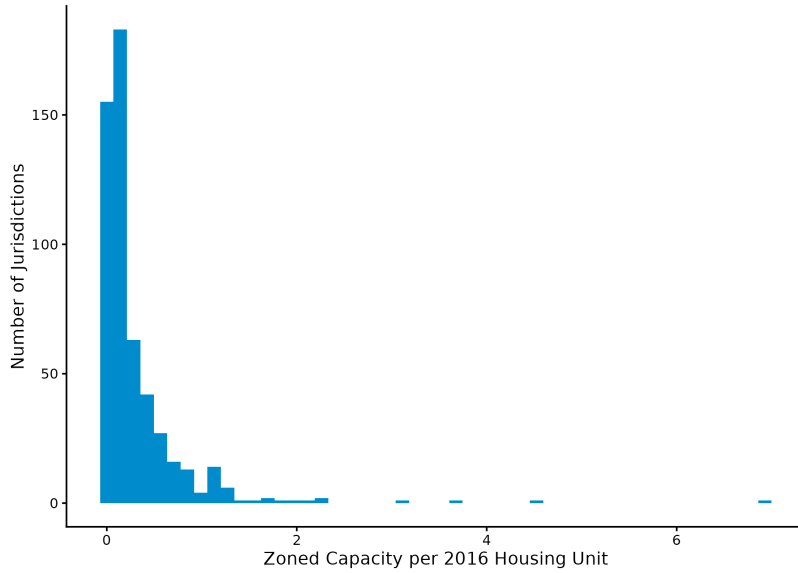
Figure 4: Binned Scatter Plots of Proposition 13 Property Tax Shelter in 2016 and the Share of Parcels Ever Transacting or Ever Building an ADU in Sample



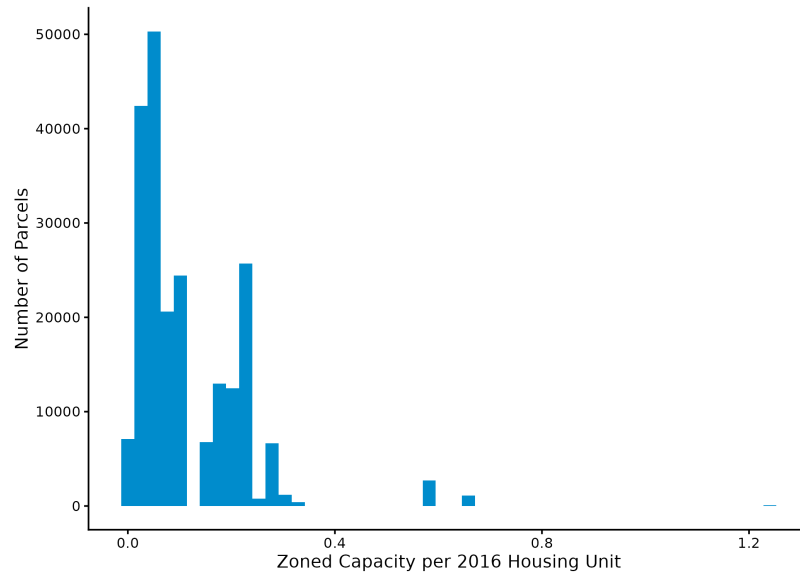
This figure uses data from [Cotality \(2025\)](#), [California Department of Housing and Community Development \(2025\)](#) to show a binned scatter plot of each parcel's property tax shelter in 2016 against the share of parcels transacting between 2010 and 2024 and the share of parcels building an ADU between 2018 and 2024. Each parcel's property tax shelter is defined as the gap between the parcel's estimated market value and the parcel's estimated assessed value divided by the parcel's estimated market value in 2016 for all parcels in the Coastal 15 counties. We build estimates of the parcel's market value using a hedonic model estimated on transaction data between 2010 and 2016. If a parcel transacts in 2016, we use that value instead of the estimate. We also build estimated assessed values by growing the parcel's last recorded assessed value by 2% per year. These values are shown for each parcel on the horizontal axis. In Panel A, the vertical axis is the share of parcels that transact between 2010 and 2024 at least one time. We include both arms length and non-arms length transactions in this variable. We show a loess smoother in orange.

Figure 5: Histogram of Pre-2016 Zoned Capacity Per Housing Unit

(a) All California Jurisdictions

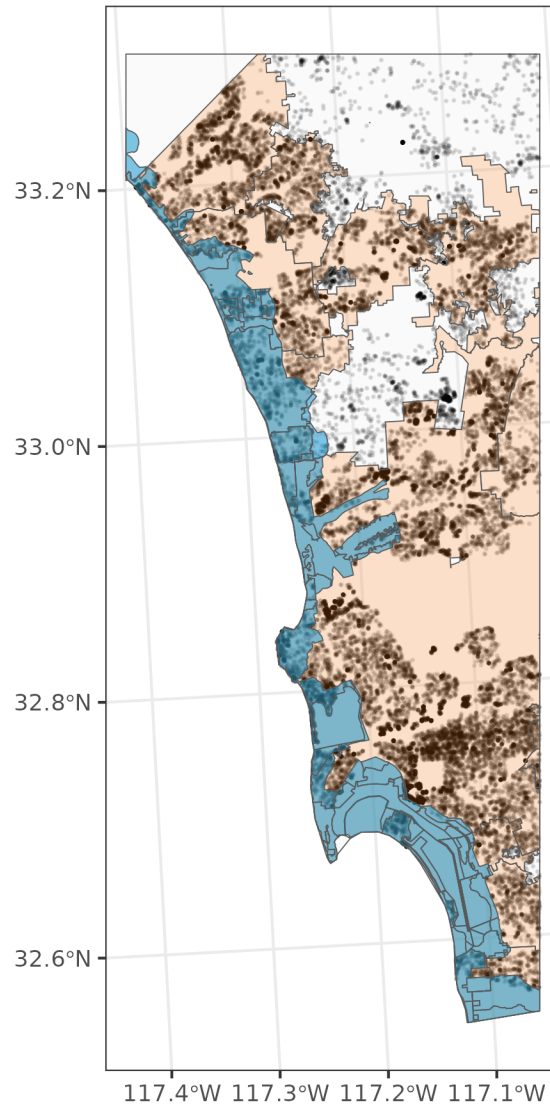


(b) Parcels Within 450m of the California Coastal Zone Boundary



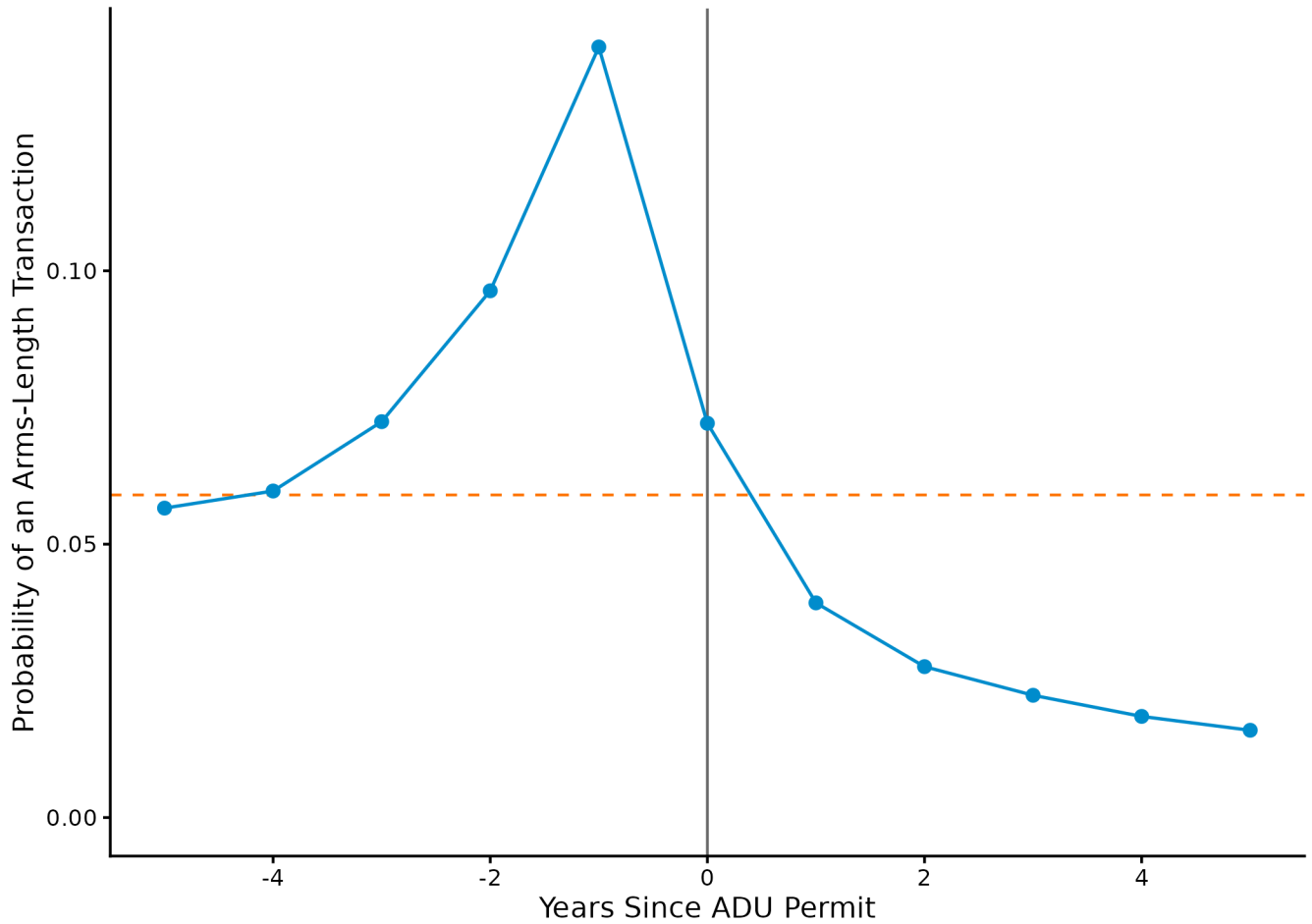
This figure uses data from [Monkkonen, Manville and Lens \(2024\)](#), [U.S. Census Bureau \(2016\)](#) and [Cotality \(2025\)](#). Panel A shows zoned capacity from the 2014 RHNA planning cycle per 2016 housing unit for all California jurisdictions (counties, cities). Panel B shows the zoned capacity per housing unit of residential parcels in the 450m bandwidth difference in discontinuity sample.

Figure 6: Map of San Diego Coastal Zone, Incorporated Cities, and Real Estate Transactions in 2016



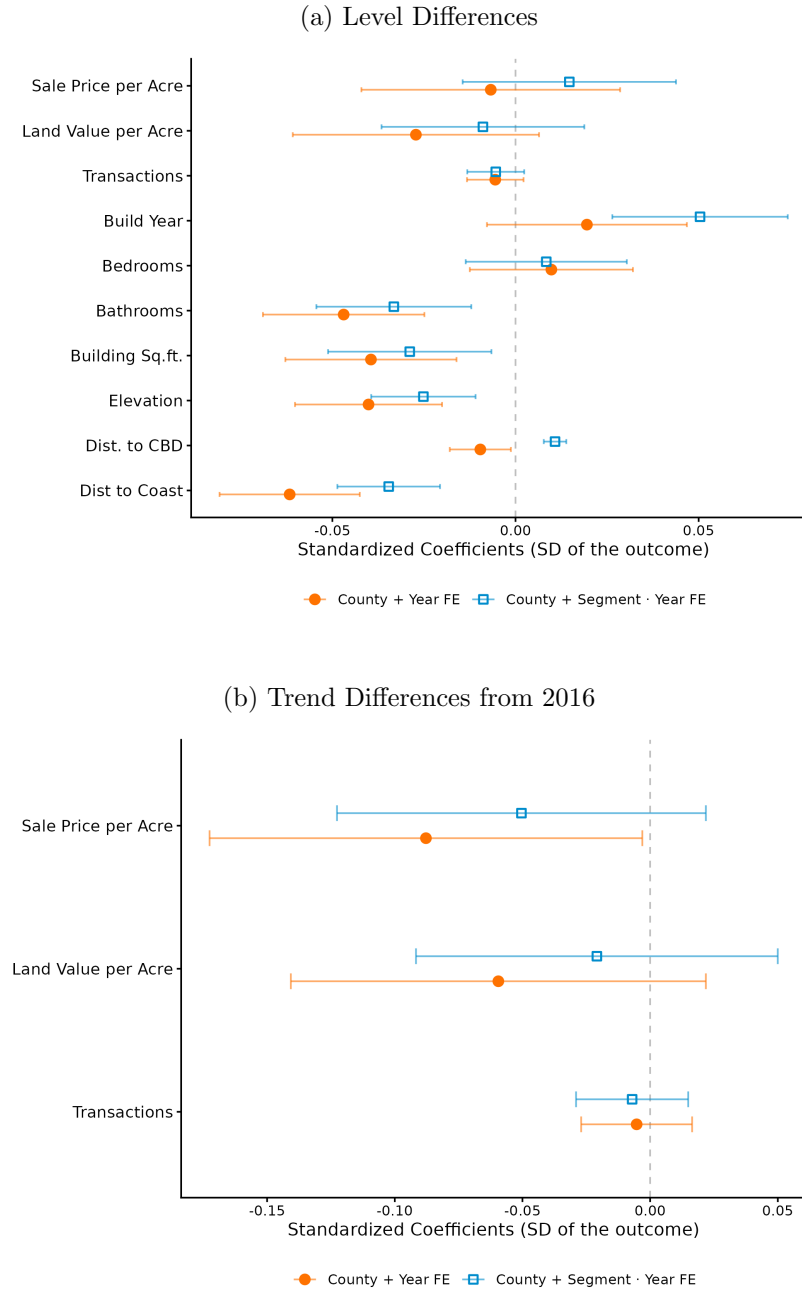
This figure shows the California Coastal Zone in San Diego County. The Coastal Zone is shown in blue. The orange area shows incorporated cities near the coast, and the black dots show all real estate transactions in 2016 from the [Cotality \(2025\)](#) data.

Figure 7: Parcel Transaction Probabilities Centered Around the Year an ADU is Permitted



This figure shows the unconditional probability that a parcel that pulls an ADU permit in year 0 transacts in each year within a 5 year window of the permit's issue date. Each parcel that pulls an ADU permit is included and event time is centered around the first ADU permit year. The probabilities of transaction are calculated from all parcels at risk of transacting to correct for short panel biases. Parcels can transact more than one time. The dashed orange horizontal line shows the probability that a parcel that builds an ADU transacts in a given year.

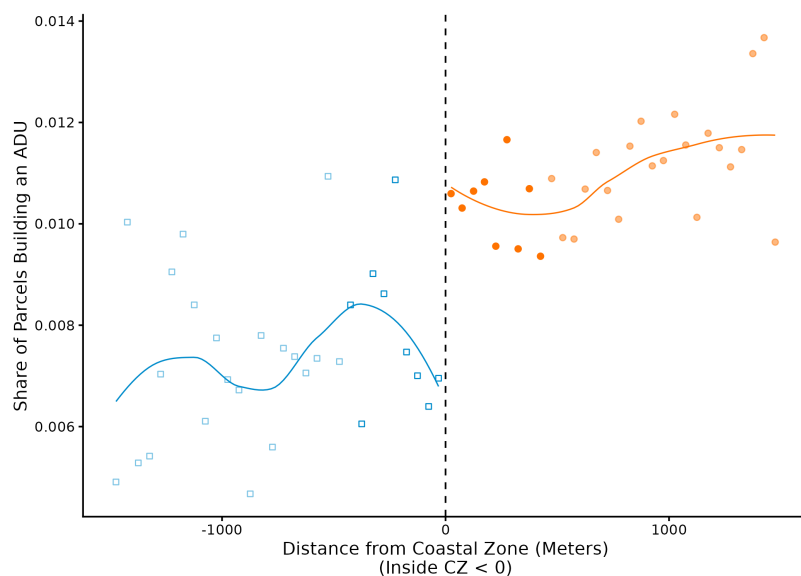
Figure 8: Baseline Pre-Reform Differences Across the Coastal Zone Boundary Using the 450 Meter Bandwidth From 2010 to 2016



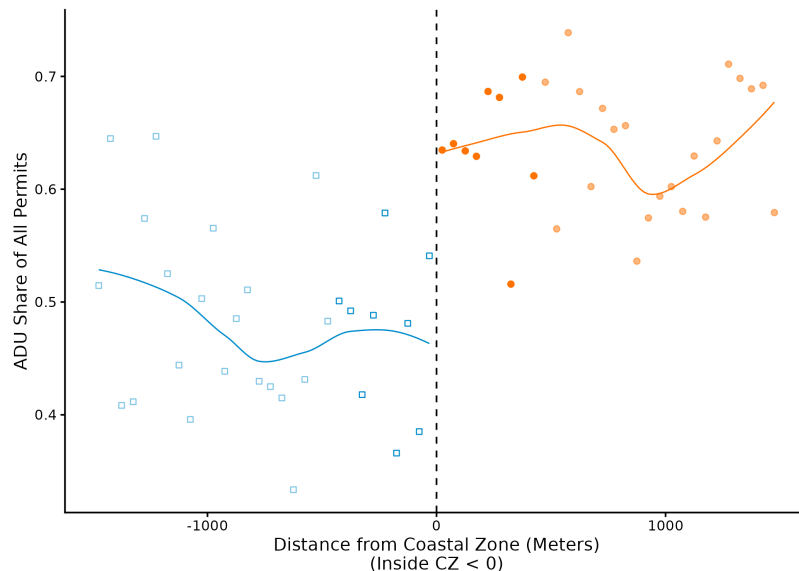
This figure uses data from [Cotality \(2025\)](#), [California Department of Housing and Community Development \(2025\)](#), [U.S. Geological Survey \(2026\)](#), [Holian \(2019\)](#), [California Department of Fish and Wildlife \(2026\)](#), and [California Coastal Commission \(2014\)](#) between 2010 and 2016. Each coefficient is scaled by the outcome's standard deviation within the 450m bandwidth between 2010 and 2016. In Panel A, we show level differences from the 1st degree polynomial regression discontinuity equation. In Panel B, we show differences in the pre-period from the baseline coefficient in 2016, similar to an event study or joint pre-trend style test in our difference in discontinuities specification. In both panels, estimates with orange circles use county and year fixed effects, and estimates in blue squares add segment by time fixed effects. Robust standard errors are clustered by parcel. The standard deviations used for scaling are: 1.11 for (log real) sale price per acre, 1.25 for (log real) land value per acre, 0.18 for the arms-length transaction dummy, 23 for build years, 1.1 for bedrooms, 1.2 for bathrooms, 914 for building square feet, 55 meters for elevation, 1.1 for (log) distance (meters) to the CBD, and 0.9 for (log) distance (meters) to the nearest coastline.

Figure 9: Accessory Dwelling Unit Permit Activity Around the Coastal Zone Boundary after 2018

(a) Accessory Dwelling Unit Permits Per Parcel by Distance Bin to the Coastal Zone

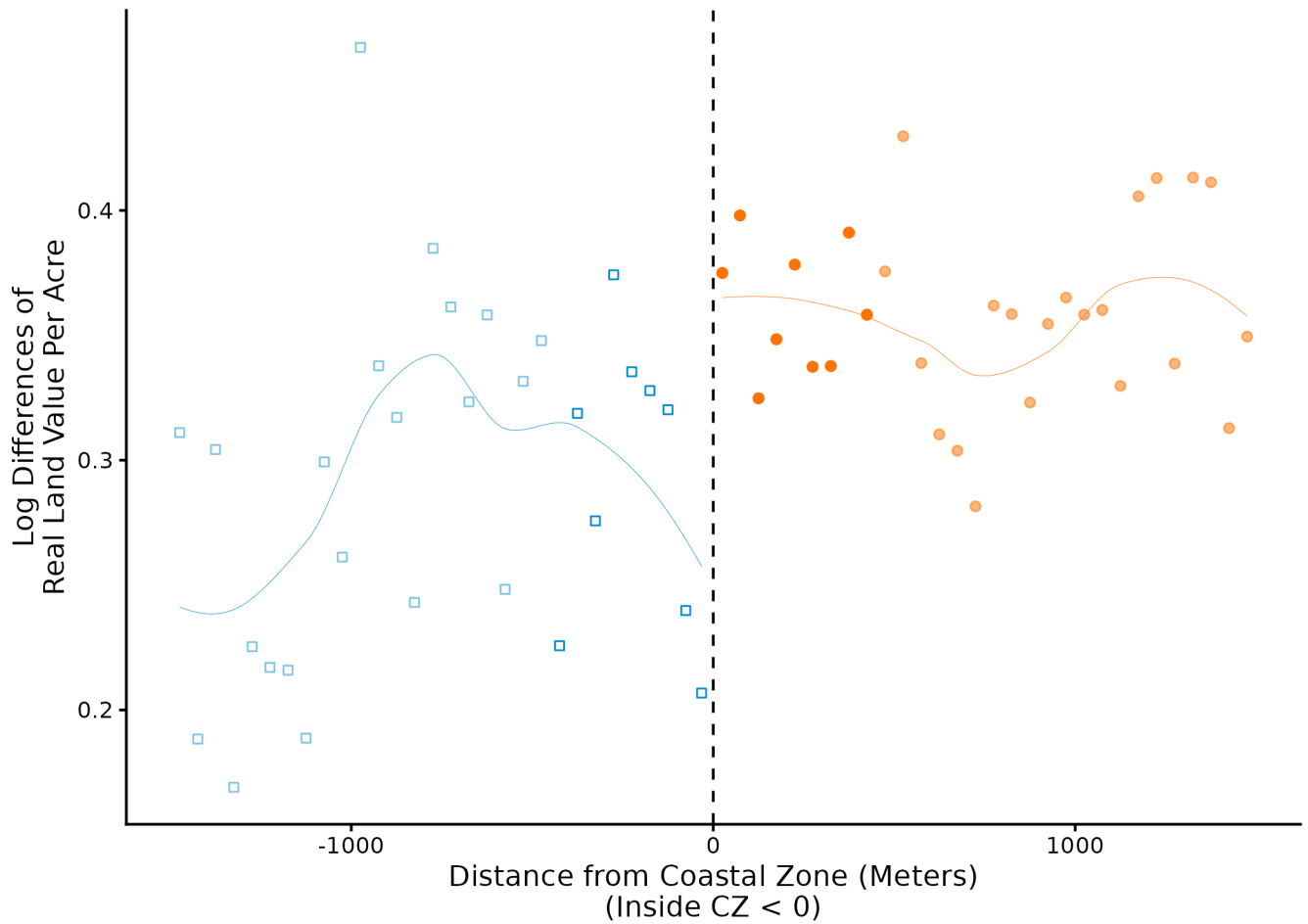


(b) Accessory Dwelling Unit Permits as a Share of All Building Permits by Distance Bin to the Coastal Zone



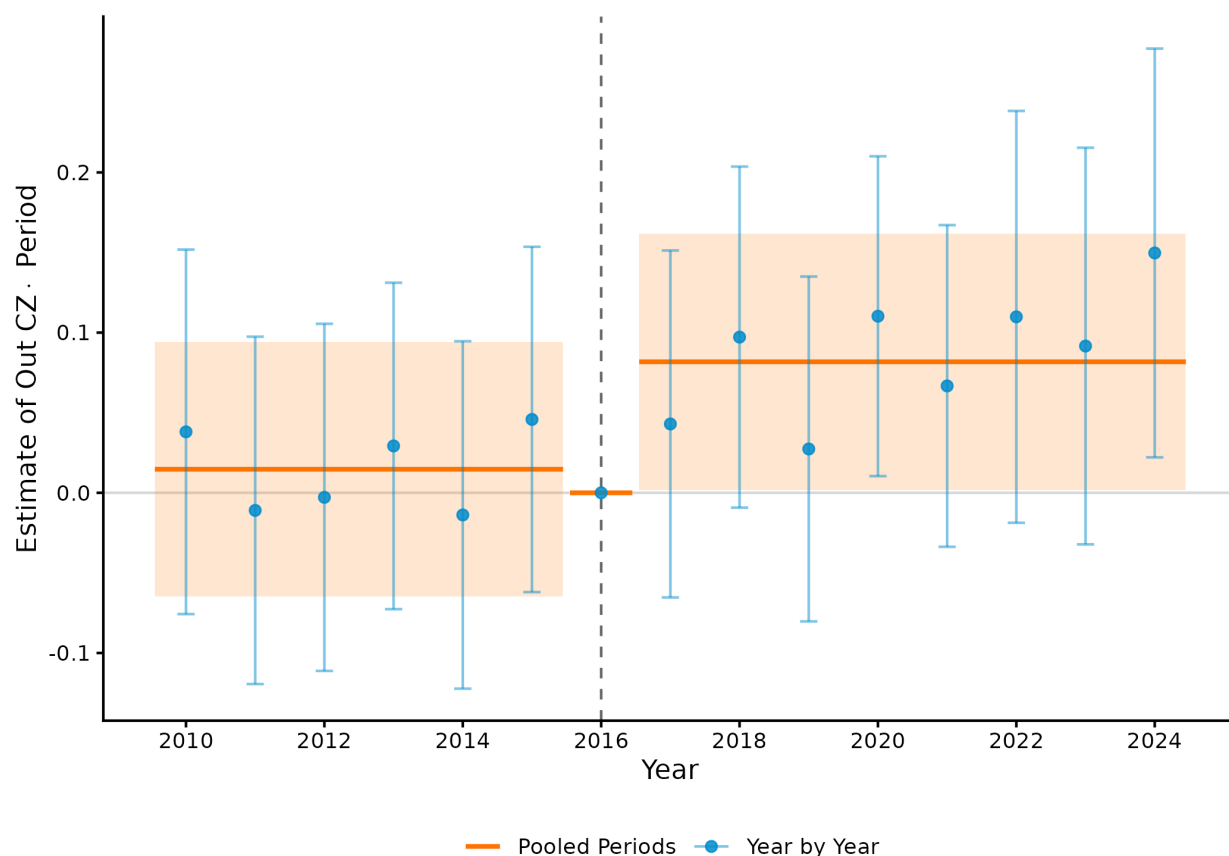
This figure uses data from [Cotality \(2025\)](#), [California Department of Housing and Community Development \(2025\)](#), and [California Coastal Commission \(2014\)](#) to show how permitting activity for ADUs changes across the Coastal Zone boundary between 2018 and 2024. The horizontal axis in both panels shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the permitting reforms (blue squares). We group parcels in 50 meter bins. In Panel A, we show the number of ADU permits on a per-parcel basis for each bin. In Panel B, we scale the permit counts by the total number of building permits issued in each bin.

Figure 10: Changes in Real Land Values per Acre Around the Coastal Zone Boundary Pre- and Post-2016 Reforms



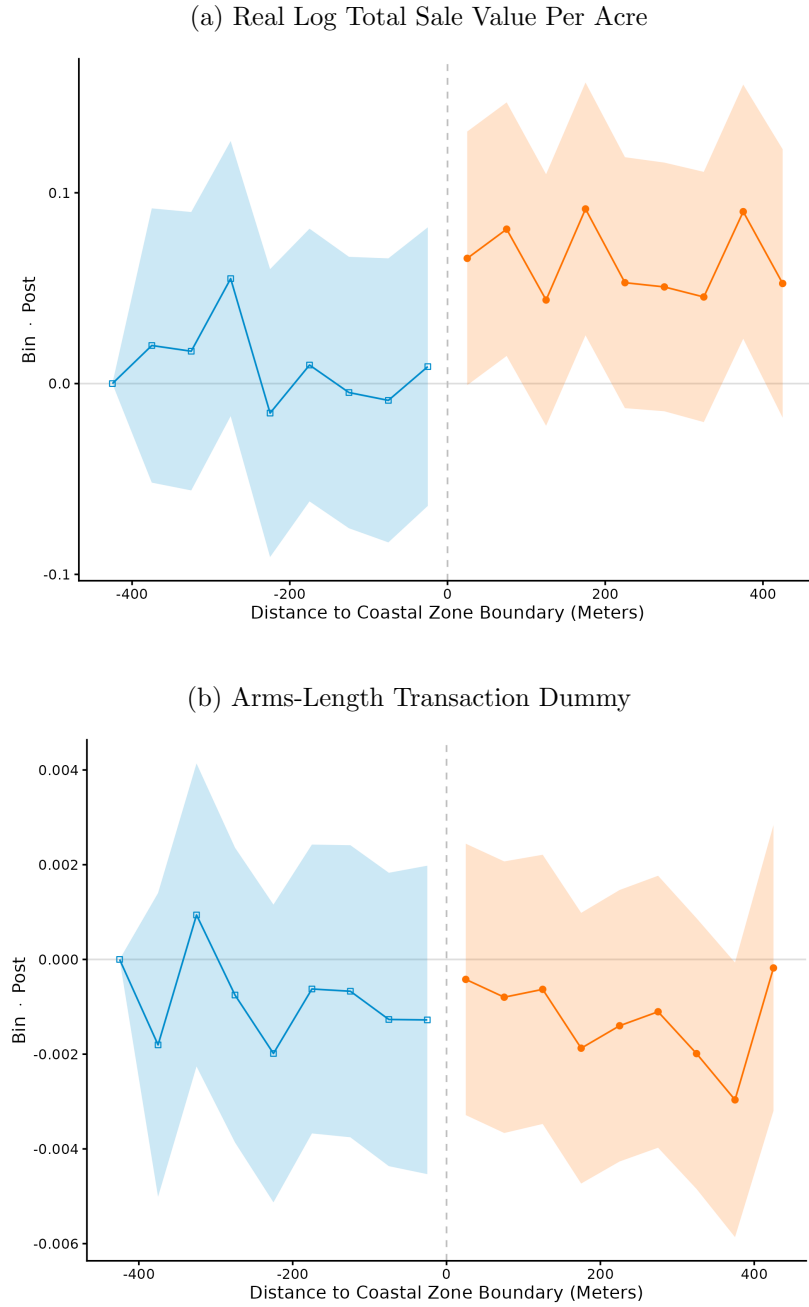
This figure uses data from [Cotality \(2025\)](#) and [California Coastal Commission \(2014\)](#) to show how residential land values change across the Coastal Zone boundary between 2010 and 2024. The horizontal axis shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the post-2016 permitting reforms. We group parcels in 50 meter bins. The vertical axis shows the log difference of the real land value per acre for each bin between the pre-reform (2010-2016) and post-reform (2017-2024) periods. The orange squares show the values inside the coastal zone. The blue circles show values outside the coastal zone boundary that experience ADU reforms. The difference in discontinuities estimate can be seen in difference in changes just before and then just after crossing the Coastal Zone boundary at 0.

Figure 11: Event Study Estimates of the Impacts of Permitting Reforms on Log Real Land Value Per Acre



This figure presents event study estimates of the impact of California’s post-2016 ADU permitting reforms on residential land values, using data from [Cotality \(2025\)](#) and [California Coastal Commission \(2014\)](#) following Equation 3. Each point shows the estimated coefficient on the interaction of the outside-Coastal-Zone indicator (*OutCZ*) with a year dummy, relative to the omitted base year of 2016. Estimates are shown for an event-study version of the repeat sale model with segment by time fixed effects corresponding to Column 7 of Table 4. We also show pooled coefficients with 2016 as the omitted year for 2010 to 2015 and 2017 to 2024. The sample is restricted to residential parcels within 450 meters of the Coastal Zone boundary. Confidence intervals are at the 95% level, with standard errors clustered at the parcel level. We use triangle weights.

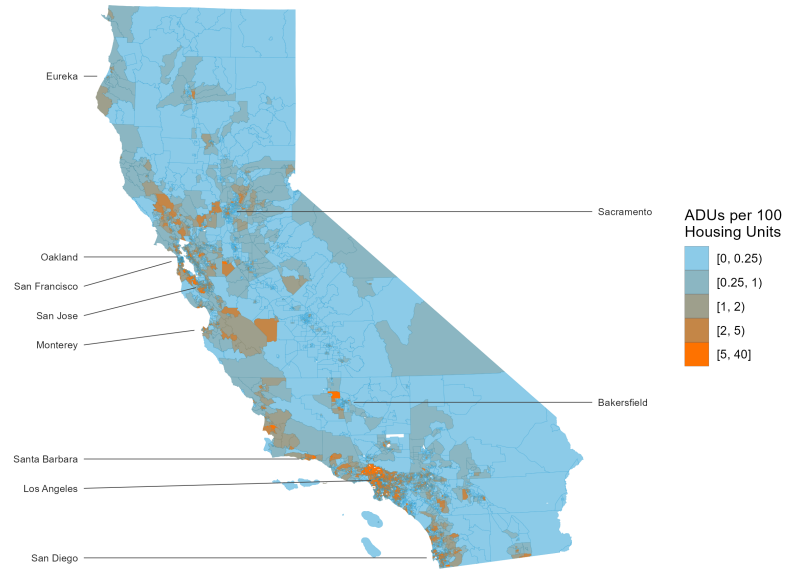
Figure 12: Spillovers Estimates By Coastal Zone Distance Bin for Changes in Real Total Sale Value Per Acre and Transactions



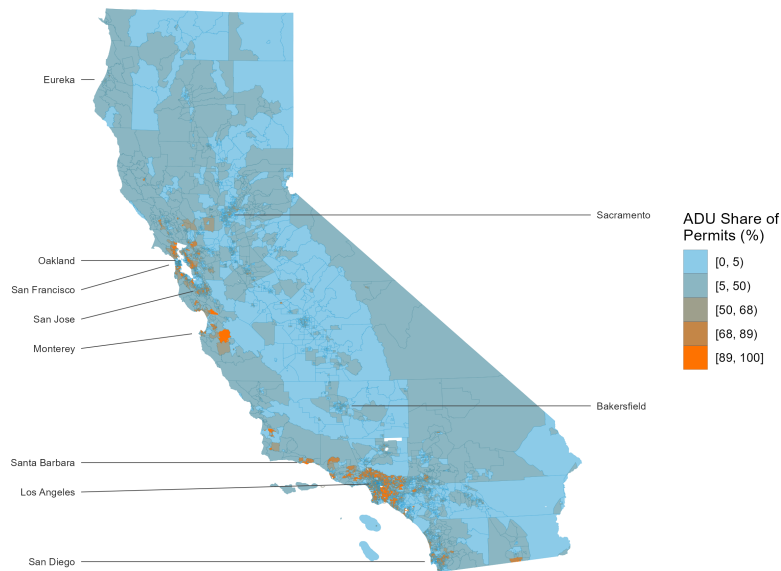
This figure uses data from [Cotality \(2025\)](#) and [California Coastal Commission \(2014\)](#) to show how changes in the total sale price per acre change across the Coastal Zone boundary between 2010 and 2024 in Panel A. The horizontal axis shows each bin's distance from the Coastal Zone boundary, where negative values, shown in blue, represent parcels inside the Coastal Zone that do not receive the post 2016 permitting reforms. We group parcels in 50 meter bins. The vertical axis shows the coefficients from Equation 4 with segment by time fixed effects included, and the log of the real total sale price per acre as the dependent variable. These coefficients are on the set of distance bin times post interactions, where the bin $[-450, -400)$ meters has been omitted to serve as the reference bin. In Panel B, we run the same regression estimates and swap the outcome with a dummy that takes a value of 1 when a parcel transacts at arms-length in a given year, and 0 otherwise. The regressions in Panel B also use a parcel fixed effect.

Figure 13: Maps of Accessory Dwelling Unit Permits Issued in California Between 2018 to 2024

(a) ADU Permits Per Housing Units in 2016



(b) ADU Permits as a Share of All Permits Between 2018 and 2024



This figure shows two maps of ADU permits from [California Department of Housing and Community Development \(2025\)](#) issued in California between 2018 and 2024. Permits are matched to census tracts using assessor parcel numbers in the [Cotality \(2025\)](#) and to block groups as a fallback using [U.S. Census Bureau \(2026\)](#).

Table 1: Sample Statistics for the Transaction Data by Coastal Zone Sample Status and Reform Period

	450 Meter Bandwidth						
	In CZ			Out CZ		Full County	
	Pre-2016 (1)	Post-2016 (2)	Pre-2016 (3)	Post-2016 (4)	Pre-2016 (5)	Post-2016 (6)	
Share SFR	0.877 (0.329)	0.893 (0.309)	0.89 (0.313)	0.899 (0.301)	0.919 (0.274)	0.928 (0.259)	
Dist. to the CZ	-220.525 (124.898)	-222.008 (125.26)	223.23 (129.231)	222.57 (129.395)	15,584 (16,014)	15,495 (16,082)	
Effective Build Year	1967 (24.322)	1972 (24.847)	1966 (23.974)	1969 (23.741)	1966 (23.181)	1969 (23.145)	
Bedrooms	3.259 (1.162)	3.303 (1.146)	3.243 (1.07)	3.277 (1.09)	3.307 (1.039)	3.349 (1.051)	
Bathrooms	2.522 (1.232)	2.72 (1.289)	2.387 (1.164)	2.514 (1.18)	2.333 (1.057)	2.453 (1.101)	
Building Sq. Ft.	2,004 (1,002)	2,217 (1,157)	1,883 (896.452)	2,041 (1,000)	1,852 (900.65)	2,001 (998.652)	
Land Value Per Acre	6,716,901 (6,805,269)	8,792,829 (7,814,689)	5,229,268 (5,101,423)	7,560,034 (6,895,375)	3,130,478 (3,746,220)	4,520,994 (4,909,490)	
Total Sale Price Per Acre	10,758,600 (9,135,531)	13,853,775 (10,455,810)	8,709,221 (7,380,480)	12,195,017 (9,519,186)	5,626,841 (5,769,792)	7,955,063 (7,265,957)	
Total Sale Price (0.2 Acre Lot)	2,151,720 (1,827,106)	2,770,755 (2,091,162)	1,741,844 (1,476,096)	2,439,003 (1,903,837)	1,125,368 (1,153,958)	1,591,012 (1,453,191)	
ADU Suitable	0.84 (0.366)	0.851 (0.356)	0.886 (0.318)	0.879 (0.327)	0.873 (0.333)	0.861 (0.346)	
Excess Lot Sq. Ft.	9,951 (63,740)	10,551 (46,012)	9,066 (93,422)	8,848 (55,098)	14,985 (678,734)	17,197 (340,596)	
Observations	17,567	18,298	33,068	34,842	1,070,702	1,090,087	

Notes: This table shows summary statistics for the transaction dataset. Standard deviations are shown in parentheses. Columns (1)-(2) show parcels within the Coastal Zone, columns (3)-(4) show parcels outside the Coastal Zone but within 450 meters of the boundary, and columns (5)-(6) show all parcels in the 15 coastal counties of California. We present means and standard deviations for the share of parcels on land zoned only for single family residences, the parcel's distance to the Coastal Zone boundary in meters, the effective build year of the structure(s) on the parcel, the number of bedrooms, number of bathrooms, building square footage, real land value per acre, the real total sale price per acre, the real total sale price scaled to a 0.2 acre lot, the share of parcels that could put down an ADU (more than 2,000 sq.ft. of excess space and a ground slope under 20 degrees), and the excess sq. ft. on the lot not covered by a building. Dollar values are in 2025 USD.

Table 2: Sample Statistics for the Balanced Parcel Panel Data by Coastal Zone Sample Status and Reform Period

	450 Meter Bandwidth						
	In CZ			Out CZ		Full County	
	Pre-2016 (1)	Post-2016 (2)	Pre-2016 (3)	Post-2016 (4)	Pre-2016 (5)	Post-2016 (6)	
Share SFR	0.852 (0.355)	0.86 (0.347)	0.887 (0.316)	0.896 (0.306)	0.908 (0.289)	0.914 (0.28)	
Dist. to the CZ	-220.385 (124.654)	-220.402 (124.643)	221.547 (128.827)	221.556 (128.82)	14,486 (14,750)	14,486 (14,749)	
Effective Build Year	1967 (23.473)	1972 (24.492)	1965 (22.773)	1969 (23.224)	1965 (21.736)	1968 (22.327)	
Bedrooms	3.295 (1.195)	3.352 (1.198)	3.257 (1.09)	3.311 (1.106)	3.338 (1.053)	3.388 (1.072)	
Bathrooms	2.535 (1.237)	2.708 (1.298)	2.33 (1.124)	2.456 (1.155)	2.317 (1.037)	2.413 (1.072)	
Building Sq. Ft.	1,994 (992.873)	2,168 (1,122)	1,865 (868.049)	2,008 (961.741)	1,856 (879.31)	1,976 (955.993)	
Property Tax Per Acre	64,999 (64,711)	79,993 (76,096)	54,104 (53,965)	67,415 (66,371)	39,121 (41,105)	48,469 (49,774)	
Land Share	0.547 (0.205)	0.569 (0.211)	0.525 (0.195)	0.55 (0.201)	0.503 (0.19)	0.527 (0.194)	
ADU Permit Per Year (2018+)		0.002 (0.041)		0.002 (0.046)		0.003 (0.053)	
Any Permit (2018+)		0.003 (0.054)		0.003 (0.056)		0.004 (0.064)	
ADU Suitable	0.572 (0.495)	0.567 (0.496)	0.631 (0.483)	0.611 (0.488)	0.678 (0.467)	0.677 (0.468)	
Excess Lot Sq. Ft.	19,234 (1,324,608)	33,053 (2,976,177)	27,385 (1,107,478)	25,806 (525,869)	25,365 (2,046,433)	28,947 (5,977,357)	
Observations	521,282	595,473	985,277	1,125,737	29,087,092	33,237,497	

Notes: This table shows summary statistics for the balanced parcel panel. Standard deviations are shown in parentheses. Columns (1)-(2) show parcels within the Coastal Zone, columns (3)-(4) show parcels outside the Coastal Zone but within 450 meters of the boundary, and columns (5)-(6) show all parcels in the 15 coastal counties of California. We present means and standard deviations for the share of parcels on land zoned only for single family residences, the parcel's distance to the Coastal Zone boundary in meters, the effective build year of the structure(s) on the parcel, the number of bedrooms, number of bathrooms, building square footage, property taxes paid per acre, the land share of assessed value, whether or not an ADU permit is associated with the parcel each year, whether or not any permit is associated with the parcel each year, the share of parcels that could put down an ADU (more than 2,000 sq.ft. of excess space and a ground slope under 20 degrees), and the excess sq. ft. on the lot not covered by a building. Dollar values are in 2025 USD.

Table 3: Spatial Regression Discontinuity Estimates of the Impacts of Permitting Reforms on Accessory Dwelling Unit Permits after 2018 within 450 Meters of the Coastal Zone Boundary

	<i>Dep. Var.: ADUs per 1,000 Parcel-Years</i>			
	(1)	(2)	(3)	(4)
<i>OutCZ</i>	0.623*** (0.170)	0.629*** (0.174)	0.530** (0.171)	0.618*** (0.173)
1 st Degree Polynomial	Y	Y	Y	Y
County FEs	Y	Y	Y	Y
Year FEs	Y	Y		
Segment · Time FEs			Y	Y
Pre-2016 Static Hedonics		Y		Y
N	1 505 988	1 485 253	1 505 974	1 485 239
Adj. R^2	0.001	0.002	0.002	0.003
Mean	2.248	2.248	2.248	2.248
SD	47.36	47.36	47.36	47.36
In-CZ Mean	1.900	1.900	1.900	1.900
In-CZ SD	43.55	43.55	43.55	43.55

Notes: This table uses data from [Cotality \(2025\)](#), [California Department of Housing and Community Development \(2025\)](#), and [California Coastal Commission \(2014\)](#) to estimate Equation 2 for a balanced set of residential parcels between 2018 and 2024: $Y_{ict} = \alpha + \beta_1 OutCZ_i + \beta_2 DistCZ_i + \beta_3 (DistCZ_i \cdot OutCZ_i) + \theta'_{(i)} \gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable is a dummy taking a value of 1 if parcel i has an ADU permit associated with it in year t , and we scale this outcome by dividing by 1,000 for display purposes. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's permitting reforms for ADUs. $DistCZ_i$ is a function of a parcel's distance to the Coastal Zone boundary, which uses a first degree polynomial here. We also include a set of county fixed effects, λ_c , and a set of year fixed effects, τ_t . We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We present summary statistics for the permit dummy for both the entire sample and just the sample within the coastal zone at the bottom of the table. In Column 1, we present a specification only with a first degree polynomial, a set of county fixed effects, and a set of year fixed effects. Then in Columns 2 and 4, we include the set of hedonic fixed effects. Then in columns 3 and 4, we replace the year fixed effects with a segment by time fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. We cluster our robust standard errors at the parcel level. We report estimates using a 450 meter bandwidth from the Coastal Zone boundary and use triangle weights. * $p < .05$, ** $p < .01$, *** $p < .001$

Table 4: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Land Values within 450 Meters of the Coastal Zone Boundary

	<i>Dep.Var.: Log of Real Land Value Per Acre</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>OutCZ · Post</i>	0.141*** (0.027)	0.125*** (0.026)	0.085*** (0.022)	0.070** (0.022)	0.088*** (0.022)	0.069** (0.021)
1 st Degree Polynomial	Y	Y	Y	Y	Y	Y
County FEs	Y	Y	Y	Y		
Parcel FEs					Y	Y
Year FEs	Y	Y			Y	
Segment · Time FEs			Y	Y		Y
Pre-2016 Static Hedonics		Y		Y		
N	103 766	102 901	103 718	102 850	49 440	49 287
Adj. R^2 .	0.524	0.546	0.662	0.678	0.903	0.908
Mean Pre-2016	15.01	15.01	15.01	15.01	15.02	15.02
SD Pre-2016	1.246	1.246	1.246	1.246	1.245	1.245
In-CZ Mean Pre-2016	15.14	15.14	15.14	15.14	15.14	15.14
In-CZ SD Pre-2016	1.317	1.317	1.317	1.317	1.316	1.316

Notes: This table uses data from [Cotality \(2025\)](#) and [California Coastal Commission \(2014\)](#) to estimate Equation 3 for a transactions on residential parcels between 2010 and 2024: $\ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3f(DistCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable the log of the real land value per acre in 2025 USD. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 demarcating the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first degree polynomial form. Distance slopes can change after the boundary. We also include a set of county fixed effects, λ_c , and a set of year fixed effects, τ_t . We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We cluster our robust standard errors at the parcel level. We report estimates using a 450 meter bandwidth from the Coastal Zone boundary. We present summary statistics for the log of the real land value per acre for both the entire sample from 2010 to 2016 and just the sample within the coastal zone from 2010 to 2016 at the bottom of the table. In Column 1, we present a specification only with a set of county fixed effects and a set of year fixed effects. In Columns 2 and 4 we add hedonic fixed effects. Then in Column 3, 4, and 6, we include the segment by time fixed effects. Finally, in columns 5 and 6, we include a parcel fixed effect and use a sample restricted to parcels that transact both between 2010 to 2016 and 2017 to 2024. All specifications use triangle weights and report parcel clustered standard errors. * $p < .05$, ** $p < .01$, *** $p < .001$

Table 5: Heterogeneity Estimates by Lot Quality of the Impacts of Permitting Reforms on Accessory Dwelling Unit Permits after 2018 and Land Values Per Acre within 450 Meters of the Coastal Zone Boundary

<i>Panel A Dep.Var.: Heterogeneity Var.:</i>	<i>ADUs per 1,000 Parcel-Years</i>			
	<i>Excess Sq.Ft.</i>		<i>Dist. to CBD (m)</i>	
	(1)	(2)	(3)	(4)
<i>OutCZ</i>	0.905** (0.295)	0.618* (0.292)	1.146*** (0.261)	0.907*** (0.262)
<i>OutCZ · H_i</i>	−0.027 (0.526)	0.367 (0.521)	−0.989** (0.341)	−0.712* (0.344)
1 st Degree Polynomial	Y	Y	Y	Y
RD Terms · <i>H_i</i>	Y	Y	Y	Y
County FEs	Y	Y	Y	Y
Year FEs	Y		Y	
Segment · Time FEs		Y		Y
N	971 458	971 447	1 505 988	1 505 974
Adj. <i>R</i> ²	0.001	0.013	0.001	0.002
Mean Dep. Var.:	3.200	3.200	2.248	2.248
Median <i>H_i</i> :	4,226	4,226	25,828	25,828
<i>Panel B Dep.Var.: Heterogeneity Var.:</i>	<i>Log of Real Land Values Per Acre</i>			
	<i>Excess Sq.Ft.</i>		<i>Dist. to CBD (m)</i>	
	(1)	(2)	(3)	(4)
<i>OutCZ · Post</i>	0.080** (0.028)	0.058* (0.028)	0.170*** (0.037)	0.064* (0.029)
<i>OutCZ · Post · H_i</i>	0.028 (0.050)	0.033 (0.042)	−0.107* (0.051)	0.010 (0.042)
1 st Degree Polynomial	Y	Y	Y	Y
RD Terms · <i>H_i</i>	Y	Y	Y	Y
County FEs	Y	Y	Y	Y
Year FEs	Y		Y	
Segment · Time FEs		Y		Y
Parcel FEs		Y		Y
N	91 266	44 299	103 766	49 287
Adj. <i>R</i> ²	0.425	0.874	0.543	0.908
Mean Dep. Var. Pre-2016:	15.22	15.22	15.01	15.02
Median <i>H_i</i> :	4,212	4,212	26,826	26,826

Notes: This table uses data from [Cotality \(2025\)](#), [California Coastal Commission \(2014\)](#), [California Department of Housing and Community Development \(2025\)](#), and [Holian \(2019\)](#) to show heterogeneity estimates following Equation 5 and reporting coefficients to characterize the CATE in Equation 6. Our heterogeneity variable in each regression is represented by H_i . Panel A shows regression discontinuity estimates of the CATE for ADU permits per 1000 parcel-years. Columns 1 and 2 use a dummy for whether a lot is above the sample median for excess square footage on the parcel. A value of 0 reflects a parcel with less than 4226 available square feet to build an ADU. Columns 3 and 4 report heterogeneity using a dummy that takes a value of 1 when a lot is farther than 25.828 kilometers from the nearest CBD, where a value of 0 reflects a lot within 25.828 kilometers from the nearest CBD. Panel B reports a parallel set of difference in discontinuity estimates for the log of real land value per acre. Columns 2 and 4 add in segment by time fixed effects in Panel A, and additionally add in a parcel fixed effect in Panel B. We report estimates using a 450 meter bandwidth from the Coastal Zone boundary and use triangle weights. * $p < .05$, ** $p < .01$, *** $p < .001$

Table 6: Heterogeneity Estimates by Property Tax Shelter and Block Group Age of the Impacts of Permitting Reforms within 450 Meters of the Coastal Zone Boundary

<i>Panel A Dep.Var.: Heterogeneity Var.:</i>	<i>Any Arms-Length Transaction</i>					
	<i>Prop. Tax. Shelter</i>			<i>Share Over 55</i>		
	<i>2010-2019</i>			<i>2010-2024</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>OutCZ · Post</i>	−0.002 (0.002)	−0.002 (0.002)	−0.015** (0.005)	−0.013* (0.005)	−0.007 (0.004)	−0.006 (0.004)
<i>OutCZ · Post · H_i</i>	0.001 (0.003)	0.000 (0.003)	0.029*** (0.008)	0.025** (0.008)	0.013* (0.006)	0.012 (0.006)
1 st Degree Polynomial	Y	Y	Y	Y	Y	Y
RD Terms · Covariate	Y	Y	Y	Y	Y	Y
Covariate Terms · Post	Y	Y	Y	Y	Y	Y
Year FEs	Y		Y		Y	
Segment · Time FEs		Y		Y		Y
Parcel FEs	Y	Y	Y	Y	Y	Y
N	3 122 313	3 122 298	2 148 934	2 148 914	3 222 639	3 222 609
Adj. R ²	0.020	0.021	0.012	0.012	0.013	0.014
Mean Dep. Var. Pre-2016:	0.036	0.036	0.036	0.036	0.036	0.036
Mean H _i :	0.508	0.508	0.614	0.614	0.614	0.614
<i>Panel B Dep.Var.: Heterogeneity Var.:</i>	<i>ADUs per 1,000 Parcel-Years</i>					
	<i>Prop. Tax. Shelter</i>			<i>Share Over 55</i>		
	<i>2018-2019</i>			<i>2018-2024</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>OutCZ</i>	0.934* (0.372)	0.890* (0.373)	1.563* (0.708)	1.734* (0.717)	1.201 (0.641)	1.459* (0.652)
<i>OutCZ · H_i</i>	−0.585 (0.625)	−0.655 (0.625)	−1.295 (1.134)	−1.750 (1.155)	−0.904 (0.990)	−1.498 (1.012)
1 st Degree Polynomial	Y	Y	Y	Y	Y	Y
RD Terms · Covariate	Y	Y	Y	Y	Y	Y
Covariate Terms (Levels)	Y	Y	Y	Y	Y	Y
County FEs	Y	Y	Y	Y	Y	Y
Year FEs	Y		Y		Y	
Segment · Time FEs		Y		Y		Y
N	1 456 892	1 456 885	429 889	429 885	1 503 594	1 503 580
Adj. R ²	0.001	0.002	0.001	0.003	0.001	0.002
Mean Dep. Var.:	2.291	2.291	1.242	1.242	2.248	2.248
Mean H _i :	0.508	0.508	0.614	0.614	0.614	0.614

Notes: This table uses data from [Cotality \(2025\)](#), [California Coastal Commission \(2014\)](#), [California Department of Housing and Community Development \(2025\)](#), and [U.S. Census Bureau \(2016\)](#) to show heterogeneity estimates following Equation 5 and reporting coefficients to characterize the CATE in Equation 6. Our heterogeneity variable in each regression is represented by H_i . Panel A shows difference in discontinuity estimates of the CATE for our any transaction dummy variable. Columns 1 and 2 use the property tax shelter in 2016 as described in the text, where a value of 0 reflects taxation on the market rate. Columns 3 through 6 report heterogeneity using the block group share of owners over 55 for a pre-COVID sample and our full sample. A block group owner share of 0 reflects a parcel within a block group where all owners of homes are younger than 55. Panel B reports a parallel set of regression discontinuity estimates for ADU permits per 1000 parcel-years. We report estimates using a 450 meter bandwidth from the Coastal Zone boundary and use triangle weights. * $p < .05$, ** $p < .01$, *** $p < .001$

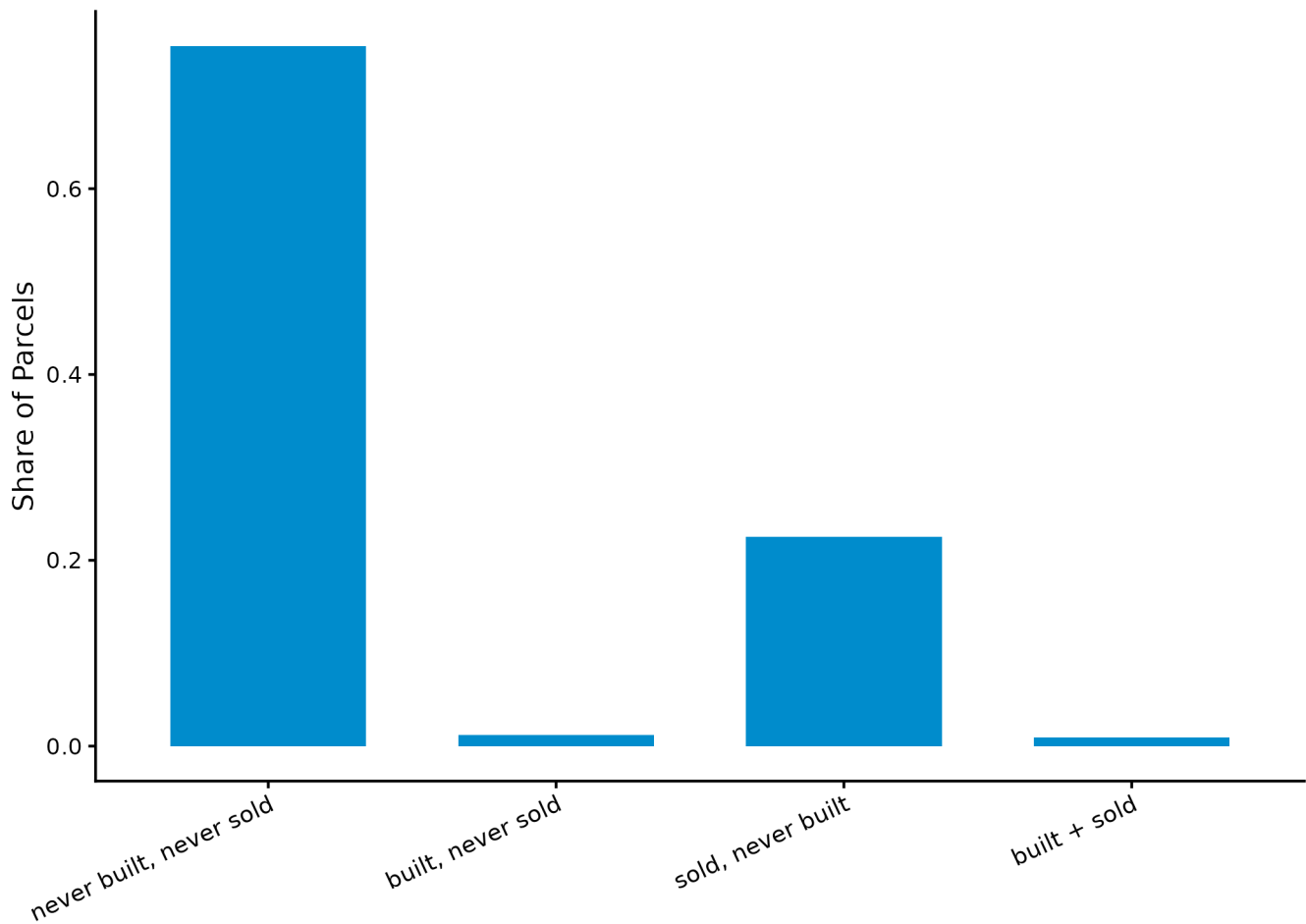
Table 7: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Property Tax Collections Per Acre within 450 Meters of the Coastal Zone Boundary

	<i>Dep.Var.: Log of Real Property Tax Per Acre</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>OutCZ · Post</i>	0.074*** (0.007)	0.041*** (0.006)	0.069*** (0.007)	0.036*** (0.006)	0.018*** (0.005)	0.013** (0.005)
1 st Degree Polynomial	Y	Y	Y	Y	Y	Y
County FEs	Y	Y	Y	Y		
Parcel FEs					Y	Y
Year FEs	Y	Y			Y	
Segment · Time FEs			Y	Y		Y
Pre-2016 Static Hedonics		Y		Y		
N	2 488 513	2 482 314	2 488 499	2 482 294	2 488 024	2 488 010
Adj. R^2	0.288	0.378	0.387	0.456	0.941	0.942
Mean Pre-2016	10.37	10.37	10.37	10.37	10.37	10.37
SD Pre-2016	1.333	1.333	1.333	1.333	1.333	1.333
In-CZ Mean Pre-2016	10.46	10.46	10.46	10.46	10.46	10.46
In-CZ SD Pre-2016	1.387	1.387	1.387	1.387	1.387	1.387

Notes: This table uses data from [Cotality \(2025\)](#) and [California Coastal Commission \(2014\)](#) to estimate Equation 3 for a panel on residential parcels between 2010 and 2024: $\ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3 f(DistCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable the log of the real property taxes paid per acre in 2025 USD. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 demarcating the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first degree or second degree polynomial form. In both polynomial specifications, distance slopes can change after the boundary. We also include a set of county fixed effects, λ_c , and a set of year fixed effects, τ_t . We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We cluster our robust standard errors at the parcel level. We report estimates using a 450 meter bandwidth from the Coastal Zone boundary. We present summary statistics for the log of the real land value per acre for both the entire sample from 2010 to 2016 and just the sample within the coastal zone from 2010 to 2016 at the bottom of the table. In Column 1, we present a specification only with a first degree polynomial, a set of county fixed effects, and a set of year fixed effects. In the even columns, we replace the first degree polynomial with the second degree polynomial. Then in Column 3 and 4, we include the set of hedonic fixed effects. Then in columns 5 through 8, we replace the year fixed effects with a segment by time fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. * $p < .05$, ** $p < .01$, *** $p < .001$

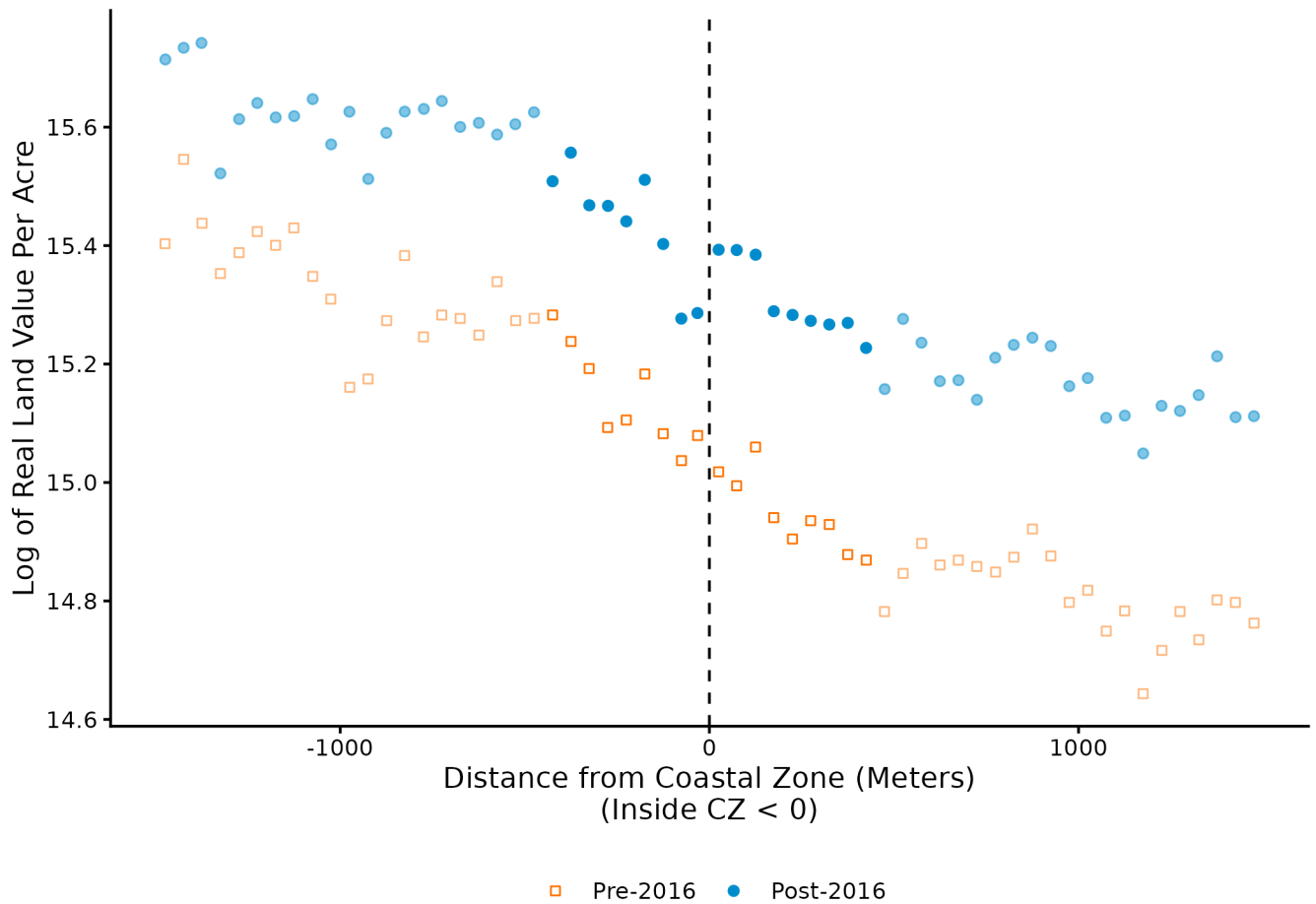
A Additional Exhibits

Figure A.1: Share of Parcels that Ever Transacted or Permitted an ADU between 2016 and 2024 in the Coastal 15 Counties



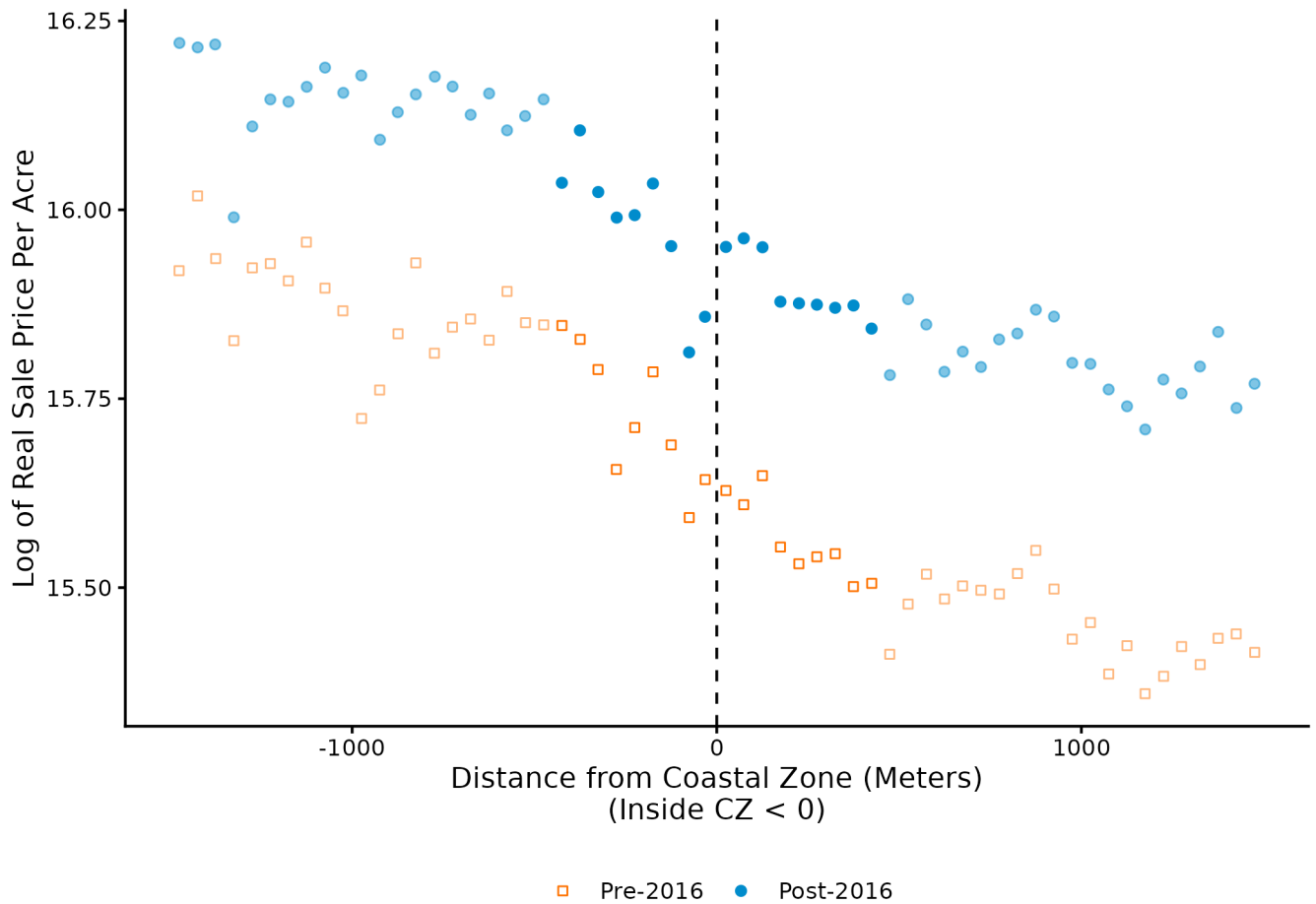
This figure uses data from [Cotality \(2025\)](#) and [California Department of Housing and Community Development \(2025\)](#) to show which parcels ever transact, ever build an ADU, or both between 2018 and 2024. Transactions here include both arms length and non-arms length transactions. 75% of parcels never build an ADU and never transact. 22.5% of parcels transact but never build an ADU, 2.5% of parcels build an ADU, with 1% both transacting and building and 1.5% building but not transacting. This figure uses data from all residential parcels within the coastal 15 counties.

Figure A.2: Trends in Real Land Values per Acre Around the Coastal Zone Boundary Pre- and Post- 2016 Reforms



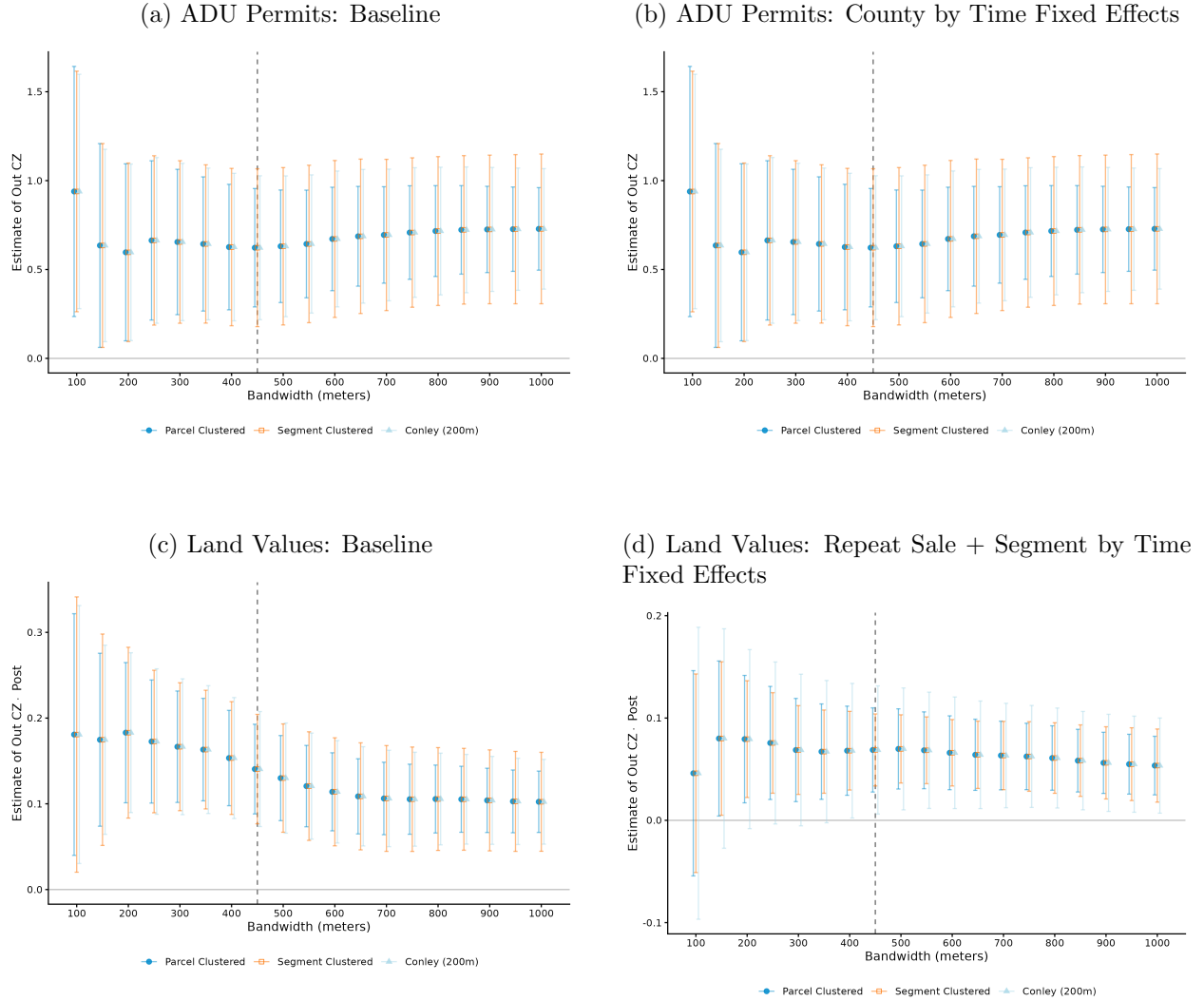
This figure uses data from [Cotality \(2025\)](#) and [California Coastal Commission \(2014\)](#) to show how residential land values change across the Coastal Zone boundary between 2010 and 2024. The horizontal axis shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the post 2016 permitting reforms. We group parcels in 50 meter bins. The vertical axis shows the log of the real land value per acre for each bin. The orange squares show the 2010-2016 distribution of land values in relation to the Coastal Zone boundary. The blue circles show the 2017 to 2024 distribution of land values in relationship to the Coastal Zone boundary. The difference in discontinuities estimate can be seen in comparing the pre- and post-2016 lines just before and then just after crossing the Coastal Zone boundary at 0.

Figure A.3: Change in Real Total Sale Prices per Acre Around the Coastal Zone Boundary Pre- and Post- 2016 Reforms



This figure uses data from [Cotality \(2025\)](#) and [California Coastal Commission \(2014\)](#) to show how residential total sale values change across the Coastal Zone boundary between 2010 and 2024. The horizontal axis shows each bin's distance from the Coastal Zone boundary, where negative values represent parcels inside the Coastal Zone that do not receive the post 2016 permitting reforms. We group parcels in 50 meter bins. The vertical axis shows the log of the real total sale price per acre for each bin. The orange squares show the 2010-2016 distribution of total sale values in relation to the Coastal Zone boundary. The blue circles show the 2017 to 2024 distribution of total sale values in relationship to the Coastal Zone boundary. The difference in discontinuities estimate can be seen in comparing the gap between the pre- and post-2016 lines just before and then just after crossing the Coastal Zone boundary at 0.

Figure A.4: Robustness of Accessory Dwelling Permitting and Real Land Value Per Acre Estimates to Bandwidth and Standard Error Choice



This figure shows estimates of Equation 2 in Panels A and B and 3 in Panel C and D for alternative choices for the sample bandwidth cutoff and standard errors. We show these (symmetric) sample bandwidths on the horizontal axis and the coefficients of interest on the vertical axis. Our mean squared error optimal bandwidth choice is marked with a vertical line. These estimates correspond to Column 1 in Table 3 in Panel A, a modified Column 4 in Table 3 in Panel B, Column 1 in Table 4 in Panel C, and Column 7 in Table 4 in Panel D. The outcome variable in Panels A and B is a dummy taking a value of 1 if a parcel gets a permit for an ADU in a particular year and 0 otherwise, divided by 1,000 for scaling purposes. The outcome variable in Panels C and D is the log of the real land value per acre in 2025 USD. All estimates use triangle weights. We present standard errors clustered by parcel in blue, clustered by Coastal Zone segment in orange, and [Conley \(1999\)](#) spatial correlation robust errors (200 meters) in light blue.

Table A.1: Cumulative Description of Sample Restrictions

Sample Restriction	Transaction Data (1)	Balanced Panel (2)
Total Observations	10,037,688	82,360,928
+ Residential Properties	8,093,225	66,608,192
+ Arms-Length Transactions	2,362,389	–
+ Assessed Value $\geq$ \$1,000	2,362,344	66,579,941
+ Valid Sale Price	2,322,650	–
+ Sale Volume Not Extreme	2,270,456	–
+ Valid Rooms/Units Ratio	2,270,456	66,482,505
+ Land Value Per Acre $>$ \$1	2,160,457	–
+ Valid Implied Age	2,160,457	66,482,424
Final Sample	2,160,457	66,482,424

Notes: This table presents the impact of sample restrictions for the full 15 coastal county samples using data from [Cotality \(2025\)](#), [California Coastal Commission \(2014\)](#), and [California Department of Housing and Community Development \(2025\)](#). We show how each restriction reduces the total number of observations cumulatively descending through the rows. To start, all parcels must have an identifier and a valid location (latitude and longitude). Column 1 shows how restrictions impact the transaction dataset. These include a restriction to small residential properties, generally single family houses but strictly just excluding large multifamily units and condos; restricting to arms-length (non family) transactions; limiting to parcels with assessed values greater than \$1000 USD; having a sale price within the 1st and 99th percentile of the sample distribution; not having an extreme number of transactions above the 99th percentile of the sample distribution per year; having a room to unit ratio within the 1st and 99th percentile of the sample distribution; having a land value per acre above \$1; and having an internally consistent structure age. In Column 2, we show these restrictions for the balanced panel dataset, which has one observation per parcel between 2010 and 2025, when the restriction is applicable since the data is only based on assessment records and not transaction records. Since we only have a single quarter of 2025 data from [Cotality \(2025\)](#), we stop our analysis sample at 2024.

Table A.2: Interaction of Sample Restrictions with the Coastal Zone Boundary in the Transaction and Parcel Datasets

	<i>Dep. Var.: Observation Dropped by a Sample Restriction</i>					
	2010–2024		2010–2016		2017–2024	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>OutCZ · Post</i>	−0.011* (0.004)	0.000 (0.000)				
<i>OutCZ</i>	0.001 (0.003)	−0.018*** (0.004)	0.002 (0.003)	−0.019*** (0.004)	−0.010** (0.003)	−0.019*** (0.004)
1 st Degree Polynomial	Y	Y	Y	Y	Y	Y
Difference in Discontinuities	Y	Y				
County FEs	Y	Y	Y	Y	Y	Y
Year FEs	Y	Y	Y	Y	Y	Y
N	600 027	4 437 135	300 983	2 070 663	299 044	2 366 472
Adj. R^2	0.017	0.104	0.017	0.104	0.017	0.104
Mean	0.827	0.273	0.832	0.272	0.822	0.273
SD	0.378	0.445	0.374	0.445	0.382	0.445
In-CZ Mean	0.841	0.318	0.845	0.318	0.837	0.318
In-CZ SD	0.366	0.466	0.362	0.466	0.370	0.466
Data	Transactions	Balanced Panel	Transactions	Balanced Panel	Transactions	Balanced Panel

Notes: The following table uses data from [Cotality \(2025\)](#) and [California Department of Housing and Community Development \(2025\)](#) to see how sample restrictions interact with our baseline regression equations. In both columns the outcome variable is a dummy variable for exclusion from the analysis sample, taking a value of 1 if an observation is trimmed from the final sample. In Column 1, we show the baseline version of Equation 3, which shows us that our sample exclusions are not changing over time. In Column 2, we show our baseline version of Equation 2 for only the 2018 and after sample in which we run our regressions. Here, we see that properties inside the Coastal Zone are more likely to be excluded. However, from Column 1 we can see that these restrictions likely don't change over time. * $p < .05$, ** $p < .01$, *** $p < .001$

Table A.3: Summary Statistics for the Full County Sample

<i>Panel A: Parcel Panel Data</i>	Mean	SD	Min	Max
	(1)	(2)	(3)	(4)
Share SFR	0.911	0.284	0.000	1.000
Dist. to the CZ	14 486	14 750	−44 394	108 133
Effective Build Year	1967	22.14	1850	2024
Bedrooms	3.366	1.063	1.000	11.00
Bathrooms	2.369	1.057	1.000	9.000
Building Sq. Ft.	1923	924.8	598.0	9999
Land Share	0.516	0.192	0.000	1.000
ADU Permit	0.001	0.038	0.000	1.000
Any Permit	0.002	0.047	0.000	1.000
ADU Suitable Pre-2016	0.880	0.325	0.000	1.000
Property Tax Per Acre	44 052	46 119	0.000	343 699
<i>Panel B: Transaction Data</i>	Mean	SD	Min	Max
	(1)	(2)	(3)	(4)
Share SFR	0.923	0.266	0.000	1.000
Dist. to the CZ	15 540	16 049	−42 121	103 660
Effective Build Year	1968	23.22	1850	2023
Bedrooms	3.328	1.045	1.000	11.00
Bathrooms	2.395	1.081	1.000	9.000
Building Sq. Ft.	1932	956.8	598.0	9999
Land Value Per Acre	3 832 004	4 426 889	7.878	49 570 267
Total Sale Price Per Acre	6 801 396	6 669 696	20 511	68 344 349
Land Share	0.530	0.194	0.000	1.000

Notes: This table presents summary statistics for the full 15 coastal county samples using data from [Cotality \(2025\)](#), [California Coastal Commission \(2014\)](#), and [California Department of Housing and Community Development \(2025\)](#). We show summary statistics from 2010 to 2024 on the full county sample using the balanced parcel panel dataset in Panel A and the transaction dataset in Panel B. We show summary statistics for the share of parcels that are zoned for single family use, the parcel’s euclidean distance to the Coastal Zone boundary, the parcel’s effective build year, the parcel structure’s number of bedroom, bathrooms, building square footage, land values per acre, total sale value per acre, and the share of value assessed to come from the land. In Panel B, we also show the share of parcels in a particular year that have an ADU permit or any building permit associated with them. We also show the share of parcels that were ADU suitable before 2016 as defined as having a ground slope under 20% and more than 2,000 sq.ft. available as excess lot space. Then, we also show the average property taxes paid per acre.

Table A.4: Second Degree Polynomial Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Land Values within 450 Meters of the Coastal Zone Boundary

	<i>Dep.Var.: Log of Real Land Value Per Acre</i>			
	(1)	(2)	(3)	(4)
<i>OutCZ · Post</i>	0.197*** (0.041)	0.176*** (0.041)	0.116*** (0.033)	0.093** (0.033)
2 nd Degree Polynomial	Y	Y	Y	Y
County FEs	Y	Y	Y	Y
Year FEs	Y	Y		
Segment · Time FEs			Y	Y
Pre-2016 Static Hedonics		Y		Y
N	103 766	102 901	103 718	102 850
Adj. R^2 .	0.524	0.546	0.662	0.678
Mean Pre-2016	15.01	15.01	15.01	15.01
SD Pre-2016	1.246	1.246	1.246	1.246
In-CZ Mean Pre-2016	15.14	15.14	15.14	15.14
In-CZ SD Pre-2016	1.317	1.317	1.317	1.317

Notes: This table uses data from [Cotality \(2025\)](#) and [California Coastal Commission \(2014\)](#) to estimate Equation 3 for a transactions on residential parcels between 2010 and 2024: $\ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(DistCZ_i \times Post_t) + \beta_3 DistCZ_i + \beta_4(DistCZ_i^2 \times Post_t) + \beta_5 DistCZ_i^2 + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable the log of the real land value per acre in 2025 USD. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 demarcating the implementation of the first ADU reforms. $DistCZ_i$ is a function of a parcel's distance to the Coastal Zone boundary, which uses a second degree polynomial here. We also include a set of county fixed effects, λ_c , and a set of year fixed effects, τ_t . We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We present summary statistics for the permit dummy for both the entire sample and just the sample within the coastal zone at the bottom of the table. In Column 1, we present a specification only with a second degree polynomial, a set of county fixed effects, and a set of year fixed effects. Then in Columns 2 and 4, we include the set of hedonic fixed effects. Then in columns 3 and 4, we replace the year fixed effects with a segment by time fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. We cluster our robust standard errors at the parcel level. We report estimates using a 450 meter bandwidth from the Coastal Zone boundary and use triangle weights. * $p < .05$, ** $p < .01$, *** $p < .001$

Table A.5: Second Degree Polynomial Spatial Regression Discontinuity Estimates of the Impacts of Permitting Reforms on Accessory Dwelling Unit Permits after 2018 within 450 Meters of the Coastal Zone Boundary

	<i>Dep. Var.: ADUs per 1,000 Parcel-Years</i>			
	(1)	(2)	(3)	(4)
<i>OutCZ</i>	0.676** (0.254)	0.616* (0.259)	0.512* (0.255)	0.481 (0.256)
2 nd Degree Polynomial	Y	Y	Y	Y
County FEs	Y	Y	Y	Y
Year FEs	Y	Y		
Segment · Time FEs			Y	Y
Pre-2016 Static Hedonics		Y		Y
N	1 505 988	1 485 253	1 505 974	1 485 239
Adj. R^2	0.001	0.002	0.002	0.003
Mean	2.248	2.248	2.248	2.248
SD	47.36	47.36	47.36	47.36
In-CZ Mean	1.900	1.900	1.900	1.900
In-CZ SD	43.55	43.55	43.55	43.55

Notes: This table uses data from [Cotality \(2025\)](#), [California Department of Housing and Community Development \(2025\)](#), and [California Coastal Commission \(2014\)](#) to estimate Equation 2 for a balanced set of residential parcels between 2018 and 2024: $Y_{ict} = \alpha + \beta_1 OutCZ_i + \beta_2 DistCZ_i + \beta_3 (DistCZ_i \cdot OutCZ_i) + \beta_4 DistCZ_i^2 + \beta_5 (DistCZ_i^2 \cdot OutCZ_i) + \theta'_{(i)} \gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable is a dummy taking a value of 1 if parcel i has an ADU permit associated with it in year t , and we scale this outcome by dividing by 1,000 for display purposes. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's permitting reforms for ADUs. $DistCZ_i$ is a function of a parcel's distance to the Coastal Zone boundary, which uses a second degree polynomial here. We also include a set of county fixed effects, λ_c , and a set of year fixed effects, τ_t . We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We present summary statistics for the permit dummy for both the entire sample and just the sample within the coastal zone at the bottom of the table. In Column 1, we present a specification only with a first degree polynomial, a set of county fixed effects, and a set of year fixed effects. Then in Columns 2 and 4, we include the set of hedonic fixed effects. Then in column 3, we replace the year fixed effects with a segment by time fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. We cluster our robust standard errors at the parcel level. We report estimates using a 450 meter bandwidth from the Coastal Zone boundary and use triangle weights. * $p < .05$, ** $p < .01$, *** $p < .001$

Table A.6: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Alternatively Measured Land Values within 450 Meters of the Coastal Zone Boundary Using Time-Varying Hedonics

	<i>Dep.Var.: Log of Real Total Sale Value Per Acre</i>			
	(1)	(2)	(3)	(4)
<i>OutCZ · Post</i>	0.111*** (0.023)	0.152*** (0.036)	0.063*** (0.019)	0.078** (0.029)
1 st Degree Polynomial	Y		Y	
2 nd Degree Polynomial		Y		Y
County FEs	Y	Y	Y	Y
Year FEs	Y	Y		
Segment · Time FEs			Y	Y
Updating Hedonics	Y	Y	Y	Y
N	103 737	103 737	103 689	103 689
Adj. R^2	0.526	0.526	0.666	0.666
Mean Pre-2016	15.64	15.64	15.64	15.64
SD Pre-2016	1.080	1.080	1.080	1.080
In-CZ Mean Pre-2016	15.75	15.75	15.75	15.75
In-CZ SD Pre-2016	1.147	1.147	1.147	1.147

Notes: This table uses data from [Cotality \(2025\)](#) and [California Coastal Commission \(2014\)](#) to estimate Equation 3 for a transactions on residential parcels between 2010 and 2024: $\ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3 f(DistCZ_i) + \theta'_{(it)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable the log of the real total sale price per acre in 2025 USD. In each specification, we use a set of binned time varying hedonic fixed effects, $\theta'_{(it)}$, to capture the share of the total value associated with the market values of the current structural characteristics of the parcel. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 demarking the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first degree or second degree polynomial form. In both polynomial specifications, distance slopes can change after the boundary. We also include a set of county fixed effects, λ_c , and a set of year fixed effects, τ_t . ε_{ict} is an error term. We cluster our robust standard errors at the parcel level. We report estimates using a 450 meter bandwidth from the Coastal Zone boundary. We present summary statistics for the log of the real total sale price per acre for both the entire sample from 2010 to 2016 and just the sample within the coastal zone from 2010 to 2016 at the bottom of the table. In Column 1, we present a specification only with a first degree polynomial, the set of time varying hedonic fixed effects, a set of county fixed effects, and a set of year fixed effects. In the even columns, we replace the first degree polynomial with the second degree polynomial. Then in Column 3 and 4, we replace the year fixed effects with a segment by time fixed effect that limits comparisons to only be within each parcel's closest segment of the Coastal Zone boundary. * $p < .05$, ** $p < .01$, *** $p < .001$

Table A.7: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Land Values within 450 Meters of the Coastal Zone Boundary Dropping Voluntarily Adopting Segments

	<i>Dep. Var.: Log of Real Land Value Per Acre</i>			
	(1)	(2)	(3)	(4)
<i>OutCZ · Post</i>	0.144*** (0.028)	0.089*** (0.024)	0.080*** (0.022)	0.066** (0.022)
1 st Degree Polynomial	Y	Y	Y	Y
County FEs	Y	Y		
Parcel FEs			Y	Y
Year FEs	Y		Y	
Segment · Time FEs		Y		Y
N	92 961	92 914	44 700	44 563
Adj. R^2	0.544	0.681	0.912	0.916
Mean Pre-2016	15.01	15.01	15.02	15.02
SD Pre-2016	1.246	1.246	1.245	1.245
In-CZ Mean Pre-2016	15.14	15.14	15.14	15.14
In-CZ SD Pre-2016	1.317	1.317	1.316	1.316

Notes: This table uses data from [Cotality \(2025\)](#) and [California Coastal Commission \(2014\)](#) to estimate Equation 3 for a transactions on residential parcels between 2010 and 2024: $\ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3f(DistCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable the log of the real land value per acre in 2025 USD. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 demarcating the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first degree or second degree polynomial form. We use a first degree polynomial, where distance slopes can change after the boundary. We also include a set of county fixed effects, λ_c , and a set of year fixed effects, τ_t . We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We cluster our robust standard errors at the parcel level. We report estimates using a 450 meter bandwidth from the Coastal Zone boundary. We present summary statistics for the log of the real land value per acre for both the entire sample from 2010 to 2016 and just the sample within the coastal zone from 2010 to 2016 at the bottom of the table. In Columns 1 we show our baseline difference in discontinuities estimates. In Columns 2 and 4, we swap the year fixed effects for segment by time fixed effects. Then, in Columns 3 and 4, we replace our pooled specification with a repeat sale model that includes a parcel fixed effect. All specifications use triangle weights. All specifications drop segments of the Coastal Zone that overlap with Santa Cruz, Santa Barbara, Encinitas, San Mateo, and Pacifica. * $p < .05$, ** $p < .01$, *** $p < .001$

Table A.8: Spatial Difference in Discontinuities Estimates of the Impacts of Permitting Reforms on Total Sale Prices within 450 Meters of the Coastal Zone Boundary

	<i>Dep.Var.: Log of Real Total Sale Price Per Acre</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>OutCZ · Post</i>	0.112*** (0.024)	0.098*** (0.023)	0.062** (0.020)	0.049* (0.019)	0.046** (0.016)	0.030* (0.015)
1 st Degree Polynomial	Y	Y	Y	Y	Y	Y
County FEs	Y	Y	Y	Y		
Parcel FEs					Y	Y
Year FEs	Y	Y			Y	
Segment · Time FEs			Y	Y		Y
Pre-2016 Static Hedonics		Y		Y		
Repeat-Sale Sample					Y	Y
N	103 766	102 901	103 718	102 850	49 440	49 287
Adj. R^2	0.496	0.526	0.643	0.665	0.933	0.937
Mean Pre-2016	15.64	15.64	15.64	15.64	15.64	15.64
SD Pre-2016	1.080	1.080	1.080	1.080	1.080	1.080
In-CZ Mean Pre-2016	15.75	15.75	15.75	15.75	15.75	15.75
In-CZ SD Pre-2016	1.147	1.147	1.147	1.147	1.145	1.145

Notes: This table uses data from [Cotality \(2025\)](#) and [California Coastal Commission \(2014\)](#) to estimate Equation 3 for a transactions on residential parcels between 2010 and 2024: $\ln(Y)_{ict} = \alpha + \beta_1(OutCZ_i \times Post_t) + \beta_2(f(DistCZ_i) \times Post_t) + \beta_3 f(DistCZ_i) + \theta'_{(i)}\gamma + \lambda_c + \tau_t + \varepsilon_{ict}$, where i indexes parcels, c denotes counties, and t denotes the year. The outcome variable the log of the real total sale price per acre in 2025 USD. $OutCZ_i$ is a dummy taking a value of 1 when a parcel is outside the Coastal Zone and therefore receives the state's post-2016 permitting reforms for ADUs. $Post_t$ is a dummy taking a value of 1 after 2016 demarcating the implementation of the first ADU reforms. $f(DistCZ_i)$ is a function of a parcel's distance to the Coastal Zone boundary, which can take a first degree polynomial form. Distance slopes can change after the boundary. We also include a set of county fixed effects, λ_c , and a set of year fixed effects, τ_t . We also include a set of hedonic fixed effects that capture pre-2016 parcel characteristics in $\theta'_{(i)}$. ε_{ict} is an error term. We cluster our robust standard errors at the parcel level. We report estimates using a 450 meter bandwidth from the Coastal Zone boundary. We present summary statistics for the log of the real land value per acre for both the entire sample from 2010 to 2016 and just the sample within the coastal zone from 2010 to 2016 at the bottom of the table. In Column 1, we present a specification only with a set of county fixed effects and a set of year fixed effects. In Columns 2 and 4 we add hedonic fixed effects. Then in Column 3, 4, and 6, we include the segment by time fixed effects. Finally, in columns 5 and 6, we include a parcel fixed effect and use a sample restricted to parcels that transact both between 2010 to 2016 and 2017 to 2024. All specifications use triangle weights and report parcel clustered standard errors. * $p < .05$, ** $p < .01$, *** $p < .001$

B Appendix: Change Log from August 2025 Filed Pre-Analysis Plan

C Appendix: A Real Option Model of ADU Construction

We propose a simple option model to help understand our results. We use a net present value framework, which is common in real estate, in a similar style of model to [Severen and Plantinga \(2018\)](#) and [McMillen and O’Sullivan \(2013\)](#). Here, the utility for a parcel is given by the equation:

$$U_t = s + lap_t \quad (C1)$$

where s is the annual rents generated by the existing structure and land, l is the parcel’s suitability for an ADU, which we will call simply quality, a is an indicator taking a value of 1 if the parcel builds an ADU, and p_t is the time-specific (net) market price for the ADU.¹⁰⁴

There are two time periods in this model in which the owner can elect to put an ADU on her parcel. The market prices for the ADU will take a known value of p_1 in the first period, and then it will realize either a high or low value in period two and forever after following $p_t = \rho p_H + (1 - \rho)p_L$. Assume that $p_L < 0 < p_1 < p_H$. For simplicity, the choice of building an ADU is not reversible and owners face a penalty if they build an ADU in the low-price state of the world that scales with their lot quality, l . ADUs can only be built in period one or period two, for simplicity. The lot then accrues rents given its structure composition.

First, let’s consider the case in which an owner builds an ADU on her parcel in period one, before she knows the long-term price realization. We can write this as a discounted flow of utilities:

$$V_Y^1 = s + alp_1 + \sum_{t=1}^{\infty} \frac{s + alE(p_t)}{(1+i)^t} - c \quad (C2)$$

Here, we discount using a known interest rate i . Building an ADU incurs the net cost of c , which can include the cost of materials and labor used for both the structure and the permitting process. The reforms considered in this paper, in general, reduce the one-time cost of building an ADU, c .

Next, we consider the case in which the owner doesn’t build an ADU, which can be written as:

$$V_N^2 = \sum_{t=0}^{\infty} \frac{s}{(1+i)^t} \quad (C3)$$

where the owner simply enjoys the amenity value of her back yard and non-ADU structures, regardless of the price of an ADU.

In period 2, the owner is able to observe the realized price, p_t , that will persist indefinitely before choosing to build her ADU. Since we have assumed that $p_L < 0$, we know the owner will only elect to build an ADU in the event of the high draw for ADU values. We can write the value of building an ADU in the second period as:

¹⁰⁴The price of an ADU could be thought of as net of any lost amenity values in the land. The ADU price could also reflect owner utility or market rents.

$$V_Y^2|p_H = \sum_{t=0}^{\infty} \frac{s}{(1+i)^t} + \sum_{t=1}^{\infty} \frac{alp_H}{(1+i)^t} - \frac{c}{1+i} \quad (C4)$$

When the owner waits and builds an ADU in period 2 after realizing high prices, she enjoys the land amenity as before and the persistent streams of ADU rents. By deferring building by one period, the cost of building is also discounted by a period. However, under certain cost-conditions, the owner will never elect to build an ADU.

$$V_Y^2|p_H - V_N^2 = \sum_{t=1}^{\infty} \frac{alp_H}{(1+i)^t} - \frac{c}{1+i} \quad (C5)$$

Which is less than zero when:

$$c > \sum_{t=1}^{\infty} \frac{alp_H}{(1+i)^{t-1}} \quad (C6)$$

An owner will never build an ADU when the cost to build exceeds the total future revenue streams generated by the ADU. We will call this $\bar{c}$. We interpret the suite of ADU reforms as effectively lowering c such that $c < \bar{c}$ for a greater number of parcels.

If $c < \bar{c}$, the value of waiting to build an ADU in period two is a weighted average of the value of building an ADU given the high prices, and the value of never building an ADU. We can write this as:

$$\begin{aligned} EV_W &= \begin{cases} \rho(V_Y^2|p_H) + (1-\rho)(V_N^2), & c < \bar{c} \\ V_N^2, & c \geq \bar{c} \end{cases} \\ &= \begin{cases} \sum_{t=0}^{\infty} \frac{s}{(1+i)^t} + \rho[\sum_{t=1}^{\infty} \frac{alp_H}{(1+i)^t} - \frac{c}{1+i}], & c < \bar{c} \\ \sum_{t=0}^{\infty} \frac{s}{(1+i)^t}, & c \geq \bar{c} \end{cases} \end{aligned} \quad (C7)$$

We can then think about the value of waiting to build an ADU in period 2 instead of building an ADU in period 1, before the prices are realized (when $c < \bar{c}$). We can write this option value as:

$$EV_W - V_Y^1 = c[1 - \frac{\rho}{1+i}] - al[p_1 + (1-\rho) \sum_{t=1}^{\infty} \frac{p_L}{(1+i)^t}] \quad (C8)$$

where the first term is the difference in build costs between building in period 1 and period 2 and the second term is the stream of payments if the low state for ADU prices is realized, representing a loss to the owner. The value of waiting exceeds the value of building an ADU in period 1 if:

$$c[1 - \frac{\rho}{1+i}] > al[p_1 + (1-\rho) \sum_{t=1}^{\infty} \frac{p_L}{(1+i)^t}] \quad (C9)$$

Therefore, it is better to wait to build an ADU if the value of the difference between the expected build costs from deferring building is greater than the rents from the ADU in the low price state of the world. Notice that as the cost of building an ADU, c , decreases, then the likelihood of deferring an ADU should decrease as well.¹⁰⁵ Similarly, the option value decreases when the probability of realizing high rents for an ADU, ρ , is higher.

Finally, the option value is a function of the parcels' lot quality, l . This value could reflect the parcel's desirability in terms of neighborhood, location to jobs, or other land-specific amenities. We can write the derivative of the value of waiting with respect to land amenities as follows:

$$\frac{\partial EV_W - V_Y^1}{\partial l} = -p_1 + (\rho - 1) \sum_{t=1}^{\infty} \frac{p_L}{(1+i)^t} \quad (C10)$$

The option value is increasing in lot quality when the risk adjusted losses from building an ADU when low rents materialize are less than the negative baseline rent in period 1:

$$(1 - \rho) \sum_{t=1}^{\infty} \frac{p_L}{(1+i)^t} < -p_1 \quad (C11)$$

Finally, we can derive the option value with respect to the probability of realizing high prices for ADUs.

$$\frac{\partial EV_W - V_Y^1}{\partial \rho} = \sum_{t=1}^{\infty} \frac{al p_L}{(1+i)^t} - \frac{c}{(1+i)} \quad (C12)$$

Since the deflated costs and losses from building an ADU when low rents materialize are both negative, the option value is decreasing in ρ . This means, as the likelihood of realizing high rents for ADUs increases, the option values diminish.

Solving this model numerically, we can show these patterns in Figure C.1.¹⁰⁶ For a given level of lot quality for ADUs, l , total (net) cost of building c , and probability of high rents ρ , we show which parcels will elect to either build an ADU today in period 1 (dark blue), wait to decide to build an ADU in period 2 (orange), and never build an ADU (light blue). In the first panel, when high rents are certain, there is no option value and parcels either build an ADU right away or never build, due to low lot quality and prohibitive building costs. When the probability of high future rents, $\rho = 0.5$, decreases in panel 2, a wedge of middle cost and high lot quality parcels find it beneficial to wait to decide to build an ADU, reflecting a positive option value. When we decrease the probability of high rents even further, $\rho = 0.1$, even more lots decide to wait before building.

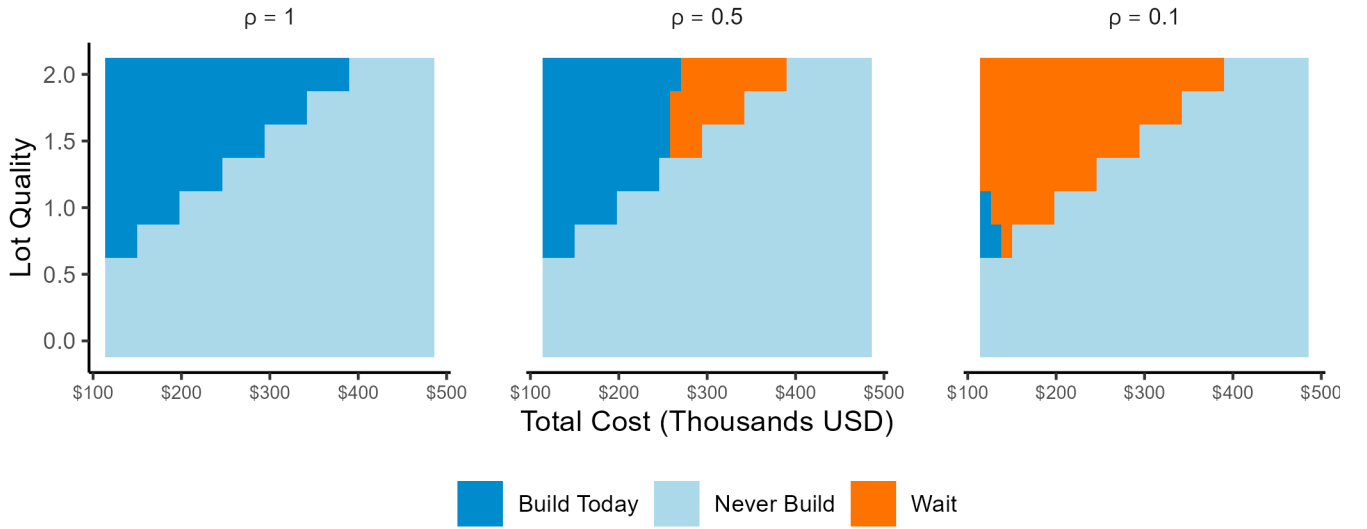
This simple model illuminates a few key elements of our results. First, a sufficiently high cost

¹⁰⁵Notice that $[1 - \frac{\rho}{(1+i)}] > 0$, indicating the derivative of Equation C8 with respect to c is positive.

¹⁰⁶The model is solved for 10 years. Interest rates are set at 10% (i) and rents from existing structures (s) are set at \$50,000 per year. We use a 1,200 square foot ADU that rents for \$1,200 per month in the base period, \$2,400 with high rents, and -\$1,200 with low rents for a base land quality l . We calculate solutions for all combinations of $l \in \{0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2\}$, $c \in \{100, 150, 200, 250, 300, 350, 400, 450, 500\}$ with c in thousands of USD, and $\rho \in \{0.1, 0.5, 1\}$. These build costs reflect industry ranges from Henson (2023).

of building an ADU, $c > \bar{c}$, can make building an ADU unattractive regardless of the realization of rents, p_t . Permitting reforms, which lower the cost of building an ADU as part of c , should increase the number of ADUs and the option value of being able to build an ADU if the reforms move $c < \bar{c}$. Second, when the value of the ADU is uncertain, whether from rental market conditions or other factors, an option value can arise since there is value in the ability to gain additional information about future market conditions. This option value arises when development is delayed. The exact share of lots that see increases in option value versus that build immediately after reform depends on the lot suitability for ADUs, the costs of building, the probability of high rents to recoup the cost of production, and the possible downside risk from low rents. Third, given that many variables, which may differ across space, impact the decision to build an ADU, wait, or never build, we believe that our results likely reflect an environment in which costs to build are prohibitively high. Construction costs for an ADU are in excess of \$200 per square foot for many parcels, and the likelihood of high rents in the future are less than certain, possibly even below 50% given that so few parcels elect to build an ADU ([Henson, 2023](#)).

Figure C.1: Model Solutions by Land Quality and Total Net Cost to Build an ADU



This figure shows solutions to the model presented in Sub-section C. The model is solved for 10 years. Interest rates are set at 10% (i) and rents from existing structures (s) are set at \$50,000 per year. We use a 1,200 square foot ADU that rents for \$1,200 per month in the base period, \$2,400 with high rents, and -\$1,200 with low rents for a base land quality l . We calculate solutions for all combinations of $l \in \{0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2\}$, $c \in \{100, 150, 200, 250, 300, 350, 400, 450, 500\}$ with c in thousands of USD, and $\rho \in \{0.1, 0.5, 1\}$. These build costs reflect industry ranges from [Henson \(2023\)](#).

D Appendix: Changes in Rents

This Appendix presents estimates of the impact of permitting reform on percentage changes in rent. We use the Coastal Zone boundary as a source of conditionally-exogenous variation in permitting rates for ADUs. The key limitation of this exercise is our reliance on rents taken from the 2016 and 2024 vintages of the American Community Survey (U.S. Census Bureau, 2016, 2024).¹⁰⁷

Our outcome variable of interest is the percent change in the real mean gross rent, which we calculate by taking the total gross rent and divide by the number of rental units in the block group to get the mean gross rent in each year.¹⁰⁸ We then calculate the percentage change in gross rent for each block group between 2016 and 2024.

We show estimates with one of two versions of our endogenous variable. We first use the cumulative amount of ADUs permitted between 2018 and 2024 from California Department of Housing and Community Development (2025), which we match to each block group. Second, we use the cumulative amount of ADUs permitted between 2018 and 2024 and scale it by the block group’s total housing unit count from 2016. The latter construction is our preferred measure of ADU permitting.

Finally, our instrument is constructed using the block group’s shapefile and the Coastal Zone shapefile from California Coastal Commission (2014). The instrument is the share of block group area that falls within the Coastal Zone. This instrument is valued between 0 and 1, with a value of 0 corresponding to no Coastal Zone overlap (1,041 block groups in our analysis sample), a value of 1 corresponding to a block group that is completely within the Coastal Zone (301 block groups in our analysis sample), and values between 0 and 1 reflecting partial overlap of the block group and the Coastal Zone (844 block groups in our analysis sample).

We restrict our final sample to the 2,186 block groups with a centroid within 2,000 meters of the Coastal Zone boundary. All 15 coastal counties represented within the final sample, with about 67% of the sample in San Diego, Los Angeles, and Orange counties. We weight our estimates by each block group’s 2016 housing unit count.¹⁰⁹

Our two stage least squares (2SLS) equations are as follows:

$$ADU_b = \pi_0 + \pi_1 InCZ_b + X'_b \gamma + u_b \quad (D1a)$$

$$\% \Delta Rent_b = \delta_0 + \delta_1 \widehat{ADU}_b + X'_b \rho + v_b \quad (D1b)$$

where b indexes block groups. $InCZ_b$ is the Coastal Zone area share for block group b . ADU_b is either cumulative ADU permits from 2018 to 2024 for block group b or the cumulative ADU permits from 2018 to 2024 per 1,000 housing units in 2016 for block group b . $\% \Delta Rent_b$ is the percentage

¹⁰⁷We work with 2020 block groups. We crosswalk 2010 block groups to 2020 geometries using IPUMS NHGIS (2026).

¹⁰⁸All monetary values are in 2025 USD (U.S. Bureau of Economic Analysis, 2025).

¹⁰⁹Due to the high frequency at which ADUs are in tracts with high ratios of owners to renters, we weight by all housing units rather than by rented housing units.

change in mean gross rent between 2016 and 2024 for block group b . We include a control set, X'_b , meant to add both add precision to our estimates and to address concerns that the Coastal Zone is correlated with other drivers of rent growth. The controls come in 5 groups: terrain and density,¹¹⁰ age structure,¹¹¹ job access,¹¹² other permitting,¹¹³ and amenities.¹¹⁴ We present all estimates with spatial correlation robust standard errors with a 1,000 meter cutoff (Conley, 1999).

Our instrumental variables estimates rest on two assumptions (Cunningham, 2021). First, the share that a block group overlaps with the Coastal Zone only impacts changes in rents through its impacts on permitting for ADUs. Given the tenuous nature of this assumption, our control set tries to partially account for the other ways that the Coastal Zone may influence rental price changes. This approach is imperfect, but local to the boundary it is more reasonable than on a sample that extends far inland. Second, the block group Coastal Zone overlap share needs to be a strong predictor of ADU permitting. We can provide some evidence in support of our second assumption below, but the exclusion restriction we cannot test directly in our just identified setting.¹¹⁵

WE show our first stage estimates from Equation D1a in Table D.1. In Columns 1 and 2 we show our unscaled endogenous variable, and in Columns 3 and 4 we show our scaled endogenous variable. Odd columns don't include controls and even Columns add the full control set. Our preferred estimates are in Column 4. The Coastal Zone overlap instrument is a strong predictor of decreased ADU permitting, with block groups completely inside of the Coastal Zone permitting permitting 5 fewer ADUs per 1,000 housing units than block groups completely outside the Coastal Zone. The F-statistic for the excluded instrument is 19.5. These estimates are large in magnitude, as the average block group in our sample has only 12 ADU permits per 1,000 housing units.

With our first stage estimates in hand, we proceed to estimate our full 2SLS model. We present estimates of Equations D1b in Table D.2. In Columns 1 and 2 we show our unscaled endogenous variable, and in Columns 3 and 4 we show our scaled endogenous variable. Odd columns don't include controls and even Columns add the full control set. Our preferred estimates are in Column 4. Our point estimates in Column 4 suggest that 1 additional ADU permit per 250 housing units in a block group reduced rents by about 1 percentage point between 2016 and 2024. So, a roughly 1 standard deviation increase in ADU permitting lowers rents by 4 percentage points, in a setting where rents grew 28% over the sample period. However, these estimates are not significant at traditional significance levels. We show a companion set of estimates that add each control set one at a time in Table D.3.

¹¹⁰Elevation, block group population per square kilometer.

¹¹¹2016 population share over age 65.

¹¹²Distance to the nearest CBD, change in jobs within 10 miles between 2016 and 2024, average commute distance for the block group.

¹¹³Cumulative non-ADU permits from 2018 to 2024 (when the endogenous variable is scaled by housing units we scale this variable as well.).

¹¹⁴A dummy variable that take a value of 1 when the closest shoreline category is a sandy beach and a value of 0 otherwise, 2016 to 2024 service sector jobs within 3 miles.

¹¹⁵We also need a monotonicity assumption on the relationship between our instrument and the endogenous variable: more overlap with the Coastal Zone needs to weakly decrease permitting activity. Controlling for coastal access helps address this.

We don't want to put too much weight on these estimates. They are too imprecise to draw strong conclusions about the impacts on ADUs and rents. However, these estimates are consistent with findings from [Bhatt \(2023\)](#). [Bhatt \(2023\)](#) uses a difference in difference approach with census tract data from the ACS and permits from San Francisco, Los Angeles, and San Diego. He compares tracts with different shares of land zoned for single family and finds no impact of ADU permitting on rents. Both of these results are likely reflective of the low levels of ADU permitting as a share of housing units and the imprecision of the ACS rental data.

Table D.1: First Stage Estimates of Block Group Coastal Zone Overlap on Cumulative Change in ADU Permitting (2018-2024)

	<i>Dep. Var.: Cumulative ADU Permits</i>			
	Count (units)		Per 1,000 Housing Units	
	(1)	(2)	(3)	(4)
Coastal Zone Area Share	-1.578*	-2.585***	-3.922**	-5.471***
	(0.714)	(0.694)	(1.215)	(1.240)
Controls		Y		Y
N	2186	2185	2186	2185
R^2	0.005	0.058	0.009	0.073
First-stage F	4.883	13.88	10.41	19.46
Mean	7.002	7.002	11.69	11.69
SD	8.995	8.995	16.53	16.53
In-CZ Mean	6.142	6.142	9.336	9.336
In-CZ SD	8.015	8.015	12.77	12.77

Notes: This table uses data from [U.S. Census Bureau \(2016\)](#), [U.S. Census Bureau \(2024\)](#), [California Department of Housing and Community Development \(2025\)](#), [California Department of Fish and Wildlife \(2026\)](#), [California Coastal Commission \(2014\)](#), [US Census Bureau \(2023\)](#), [U.S. Geological Survey \(2026\)](#), and [Holian \(2019\)](#) to estimate Equation D1a. In Columns 1 and 2 we show our unscaled endogenous variable, and in Columns 3 and 4 we show our scaled endogenous variable. Odd columns don't include controls and even Columns add the full control set. Our preferred estimates are in Column 4. We show first stage F-statistics on the excluded instrument, dependent variable means and standard deviations for the analysis sample, and dependent variable means and standard deviations calculated on the share of block groups with more than half of their area within the Coastal Zone. Standard errors follow [Conley \(1999\)](#), computed on the block-group centroid with a distance cutoff of 1,000 m. We weight estimates by 2016 ACS housing units. * $p < .05$, ** $p < .01$, *** $p < .001$

Table D.2: 2SLS Estimates of ADU Permitting Effects on Percent Changes in Block Group Gross Rent (2016-2024)

<i>Dep. Var.:</i>	<i>Per. Ch. Mean Gross Rent (2016-2024)</i>			
	Count		Per 1,000 Units	
	(1)	(2)	(3)	(4)
Cumulative ADU Permits	-1.551 (1.446)	-0.595 (0.848)	-0.627 (0.558)	-0.254 (0.402)
Controls	Y		Y	
N	2126	2126	2126	2126
First-stage F	4.883	13.88	10.41	19.46
Mean	27.72	27.72	27.72	27.72
SD	35.83	35.83	35.83	35.83
In-CZ Mean	29.66	29.66	29.66	29.66
In-CZ SD	34.57	34.57	34.57	34.57
ADU Permits Mean	7.002	7.002	11.69	11.69
ADU Permits SD	8.995	8.995	16.53	16.53

Notes: This table uses data from [U.S. Census Bureau \(2016\)](#), [U.S. Census Bureau \(2024\)](#), [California Department of Housing and Community Development \(2025\)](#), [California Department of Fish and Wildlife \(2026\)](#), [California Coastal Commission \(2014\)](#), [US Census Bureau \(2023\)](#), [U.S. Geological Survey \(2026\)](#), and [Holian \(2019\)](#) to estimate Equation D1b. In Columns 1 and 2 we show our unscaled endogenous variable, and in Columns 3 and 4 we show our scaled endogenous variable. Odd columns don't include controls and even Columns add the full control set. Our preferred estimates are in Column 4. We show first stage F-statistics on the excluded instrument, dependent variable means and standard deviations for the analysis sample, dependent variable means and standard deviations calculated on the share of block groups with more than half of their area within the Coastal Zone, and mean and standard deviations for our endogenous regressors. Standard errors follow [Conley \(1999\)](#), computed on the block-group centroid with a distance cutoff of 1,000 m. We weight estimates by 2016 ACS housing units. * $p < .05$, ** $p < .01$, *** $p < .001$

Table D.3: Sensitivity of 2SLS Estimates of ADU Permitting Effects on Percent Changes in Block Group Gross Rent (2016-2024)

<i>Dep. Var.:</i>	<i>Per. Ch. Mean Gross Rent (2016-2024)</i>						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Cumulative ADU Permits	-0.627 (0.558)	-0.386 (0.399)	-0.462 (0.541)	-0.682 (0.617)	-0.656 (0.566)	-0.571 (0.577)	-0.254 (0.402)
Terrain & Density		Y					Y
Age Structure			Y				Y
Job Access				Y			Y
Non-ADU Permitting					Y		Y
Amenities						Y	Y
N	2126	2126	2126	2126	2126	2126	2126
First-stage F	10.41	18.46	10.59	9.194	10.38	10.07	19.46
Mean	27.72	27.72	27.72	27.72	27.72	27.72	27.72
SD	35.83	35.83	35.83	35.83	35.83	35.83	35.83
In-CZ Mean	29.66	29.66	29.66	29.66	29.66	29.66	29.66
In-CZ SD	34.57	34.57	34.57	34.57	34.57	34.57	34.57
ADU Permits Mean	11.69	11.69	11.69	11.69	11.69	11.69	11.69
ADU Permits SD	16.53	16.53	16.53	16.53	16.53	16.53	16.53

Notes: This table uses data from [U.S. Census Bureau \(2016\)](#), [U.S. Census Bureau \(2024\)](#), [California Department of Housing and Community Development \(2025\)](#), [California Department of Fish and Wildlife \(2026\)](#), [California Coastal Commission \(2014\)](#), [US Census Bureau \(2023\)](#), [U.S. Geological Survey \(2026\)](#), and [Holian \(2019\)](#) to estimate Equation D1b. In Columns 1 and 2 we show our unscaled endogenous variable, and in Columns 3 and 4 we show our scaled endogenous variable. Odd columns don't include controls and even Columns add the full control set. Our preferred estimates are in Column 4. We show first stage F-statistics on the excluded instrument, dependent variable means and standard deviations for the analysis sample, dependent variable means and standard deviations calculated on the share of block groups with more than half of their area within the Coastal Zone, and mean and standard deviations for our endogenous regressors. Standard errors follow [Conley \(1999\)](#), computed on the block-group centroid with a distance cutoff of 1,000 m. We weight estimates by 2016 ACS housing units. * $p < .05$, ** $p < .01$, *** $p < .001$

E Appendix: Additional Data Details

E.1 Data Linking

In this Appendix, we describe how we link project-level permit data from [California Department of Housing and Community Development \(2025\)](#) (hereafter just “permit data”) with our assessment and transaction records from [Cotality \(2025\)](#) (hereafter just “property data”). We do permit-property linking for the 15 coastal counties only.

First, we process and separate instances in the permit data where multiple addresses are entered under a single row. Examples could be entries with multiple street addresses such as ”100-200 UCSD Econ. Way” or ”10 Taco Drive; 20 Ferrari Parkway.” Single rows could have multiple APNs as well. We parse these APNs and street addresses to get a single row per unique location. Approximately 28,000 of the rows covering multiple locations involved ADU permits.

We then implement the following hierarchical matching process. After a permit address matches to a property, we count it as matched and remove it from the pool of unmatched permits for the subsequent step. All unique permit and property entries start in step 1. If a permit matches in step 1, it doesn’t proceed to step 2, and so on. Permits can only match to one [Cotality \(2025\)](#) property, but properties can be associated with multiple permits.

We are able to match 94.1% of all permit rows and 98% of all ADU permit rows for the 15 coastal county permit data to appropriate property data.

Table E.1: [California Department of Housing and Community Development \(2025\)](#) Permit to [Cotality \(2025\)](#) Parcel Hierarchical Matching Algorithm and Match Rates

Step	Key	Rule	Matches	
			All	ADU
1	APN	Exact match on the cleaned APN.	329,332	184,312
2	House number, street, quadrant, (city or ZIP)	Exact match on house number $\times$ street $\times$ quadrant, further requiring agreement on city or ZIP. City or zip can be missing, but not both.	6,882	413
3	Google Places API	Both sides' addresses are sent to the Google Places <code>findplacefromtext</code> endpoint via <code>googleway::google_find_place</code> . A permit is matched to a property when both return the same <code>place_id</code> . Permit results are tried in Google's rank order (ranks 1–5); property results use rank 1 only.	40,570	8,7656
4	Previous APN	Match permit APN to CoreLogic <code>PREVIOUS PARCEL NUMBER FORMATTED</code> , and permit <code>PRIOR_APN</code> to the current CoreLogic APN. This captures renumbered parcels.	261	21
Matched			277,045	193,511
Un-matched			23,548	3,738

Notes: All unique permit and parcel entries start in step 1. If a permit matches in step 1, it doesn't proceed to step 2, and so on. Permits can only match to one [Cotality \(2025\)](#) parcel, but parcels can be associated with multiple permits.

E.2 Estimating Property Tax Shelter

In this Appendix, we describe how we construct market value estimates for each residential parcel and construct the tax shelter variable used in the main text.

Our tax shelter variable is meant to capture the degree to which a parcel is benefiting from Proposition 13’s assessment rules. Our target is the property tax shelter variable from the text:

$$\begin{aligned} PropTaxShelter_{i,2016} &= \frac{M_i - A_i}{M_i} \\ &= \frac{Prop13Gap_i}{M_i} \end{aligned} \tag{E1}$$

where i indexes parcels and all values are for 2016. M_i is the parcel’s market value in 2016, which in most cases must be estimated. A_i is the assessed value provided by [Cotality \(2025\)](#), which is the assessor’s valuation of the property for levying property taxes, which follows the rules of Proposition 13. Under Proposition 13, a newly purchased house is assessed at market value. Each year, the amount the assessed value A_i can grow is capped at 2%. Given the increase in home prices in California over the past decades, if an owner doesn’t sell their property they can accrue substantial property tax shelter as the market value of their home exceeds the assessed valuation that they are taxed on ([Legislative Analyst’s Office, 2015](#); [Ferreira, 2010](#); [Wasi and White, 2005](#); [Green et al., 2026](#)).

For the vast majority of parcels, we have their 2016 assessment directly and use that directly when available for A_i . If we are missing an assessment, then we take the last available assessment and grow the nominal value by 2% per year until we reach 2016. Only about 5% of parcels need to have their 2016 assessed value imputed.

To estimate the market value of a parcel in 2016, we build a hedonic pricing model. We estimate one model per county for our 15 coastal county transaction sample. We estimate equations of the following form on data from 2010 to 2016:

$$\begin{aligned} \ln(P_{it}) &= \alpha_{tract(i)} + \theta_{hed(i,t)} + \lambda_{luse(i)} + \psi_{parking(i)} + \phi_{stories(i)} + \omega_{coast(i)} + \delta_t \\ &\quad + \vec{X}_{it}' \gamma + \vec{M}_{it}' \zeta + \varepsilon_{it} \end{aligned} \tag{E2}$$

where i indexes parcels and t indexes years. $\ln(P_{it})$ is the log of the nominal transaction price. First, we include a set of fixed effects, each with an added “missing” bin to account for incomplete records. We include fixed effect sets for the following variables: 2020 census tracts ($\alpha_{tract(i)}$), the updating hedonic fixed effect set as described in the main text ($\theta_{hed(i,t)}$)¹¹⁶, land use class ($\lambda_{luse(i)}$),

¹¹⁶We construct fixed effects in a similar style to [Yagan \(2019\)](#), which include the full interaction of the following groups, with bins for missing values in each category. The bins are: above or below county median effective age (and a missing bin), land use category (single family, other, or missing), number of bedrooms (0-2, 3-4, 5 and up, and missing), number of bathrooms (0-2, 2-4, 4 and up, and missing), and within county tertiles of building square footage (and a missing bin).

number of available parking spaces top coded at 10 ($\psi_{parking(i)}$), the number of stories top coded at 4 ($\phi_{stories(i)}$), type of nearest coastline from [California Department of Fish and Wildlife \(2026\)](#) ($\omega_{coast(i)}$)¹¹⁷, and a year fixed effect (δ_t). Then, we include continuous covariates in the vector $\vec{X}_{it}$. Given the frequency of missing variables, we include a parallel set of dummies ($\vec{M}_{it}$) that take a value of 1 when a continuous variable is missing and zero otherwise. We then set the value of the missing variable to 0 to preserve observations. The continuous variables we include are: ground slope from [U.S. Geological Survey \(2026\)](#), elevation from [U.S. Geological Survey \(2026\)](#), distance to the coastline from [California Department of Fish and Wildlife \(2026\)](#), distance to the nearest CBD from [Holian \(2019\)](#), total lot acreage, ground floor building square feet, jobs (by workplace) within 3 miles and 10 miles of the parcel’s block group centroid from [US Census Bureau \(2023\)](#), and resident area jobs in the block group with parcel i from [US Census Bureau \(2023\)](#).¹¹⁸

We run these estimates on our 2010-2016 transaction data to fit each of the 15 county models. Then, we project these estimates onto the full balanced parcel panel for each unique parcel in 2016. If the parcel transacts in 2016, then we use that as its market price. To undo the logarithm, we follow [Duan \(1983\)](#). If a parcel doesn’t have a transaction and doesn’t have an estimate do to missing data, then we use a nearest-neighbors imputation. This is necessary in tracts with low transaction volumes. We find the nearest 3 parcels geographically within the same bedroom count, a square footage within 5% tolerance of the target, and lot acreage within 5% tolerance of the target. About 10% of parcels have an estimated tax gap outside of the unit interval, and we truncate these values to fall within that interval.

¹¹⁷The types are: rocky shores, coastal marsh, beaches, hardened shores, tidal flats, and unclassified.

¹¹⁸We attach all data to 2020 block groups using the 2010 to 2020 crosswalk from [IPUMS NHGIS \(2026\)](#).

F Appendix: Theoretical Model

F.1 The City's Problem

The city here takes the permitting and property tax regimes as givens, and chooses the level of head taxes to maximize the welfare of the incumbent parcel owners.¹¹⁹ The city solves the following optimization problem:¹²⁰

$$\begin{aligned} \max_h CW(h, g) = & \frac{1}{2} \sum_m \iiint \left[\overbrace{F_\alpha(\alpha_m^*)(P_{0,m}(l) + \Omega_m(l) + O)}^{\text{Incumbent Sellers}} \right. \\ & \left. + \underbrace{\int_{\alpha_m^*}^{\infty} (w + \text{OwnOpt}_m(l, k)) f_\alpha(\alpha) d\alpha}_{\text{Incumbent Stayers}} \right] dF_k dF_B dF_l \end{aligned} \quad (\text{F1})$$

Here, city welfare is a function of four pieces: the sale price and outside option for incumbents who sell from both permitting jurisdictions and the own-welfare and own-ADU build option for incumbents who don't sell their lots from both permitting jurisdictions.¹²¹ The city picks the head tax subject to the balanced budget, given by:

$$\begin{aligned} g = h + \tau \left(\frac{1}{2} \sum_m \iiint \left[\overbrace{(1 - F_\alpha(\alpha_m^*))(B + G(l) \mathbb{1}\{(1 - \frac{\tau}{r})G(l) > k + \kappa_m\})}^{\text{Incumbents benefiting from Prop 13}} \right. \right. \\ \left. \left. + \underbrace{F_\alpha(\alpha_m^*) P_m^{Fin}(l)}_{\text{Entrants Taxed at Full Value}} \right] dF_k dF_B dF_l \right) \\ = h + \tau \text{Base}(h, g) \end{aligned} \quad (\text{F2})$$

The property tax component of the balanced budget constraint highlights a feedback loop in which cities harvest additional taxes from *entrants* compared to incumbents under Proposition 13. The property tax base responds to the chosen head tax through two channels. First, as the head tax increases, the marginal entrant values the properties in the city less, eroding the sales price of parcels. Then, through a smaller channel, a higher head tax churns parcels, flipping them from Proposition 13 advantaged parcels held by incumbents to fully taxed and developed lots held by entrants.¹²²

The city's optimal choice of the head tax, h^* follows a simple Samuelson Rule.

¹¹⁹Notice that the entrants are indifferent with the outside option, so their welfare is policy invariant.

¹²⁰For convenience, we define $w \equiv \frac{v(l) + \alpha + u(g) - h - \theta D - \tau B}{r}$, the incumbent's welfare from staying on her own parcel.

¹²¹We have given each jurisdiction equal weight, but this could be relaxed.

¹²²We can write this expression as $T \equiv \tau \frac{\partial \text{Base}}{\partial h} = \frac{\tau}{r + \tau} \left[\tau \left(\frac{1}{2} \sum_m \iiint F_\alpha(\alpha_m^*) dF_k dF_B dF_l \right) - \left(\frac{1}{2} \sum_m \iiint f_\alpha(\alpha_m^*) (P_m^{Fin}(l) - B - G \mathbb{1}\{(1 - \frac{\tau}{r})G(l) > k + \kappa_m\}) dF_k dF_B dF_l \right) \right]$.

$$g^* = u'^{-1}(1) \implies h^* = u'^{-1}(1) - \tau Base(h^*, u'^{-1}(1)) \quad (\text{F3})$$

The government selects a head tax up and until the residents value a dollar of service equal to a dollar of tax relief.¹²³

¹²³If, as mentioned earlier, entrants value services more than incumbents, then the city's equilibrium solution overprovides the service relative to the Samuelson Rule, and the marginal cost of public funds is less than 1.